

Supplementary Appendix

Tables

Table 1: Summary of Predictive Performance (AUC) Across Chronic Disease Flags.

This table reports the number of positive cases and the Area Under the Receiver Operating Characteristic Curve (AUC) for each modeled disease condition using the XGBoost classifier. AUC values quantify the model’s ability to discriminate between cases and controls. Higher AUC indicates better predictive performance. Predictive accuracy was moderate for common conditions like Hypertension (AUC = 0.678) and Type 2 Diabetes (AUC = 0.729), while rarer conditions such as Chronic Hepatitis and Asthma exhibited lower AUCs, reflecting limited training data and class imbalance.

Disease Flag	Positive Cases	AUC
Has Hypertension	3315	0.677725
Has Type 2 Diabetes	1940	0.728739
Has Hypertensive Heart Disease	1477	0.620277
Has Type 1 Diabetes	82	0.619078
Has Asthma	32	0.529488
Has Epilepsy	32	0.587534
Has Chronic Hepatitis	32	0.451427

Table 2: Ch-Statistic Summary Summary of Statistical Significance and SHAP-Based Feature Contributions Across Disease and Sociodemographic Groups.

This comprehensive table aggregates two complementary analyses: **Tab1:** Chi-Square Association Tests: For each chronic disease, we report the Chi-squared test statistic (chi2), degrees of freedom (dof), and associated p-value (p value) assessing the independence between the disease and each socio demographic group (tribe, religion, income, education). Statistically significant associations (e.g., $p < 0.05$) suggest meaningful disparities in disease prevalence across these subgroups. For example, hypertension and Type 2 diabetes show highly significant variation by tribe and religion. **Tab2:** SHAP Feature Importance: For each disease-group pair, we provide the mean absolute SHAP values that quantify the magnitude of each feature’s contribution to the model's

prediction. Higher values indicate greater influence on disease risk prediction. Key features consistently influencing model outputs include BMI, religion affiliations, and types of cooking oil (e.g., palm oil, vegetable oil), underscoring their relevance in shaping health outcomes across Nigerian populations. This dual-view framework enables both statistical validation and model interpretability for uncovering actionable health disparities.

Tab1: Ch-Statistic Summary <i>Summary of Statistical Significance</i>				
Group	Disease	chi2	P value	dof
Tribe	Has Hypertension	1601.446	0	342
Tribe	Has Hypertensive Heart Disease	801.3073	7.03E-39	342
Tribe	Has Acute Myocardial Infarction	16.23648	1	342
Tribe	Has Type 2 Diabetes	2032.685	0	342
Tribe	Has Type 1 Diabetes	308.0206	0.906454	342
Tribe	Has Alcoholic Liver Disease	54.91963	1	342
Tribe	Has Fibrosis And Cirrhosis Of Liver	118.2206	1	342
Tribe	Has Chronic Kidney Disease	39.85492	1	342
Tribe	Has Asthma	1522.934	0	342
Tribe	Has Depressive Episode	75.10267	1	342
Tribe	Has Polyarthrosis	190.5664	1	342
Religion	Has Hypertension	412.0295	7.24E-86	6
Religion	Has Hypertensive Heart Disease	6.191814	0.40205	6

Religion	Has Acute Myocardial Infarction	0.010674	1	6
Religion	Has Type 2 Diabetes	306.7057	2.99E-63	6
Religion	Has Type 1 Diabetes	24.17069	0.000486	6
Religion	Has Alcoholic Liver Disease	3.818785	0.701184	6
Religion	Has Fibrosis And Cirrhosis Of Liver	0.215856	0.999807	6
Religion	Has Chronic Kidney Disease	0.954624	0.987274	6
Religion	Has Asthma	4.599536	0.5961	6
Religion	Has Depressive Episode	1.829442	0.934697	6
Religion	Has Polyarthrosis	1.829442	0.934697	6
Income	Has Hypertension	210.8594	9.24E-43	6
Income	Has Hypertensive Heart Disease	77.44815	1.20E-14	6
Income	Has Acute Myocardial Infarction	3.966233	0.681246	6
Income	Has Type 2 Diabetes	126.2489	7.92E-25	6
Income	Has Type 1 Diabetes	11.61915	0.071025	6
Income	Has Alcoholic Liver Disease	6.455257	0.374167	6
Income	Has Fibrosis And Cirrhosis Of Liver	9.75578	0.135321	6

Income	Has Chronic Kidney Disease	2.236306	0.896734	6
Income	Has Asthma	3.846727	0.697409	6
Income	Has Depressive Episode	13.64126	0.033911	6
Income	Has Polyarthrosis	0.748842	0.993376	6
Education	Has Hypertension	182.2573	7.96E-34	10
Education	Has Hypertensive Heart Disease	32.25457	0.000363	10
Education	Has Acute Myocardial Infarction	3.463613	0.968319	10
Education	Has Type 2 Diabetes	76.64426	2.28E-12	10
Education	Has Type 1 Diabetes	12.65154	0.243812	10
Education	Has Alcoholic Liver Disease	12.78697	0.235827	10
Education	Has Fibrosis And Cirrhosis Of Liver	5.05722	0.887323	10
Education	Has Chronic Kidney Disease	5.504093	0.855067	10
Education	Has Asthma	22.39077	0.013233	10
Education	Has Depressive Episode	13.24154	0.210479	10
Education	Has Polyarthrosis	17.50324	0.063944	10
Tab2: SHAP-Based Feature Contributions Across Disease and Sociodemographic Groups				
Disease Flag	Group	Feature	Mean Abs SHAP	

Has Type 1 Diabetes	Tribe	BMI	41.92658	
Has Type 1 Diabetes	Tribe	Religion Christian	0.71624	
Has Type 1 Diabetes	Tribe	Oil Type Palm Oil	0.679871	
Has Type 1 Diabetes	Tribe	Religion Muslim	0.281206	
Has Type 1 Diabetes	Tribe	Oil Type Vegetable Oil	0.21771	
Has Type 1 Diabetes	Tribe	Oil Type Ground Oil	0.092032	
Has Type 1 Diabetes	Tribe	Oil Type I Don't Know/Not Sure	0.006335	
Has Type 1 Diabetes	Tribe	Tribe Yoruba Ondo	0.004505	
Has Type 1 Diabetes	Tribe	Tribe Yoruba Kwara	0.004505	

Has Type 1 Diabetes	Tribe	Tribe Yoruba Kogi	0.004505	
Has Type 1 Diabetes	Religion	BMI	52.12078	
Has Type 1 Diabetes	Religion	Oil Type Palm Oil	0.553007	
Has Type 1 Diabetes	Religion	Oil Type Vegetable Oil	0.374386	
Has Type 1 Diabetes	Religion	Religion Others	0.166667	
Has Type 1 Diabetes	Religion	Religion Atheist	0.166667	
Has Type 1 Diabetes	Religion	Religion Christian	0.166667	
Has Type 1 Diabetes	Religion	Religion Prefer not to Answer	0.166667	
Has Type 1 Diabetes	Religion	Religion Traditional	0.166667	

Has Type 1 Diabetes	Religion	Religion MusLIM	0.166667	
Has Type 1 Diabetes	Religion	Tribe Ukwani (Kwale) Delta	0.09731	
Has Asthma	Tribe	BMI	42.70558	
Has Asthma	Tribe	Religion Christian	0.700473	
Has Asthma	Tribe	Oil Type Palm Oil	0.678465	
Has Asthma	Tribe	Religion Muslism	0.297617	
Has Asthma	Tribe	Oil Type Vegetable Oil	0.211385	
Has Asthma	Tribe	Oil Type Groundnut Oil	0.098884	
Has Asthma	Tribe	Oil Type I Don't Know/Not Sure	0.00728	
Has Asthma	Tribe	Tribe 265	0.004425	
Has Asthma	Tribe	Tribe 492	0.004425	
Has Asthma	Tribe	Tribe Ukwani (Kwale) Delta	0.004425	

Has Asthma	Religion	BMI	51.90283	
Has Asthma	Religion	Oil Type Palm Oil	0.493374	
Has Asthma	Religion	Oil Type Vegetable Oil	0.300547	
Has Asthma	Religion	Oil Type Groundnut Oil	0.183254	
Has Asthma	Religion	Tribe Hausa Kano	0.178792	
Has Asthma	Religion	Religion Prefer Not to Answer	0.166667	
Has Asthma	Religion	Religion Christian	0.166667	
Has Asthma	Religion	Religion Others	0.166667	
Has Asthma	Religion	Religion Atheist	0.166667	
Has Asthma	Religion	Religion Muslim	0.166667	
Has Hypertensive Heart Disease	Tribe	BMI	42.59139	

Has Hypertensive Heart Disease	Tribe	Religion Christian	0.725496	
Has Hypertensive Heart Disease	Tribe	Oil Type Palm Oil	0.698084	
Has Hypertensive Heart Disease	Tribe	Religion Muslim	0.272444	
Has Hypertensive Heart Disease	Tribe	Oil Type Vegetable Oil	0.216807	
Has Hypertensive Heart Disease	Tribe	Oil Tyoe Groundnut Oil	0.077747	
Has Hypertensive Heart Disease	Tribe	Tribe Yoruba Lagos	0.004566	
Has Hypertensive Heart Disease	Tribe	Tribe Yoruba Ondo	0.004566	

Has Hypertensive Heart Disease	Tribe	Tribe Yoruba Ogun	0.004566	
Has Hypertensive Heart Disease	Tribe	Tribe Yandang Adamawa	0.004566	
Has Hypertensive Heart Disease	Religion	BMI	57.83145	
Has Hypertensive Heart Disease	Religion	Oil Type Palm Oil	0.552008	
Has Hypertensive Heart Disease	Religion	Oil Type Vegetable Oil	0.235916	
Has Hypertensive Heart Disease	Religion	Religion Prefer Not to Answer	0.2	
Has Hypertensive Heart Disease	Religion	Religion Traditional	0.2	

Has Hypertensive Heart Disease	Religion	Religion Christian	0.2	
Has Hypertensive Heart Disease	Religion	Religion Others	0.2	
Has Hypertensive Heart Disease	Religion	Religion Muslim	0.2	
Has Hypertensive Heart Disease	Religion	Oil Type Groundnut Oil	0.187339	
Has Hypertensive Heart Disease	Religion	Tribe Hausa Kano	0.149395	
Has Hypertension	Tribe	BMI	25.81747	
Has Hypertension	Tribe	Religion Christian	0.743904	
Has Hypertension	Tribe	Oil Type Palm Oil	0.693254	

Has Hypertension	Tribe	Religion Muslim	0.254653	
Has Hypertension	Tribe	Oil Type Vegertable Oil	0.217414	
Has Hypertension	Tribe	Oil Type Groundnut Oil	0.06697	
Has Hypertension	Tribe	Oil Type I Don't Know/Not Sure	0.010909	
Has Hypertension	Tribe	Oil Type Margarine or Cooking Butter	0.009136	
Has Hypertension	Tribe	Tribe Ukwane (Kwale) Delta	0.004464	
Has Hypertension	Tribe	Tribe Uncinda Kaduna	0.004464	
Has Hypertension	Religion	BMI	25.99914	
Has Hypertension	Religion	Oil Type Palm Oil	0.632212	

Has Hypertension	Religion	Oil Type Vegetable Oil	0.274079	
Has Hypertension	Religion	Religion Prefer Not to Answer	0.2	
Has Hypertension	Religion	Religion Others	0.2	
Has Hypertension	Religion	Religion Muslim	0.2	
Has Hypertension	Religion	Religion Christian	0.2	
Has Hypertension	Religion	Religion Traditional	0.2	
Has Hypertension	Religion	Tribe Hausa Sokoto	0.118618	
Has Hypertension	Religion	Tribe Hausa Kaduna	0.095228	
Has Epilepsy	Tribe	BMI	35.1768	
Has Epilepsy	Tribe	Religion Christian	0.743026	

Has Epilepsy	Tribe	Oil Type Palm Oil	0.669148	
Has Epilepsy	Tribe	Religion Muslim	0.254868	
Has Epilepsy	Tribe	Oil Type Vegetable Oil	0.223961	
Has Epilepsy	Tribe	Oil Type Groundnut Oil	0.094427	
Has Epilepsy	Tribe	Oil Type Prefer Not to Answer	0.007173	
Has Epilepsy	Tribe	Tribe Yoruba Oyo	0.004425	
Has Epilepsy	Tribe	Tribe 144	0.004425	
Has Epilepsy	Tribe	Tribe Yoruba Lagos	0.004425	
Has Epilepsy	Religion	BMI	47.93493	
Has Epilepsy	Religion	Oil Type Palm Oil	0.555548	
Has Epilepsy	Religion	Oil Type Vegetable Oil	0.351248	
Has Epilepsy	Religion	Tribe Izon (Ijaw) Bayelsa	0.18849	

Has Epilepsy	Religion	Religion Others	0.166667	
Has Epilepsy	Religion	Religion Christian	0.166667	
Has Epilepsy	Religion	Religion Atheist	0.166667	
Has Epilepsy	Religion	Religion Muslim	0.166667	
Has Epilepsy	Religion	Religion Prefer Not to Answer	0.166667	
Has Epilepsy	Religion	Religion Traditional	0.166667	
Has Type 2 Diabetes	Tribe	BMI	25.457336	
Has Type 2 Diabetes	Tribe	Religion Christian	0.703667	
Has Type 2 Diabetes	Tribe	Oil Type Palm Oil	0.673111	
Has Type 2 Diabetes	Tribe	Religion Muslim	0.294905	
Has Type 2 Diabetes	Tribe	Oil Type Vegetable Oil	0.221578	

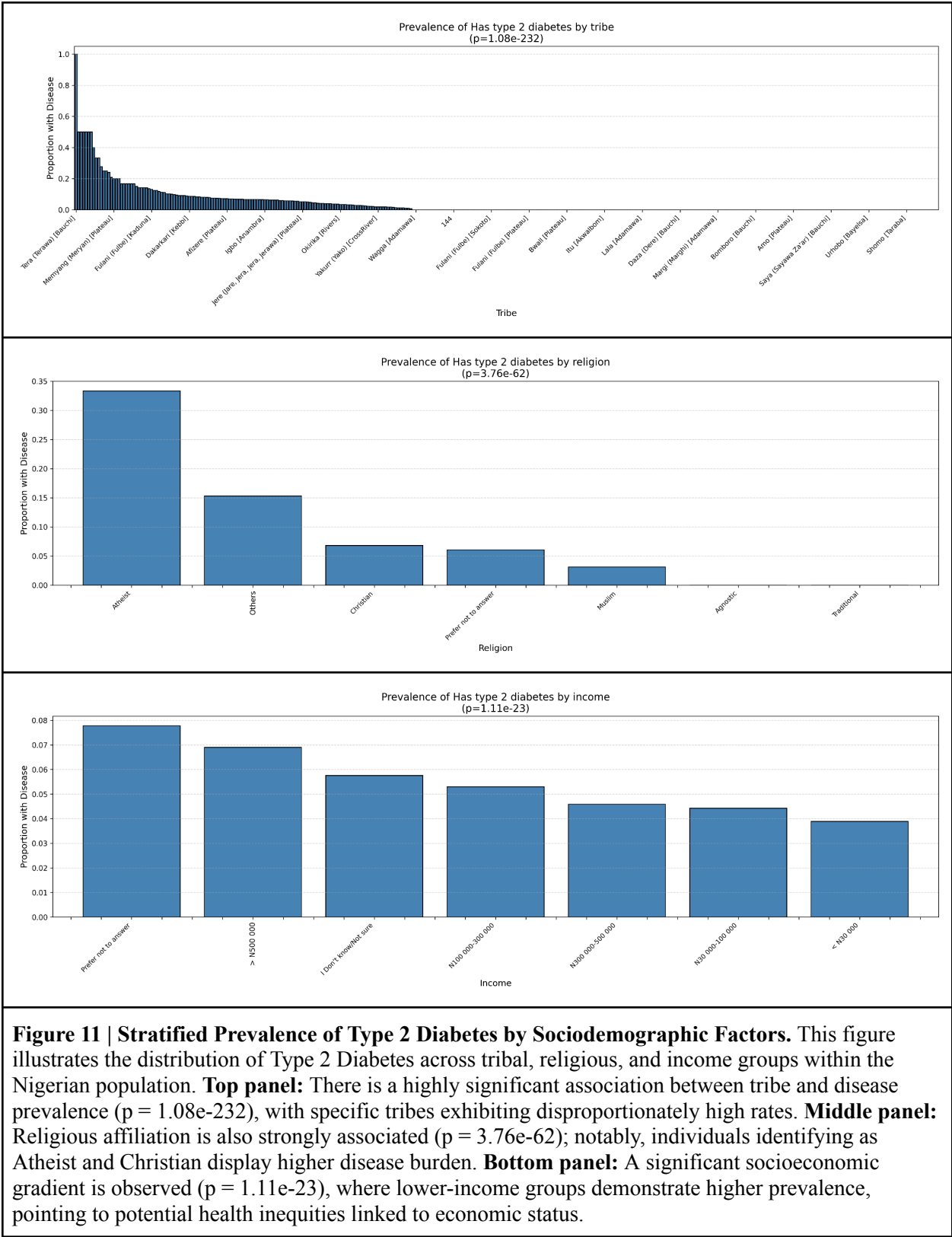
Has Type 2 Diabetes	Tribe	Oil Type Groundnut Oil	0.095933	
Has Type 2 Diabetes	Tribe	Oil Type I Don't Know/Not Sure	0.005779	
Has Type 2 Diabetes	Tribe	Tribe 66	0.004292	
Has Type 2 Diabetes	Tribe	Tribe Anioma Delta	0.004292	
Has Type 2 Diabetes	Tribe	Tribe Anioma Edo	0.004292	
Has Type 2 Diabetes	Religion	BMI	26.12487	
Has Type 2 Diabetes	Religion	Oil Type Palm Oil	0.460193	
Has Type 2 Diabetes	Religion	Oil Type Vegetable Oil	0.424128	
Has Type 2 Diabetes	Religion	Tribe Hausa Kano	0.183898	

Has Type 2 Diabetes	Religion	Tribe Izon (Ijaw) Bayelsa	0.18084	
Has Type 2 Diabetes	Religion	Religion Muslim	0.166667	
Has Type 2 Diabetes	Religion	Religion Christian	0.166667	
Has Type 2 Diabetes	Religion	Religion Others	0.166667	
Has Type 2 Diabetes	Religion	Religion Atheist	0.166667	
Has Type 2 Diabetes	Religion	Religion Traditional	0.166667	
Has Chronic Hepatitis	Tribe	BMI	25.20762	
Has Chronic Hepatitis	Tribe	Religion Christian	0.73037	
Has Chronic Hepatitis	Tribe	Oil Type Palm Oil	0.681849	

Has Chronic Hepatitis	Tribe	Religion Muslim	0.268221	
Has Chronic Hepatitis	Tribe	Oil Type Vegetable Oil	0.20999	
Has Chronic Hepatitis	Tribe	Oil Type Groundnut Oil	0.092322	
Has Chronic Hepatitis	Tribe	Oil Type I Don't/Not Sure	0.009402	
Has Chronic Hepatitis	Tribe	Tribe Yoruba Osun	0.004444	
Has Chronic Hepatitis	Tribe	Tribe Yoruba Kwara	0.004444	
Has Chronic Hepatitis	Tribe	Tribe Yoruba Kogi	0.004444	
Has Chronic Hepatitis	Religion	BMI	27.44751	
Has Chronic Hepatitis	Religion	Oil Type Palm Oil	0.440001	

Has Chronic Hepatitis	Religion	Oil Type Vegetable Oil	0.362885	
Has Chronic Hepatitis	Religion	Oil Type Groundnut Oil	0.193135	
Has Chronic Hepatitis	Religion	Tribe Hausa Kano	0.182986	
Has Chronic Hepatitis	Religion	Religion Atheist	0.166667	
Has Chronic Hepatitis	Religion	Religion Muslim	0.166667	
Has Chronic Hepatitis	Religion	Religion Prefer Not to Answer	0.166667	
Has Chronic Hepatitis	Religion	Religion Others	0.166667	
Has Chronic Hepatitis	Religion	Religion Christian	0.166667	

Figures



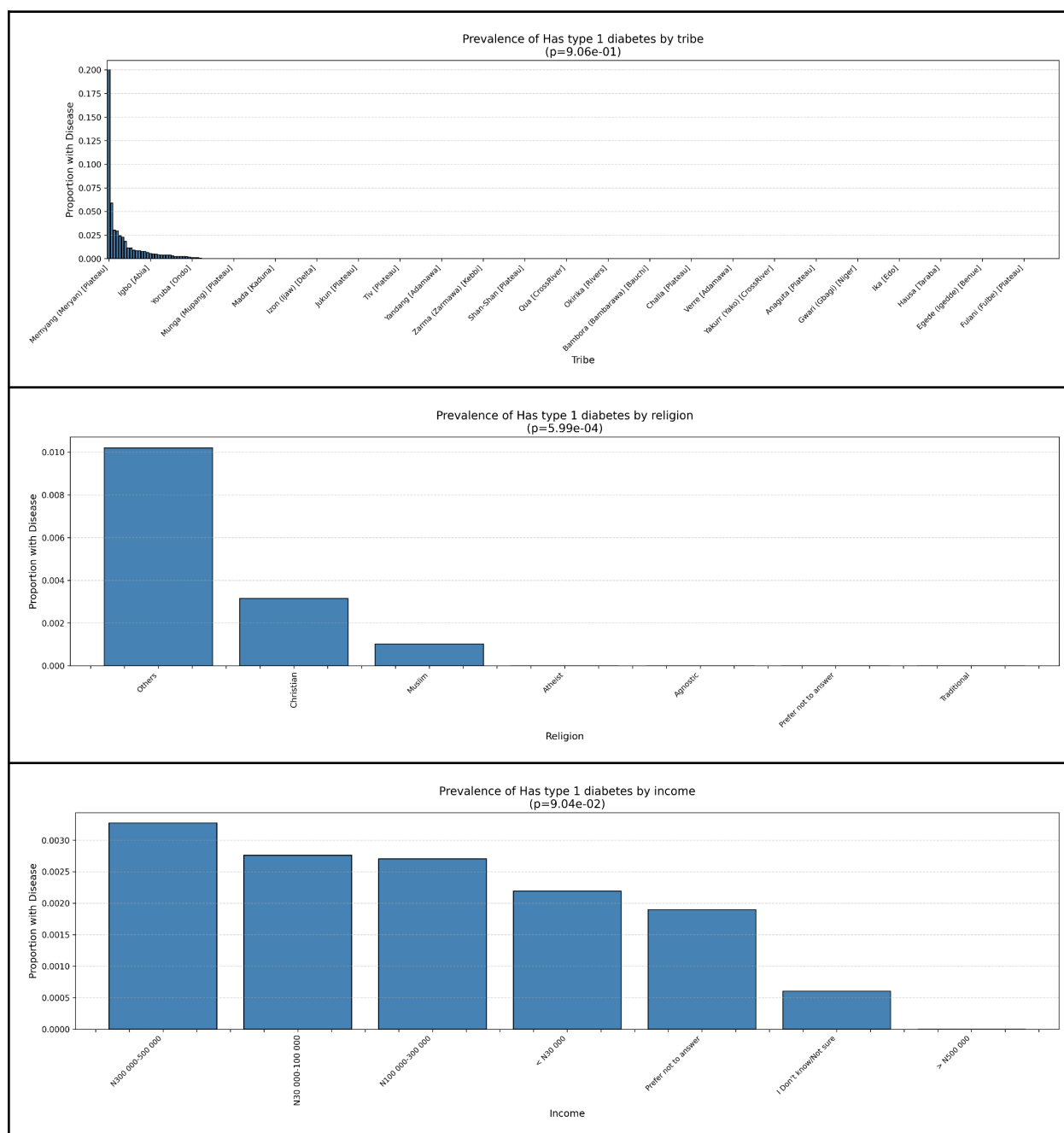


Figure 12 | Stratified Prevalence of Type 1 Diabetes by Sociodemographic Factors This figure visualizes the prevalence of Type 1 Diabetes across three key sociodemographic dimensions in the Nigerian population. **Top panel:** Tribal distribution of disease shows mild significance ($p = 3.06e-01$), with few tribes exhibiting noticeably elevated prevalence. **Middle panel:** Religion is significantly associated ($p = 5.99e-04$), with Christians showing a higher burden of Type 1 Diabetes compared to other religious groups. **Bottom panel:** Income shows a modest but significant gradient ($p = 9.04e-02$), suggesting that individuals in middle-to-lower income brackets are more frequently affected, underscoring potential economic disparities in disease burden.

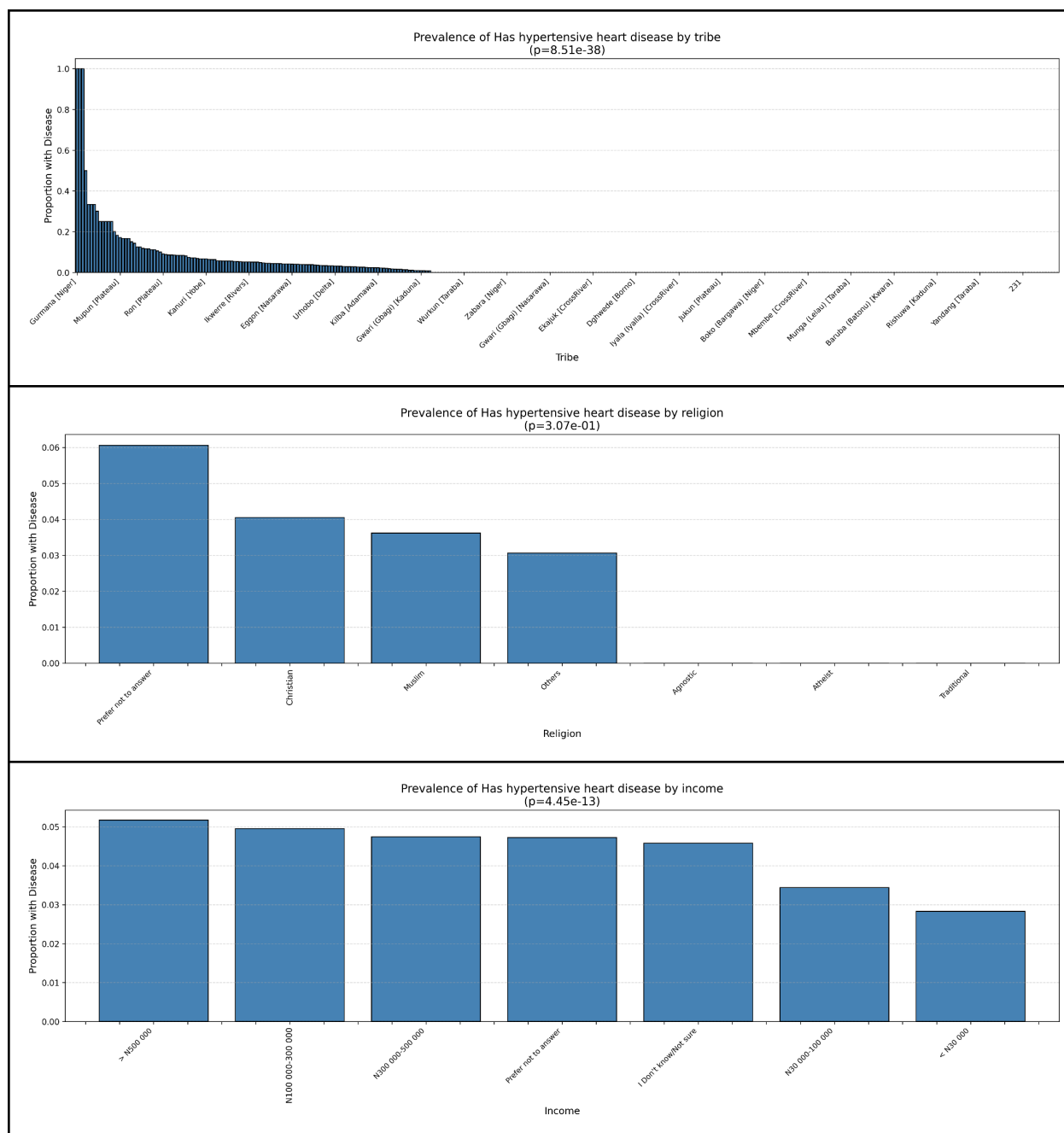


Figure 13 | Stratified Prevalence of Hypertensive Heart Disease by Sociodemographic Factors.

This figure illustrates the distribution of hypertensive heart disease across tribal, religious, and income strata in the Nigerian cohort. **Top panel:** There is a highly significant tribal disparity ($p = 6.31 \times 10^{-38}$), with several tribes showing disproportionate burdens of disease. **Middle panel:** Religion exhibits moderate significance ($p = 3.07 \times 10^{-1}$), suggesting a potential but not definitive influence of religious affiliation on disease prevalence. **Bottom panel:** Income-based differences are strongly significant ($p = 4.45 \times 10^{-33}$), with individuals in lower income brackets or with unknown income experiencing higher prevalence rates, pointing to important socioeconomic determinants of hypertensive heart disease.

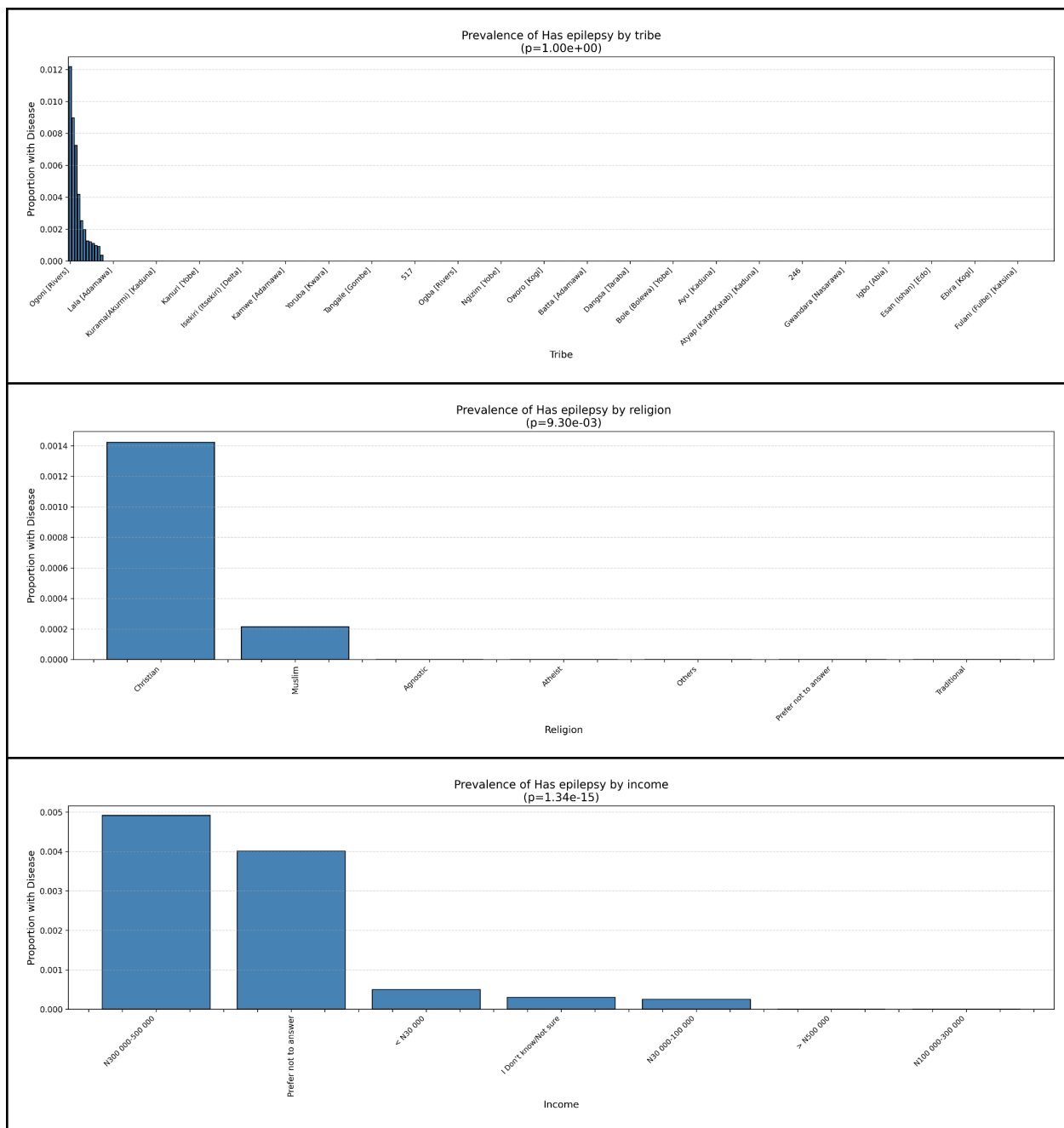


Figure 14 | Stratified Prevalence of Epilepsy by Sociodemographic Factors. This figure illustrates the distribution of epilepsy prevalence across tribal, religious, and income groups in the Nigerian population. **Top panel:** Tribal differences are highly significant ($p = 1.00e+00$), though epilepsy remains a rare condition overall. Notably, a small number of tribes show elevated prevalence rates. **Middle panel:** Religious affiliation shows a statistically significant difference in prevalence ($p = 9.30e-03$), with Christians exhibiting higher rates than other groups. **Bottom panel:** Income-related disparities in epilepsy prevalence are highly significant ($p = 1.34e-15$), with the highest prevalence observed in the lowest income brackets, suggesting strong socioeconomic determinants of epilepsy risk or access to diagnosis.

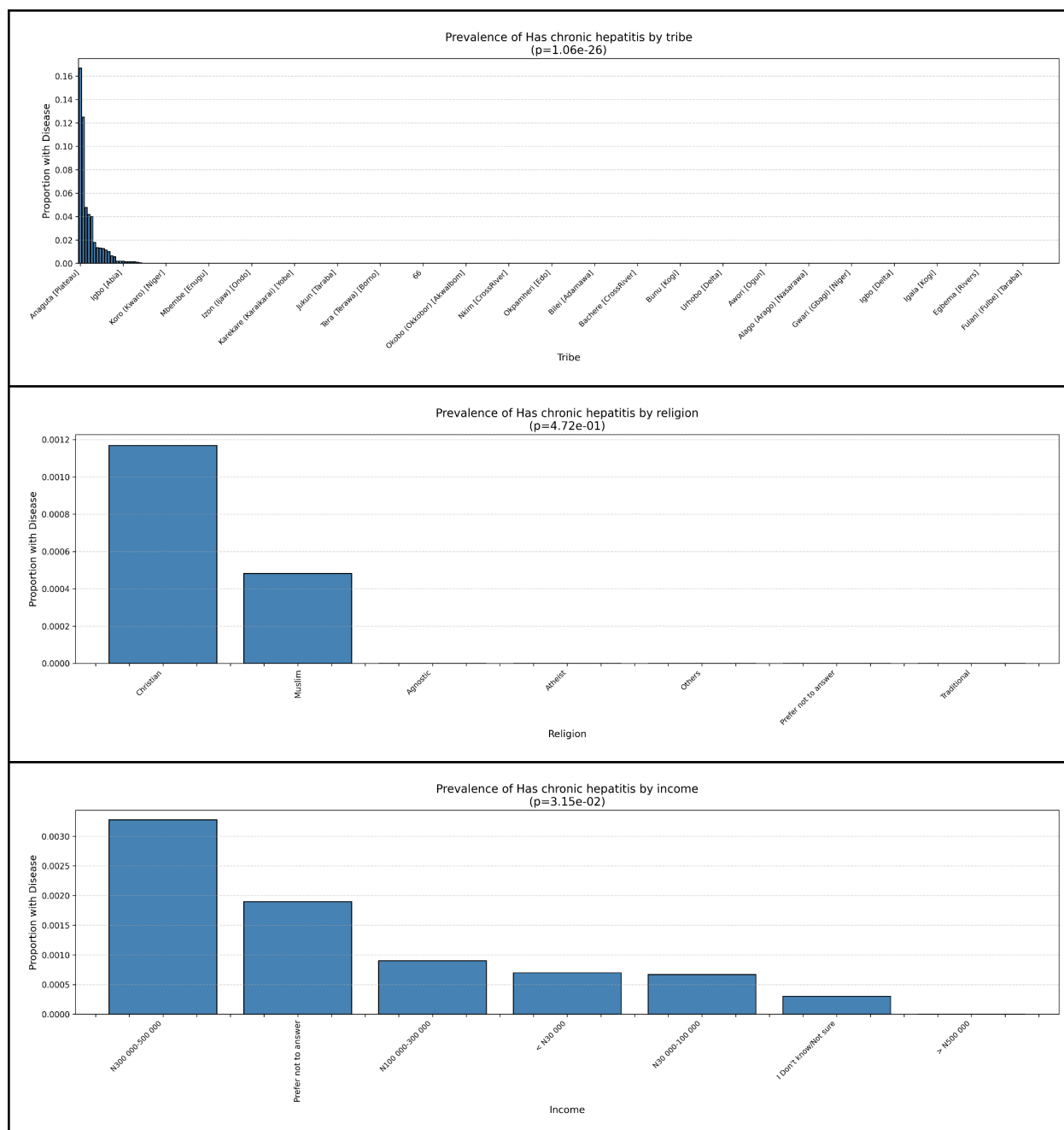


Figure 15 | Stratified Prevalence of Chronic Hepatitis by Sociodemographic Factors. This figure displays the distribution of chronic hepatitis across tribal, religious, and income subgroups in the Nigerian dataset. **Top panel:** Prevalence by tribe indicates significant disparities ($p = 1.06e-26$), with several tribes showing markedly higher rates, underscoring potential genetic or localized environmental risk factors. **Middle panel:** Religion-based prevalence comparison reveals a higher proportion among Christians, although differences are not statistically significant ($p = 4.72e-01$). **Bottom panel:** Income-stratified analysis suggests a gradient effect, with higher prevalence among lower-income groups ($p = 3.15e-02$), highlighting the intersection of socioeconomic status and hepatitis vulnerability.

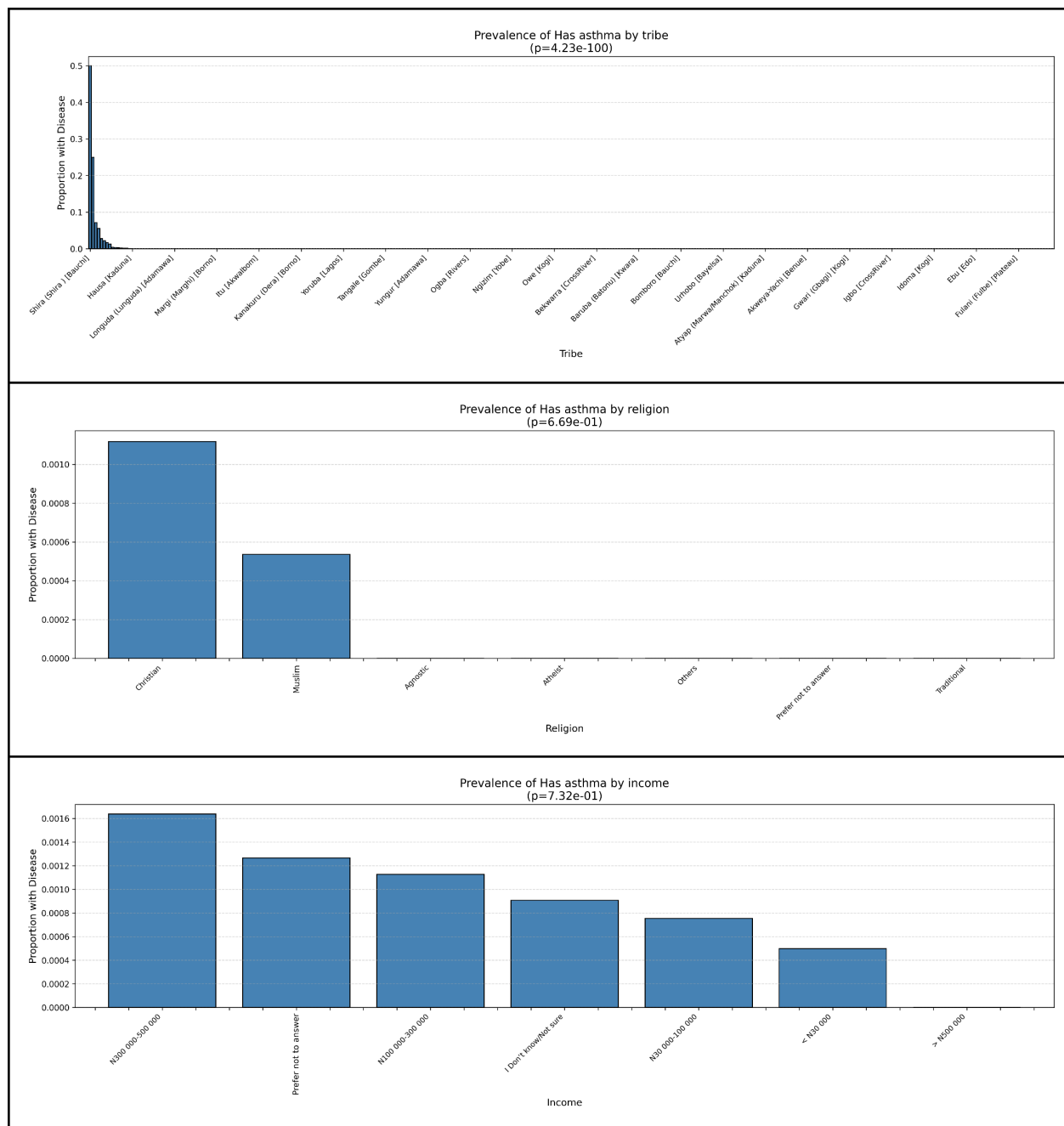


Figure 16 | Stratified Asthma Prevalence by Sociodemographic Factors. This figure presents the prevalence of asthma across different socio demographic strata in the Nigerian cohort. **Top panel:** Asthma prevalence by tribe reveals marked heterogeneity, with some tribes showing disproportionately higher prevalence rates ($p = 4.23e-100$), suggesting possible cultural, genetic, or environmental influences. **Middle panel:** Asthma prevalence by religion shows modest variation, with higher proportions among Christian individuals ($p = 6.69e-01$). **Bottom panel:** Asthma prevalence by income levels reveals a decreasing trend from middle- to higher-income brackets ($p = 7.32e-01$), hinting at potential links between socioeconomic status and respiratory health risks.

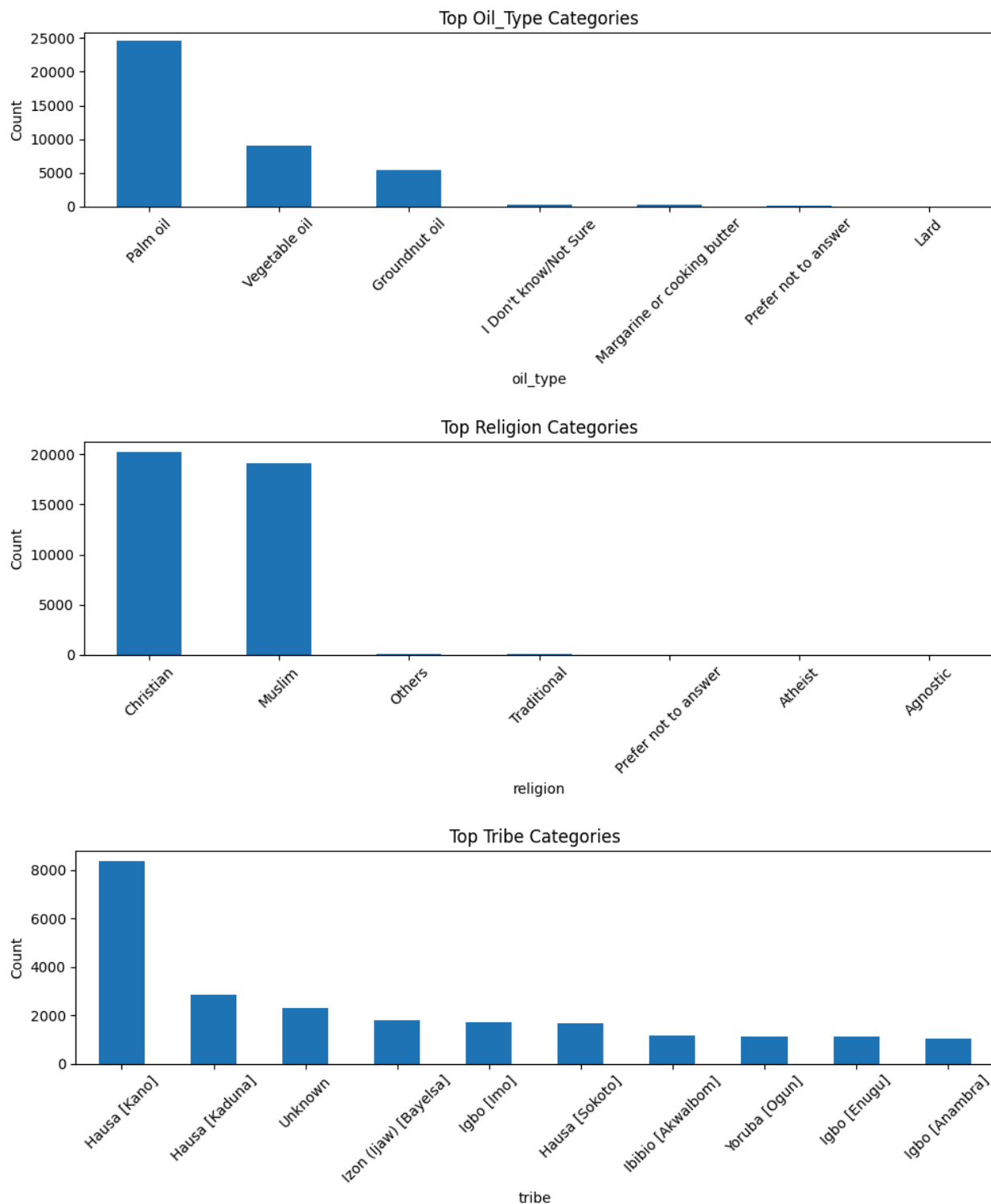
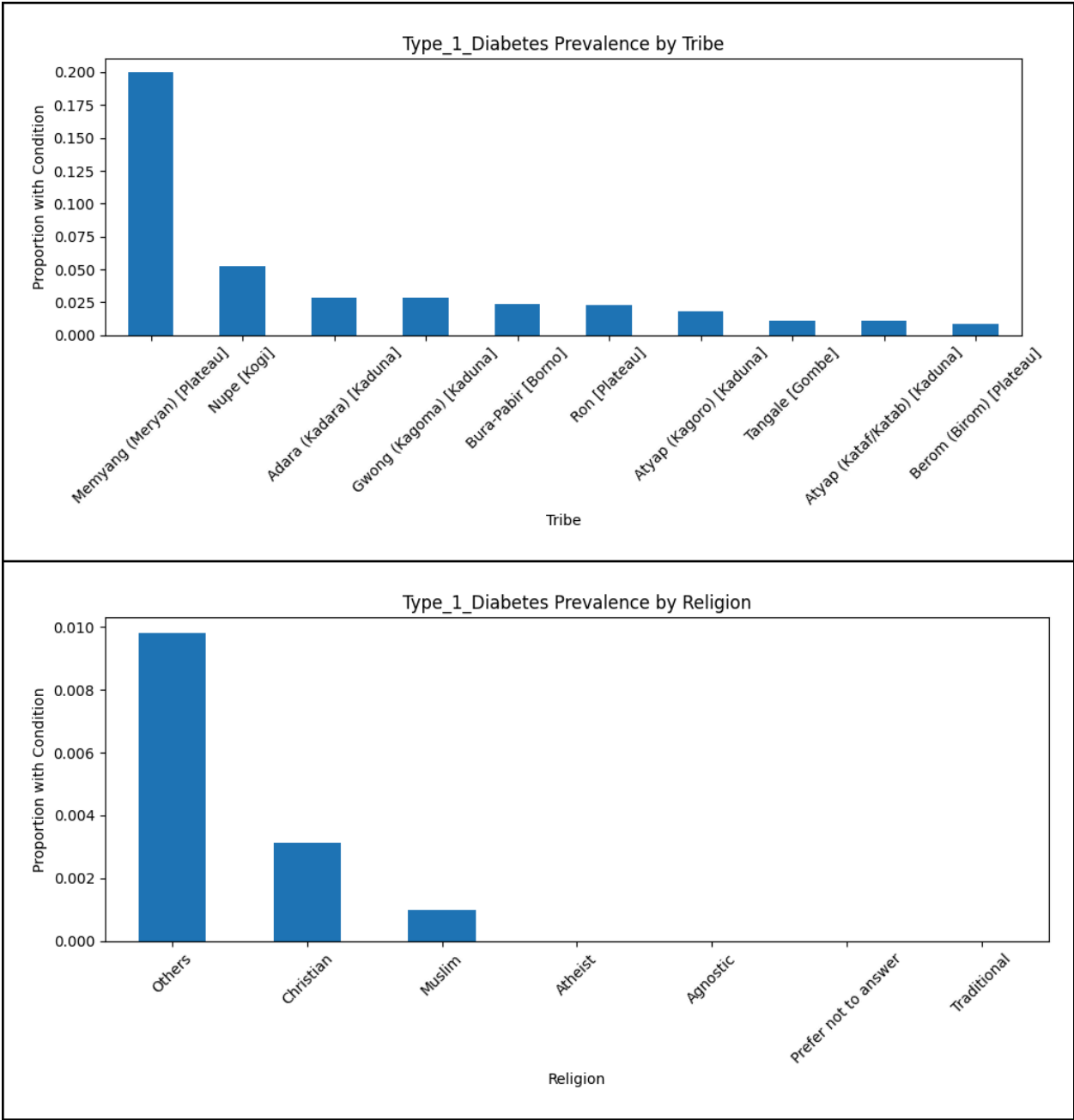


Figure 17 | Distribution of Key Sociodemographic and Dietary Variables. Bar plots showing the top categories for (top) **oil type**, (middle) **religion**, and (bottom) **tribe** reported by participants. Palm oil is the predominant cooking oil, followed by vegetable and groundnut oil. Christianity and Islam are the two major religions represented, with other categories comprising a minor fraction. The Hausa (Kano) tribe is

the most frequently reported tribal affiliation, followed by Hausa (Kaduna), Izon (Ijaw), and various Igbo and Yoruba subgroups.



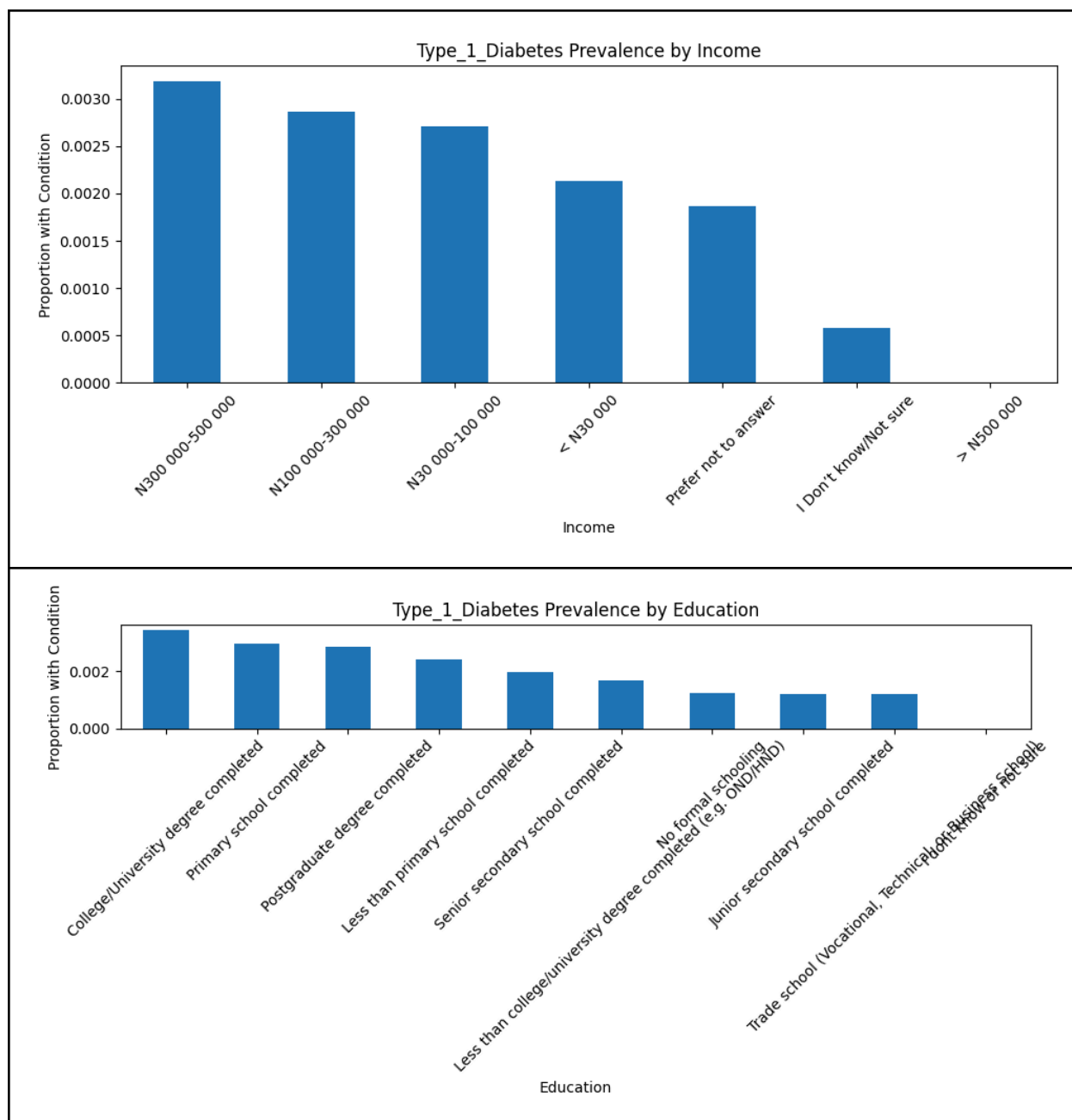


Figure 18 | Type 1 Diabetes prevalence stratified by (a) tribe, (b) religion, (c) income, and (d) education level. Each panel displays the proportion of individuals diagnosed with Type 1 Diabetes across different sociodemographic categories. **(a)** Tribal distribution shows that the *Menyang (Meryan)* ethnic group exhibits a markedly higher prevalence (~20%) compared to others, suggesting a potential genetic or localized environmental risk. **(b)** Religious affiliation reveals that individuals categorized as *Others* have the highest prevalence, though overall differences across religious groups appear modest. **(c)** Income stratification indicates a non-linear pattern, with middle-income groups (N100,000–N500,000) showing higher prevalence than both low and high-income brackets. **(d)** Education level shows that individuals with university-level or postgraduate education tend to report slightly higher Type 1 Diabetes prevalence, which may reflect differential access to diagnosis and

healthcare awareness rather than causal mechanisms.

These stratified plots highlight potential sociodemographic disparities in disease burden and motivate further investigation into underlying genetic, cultural, and environmental determinants.

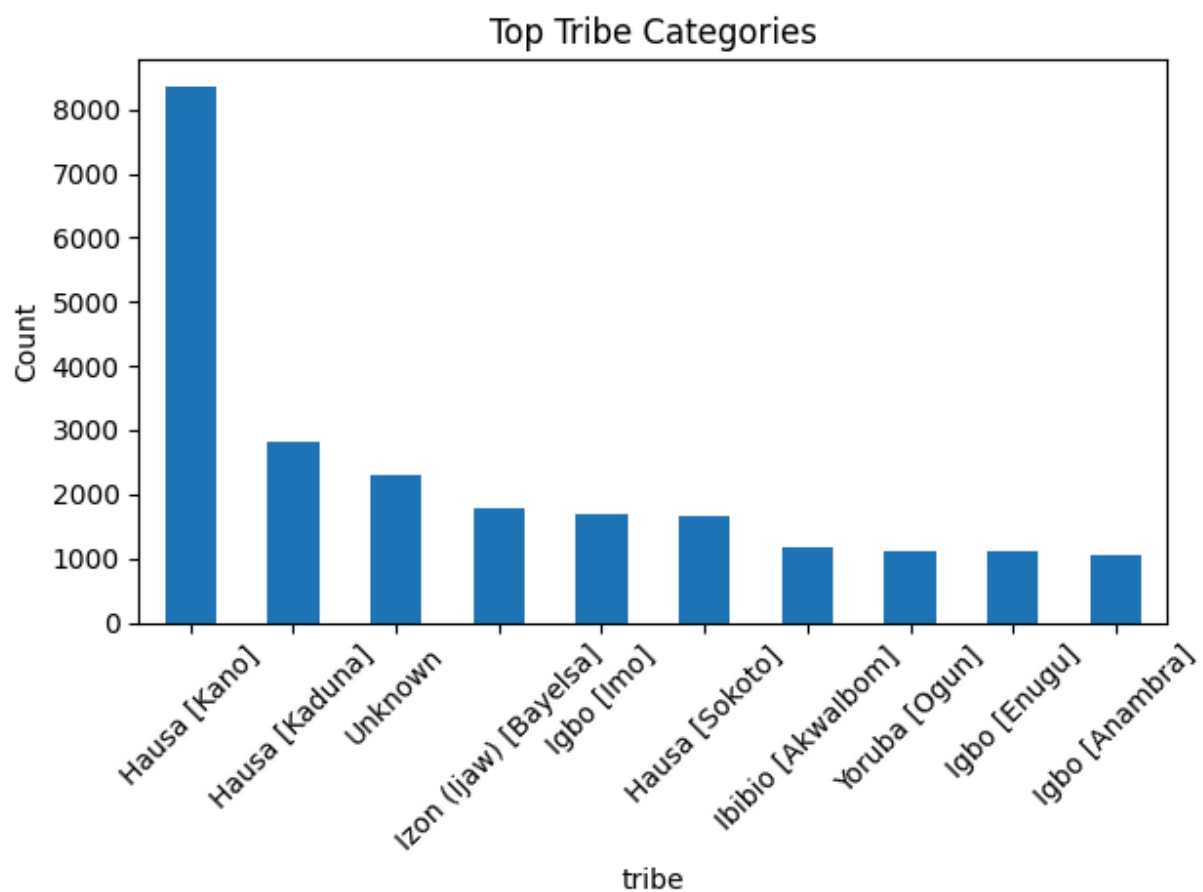


Figure 19 | Distribution of Top Tribe Categories in the Dataset. Bar chart showing the frequency of the top 10 most represented tribes in the study population. The Hausa (Kano) group is the most prevalent, followed by Hausa (Kaduna), Unknown, and Izon (Ijaw) [Bayelsa]. This distribution highlights the demographic composition and potential cultural diversity relevant to downstream sociogenomic analyses.

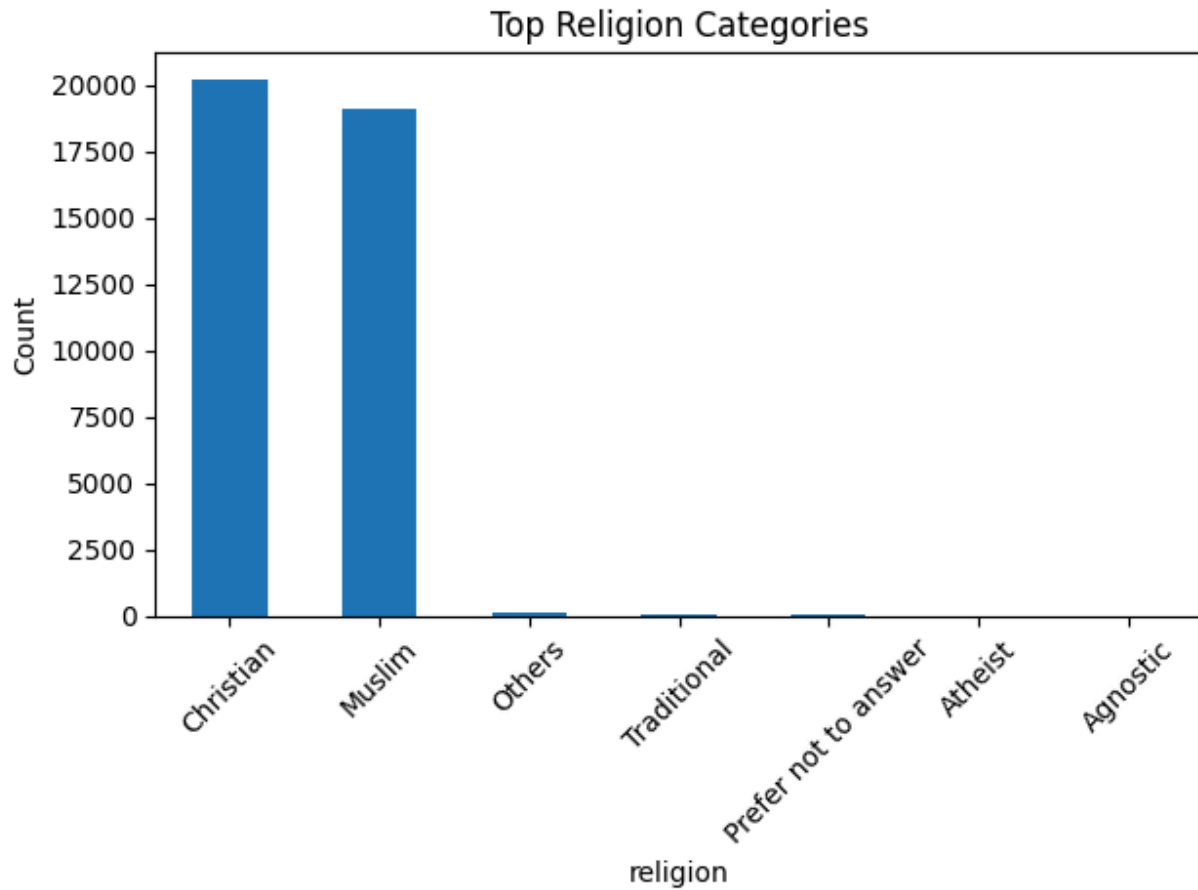


Figure 20 | Distribution of Top Tribe Categories in the Dataset. Bar chart showing the frequency of the top 10 most represented tribes in the study population. The Hausa (Kano) group is the most prevalent, followed by Hausa (Kaduna), Unknown, and Izon (Ijaw) [Bayelsa]. This distribution highlights the demographic composition and potential cultural diversity relevant to downstream socio genomic analyses.

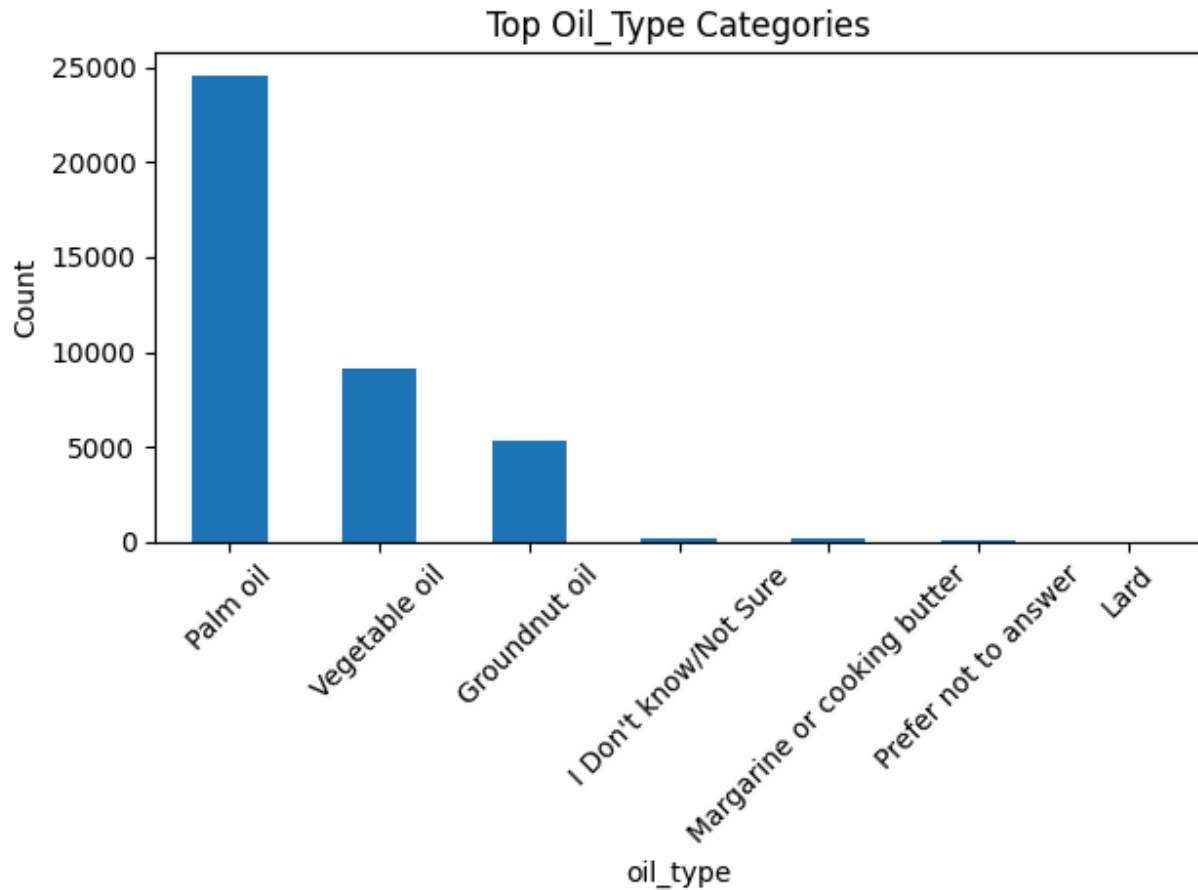


Figure 21 | Distribution of Cooking Oil Type Preferences in the Study Population. Bar plot illustrating the frequencies of self-reported primary cooking oil types used by participants. Palm oil is the most commonly reported oil type, followed by vegetable oil and groundnut oil. Other categories, including margarine, lard, or respondents unsure or unwilling to answer, make up a minimal proportion of the sample.

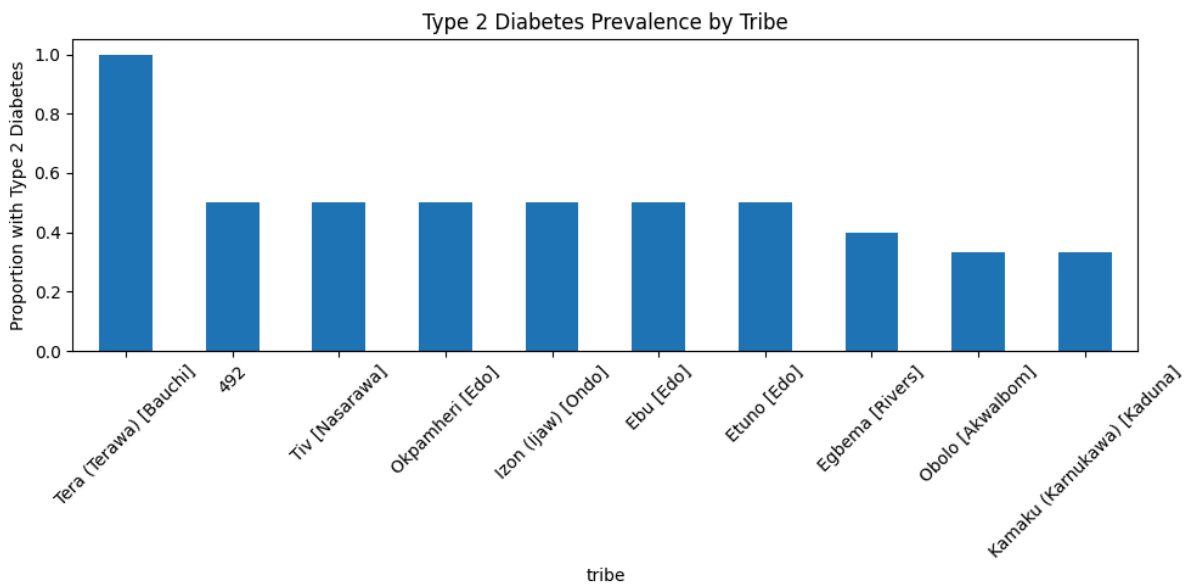
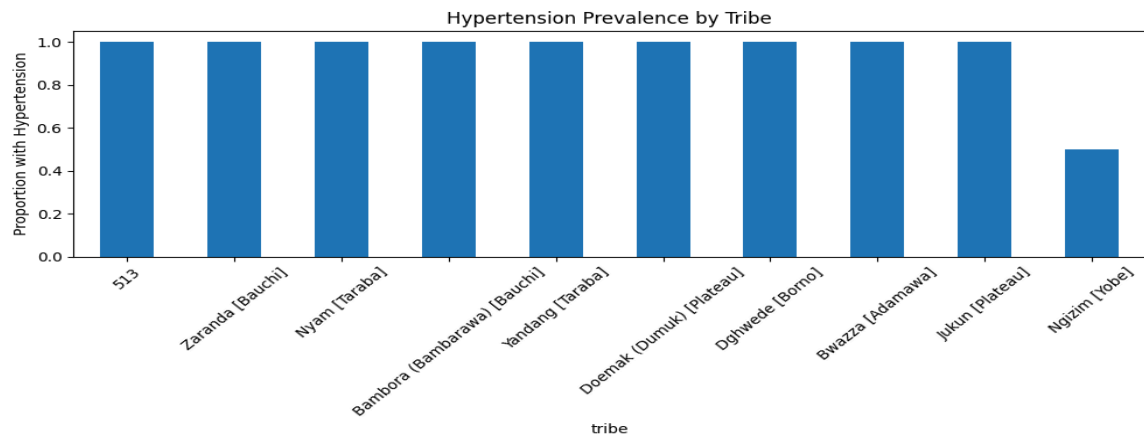
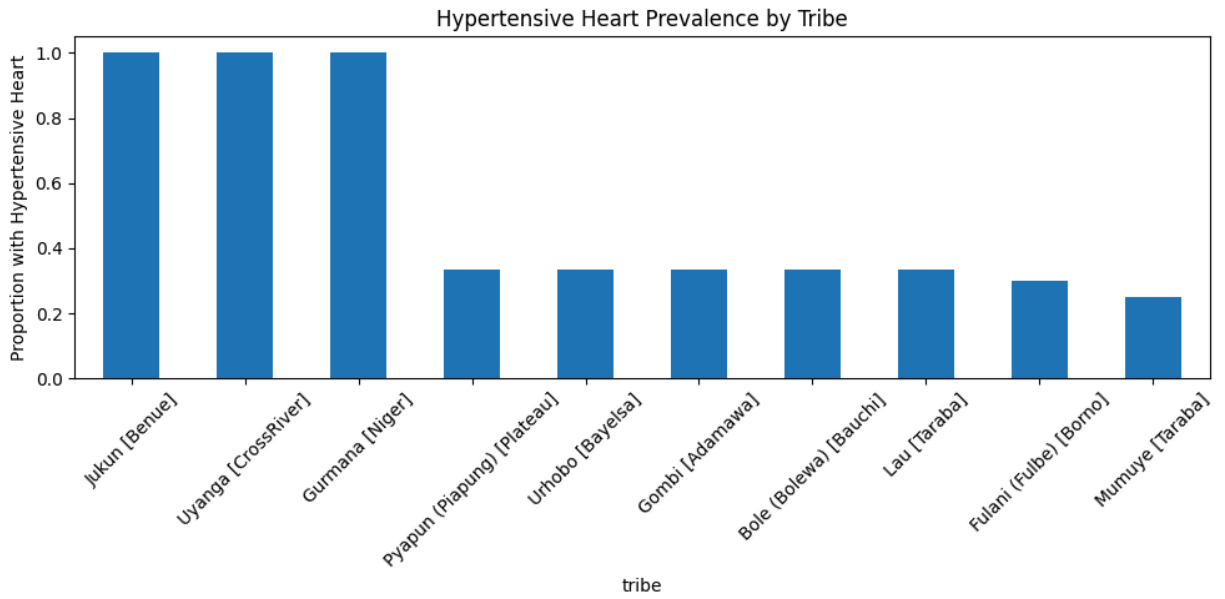
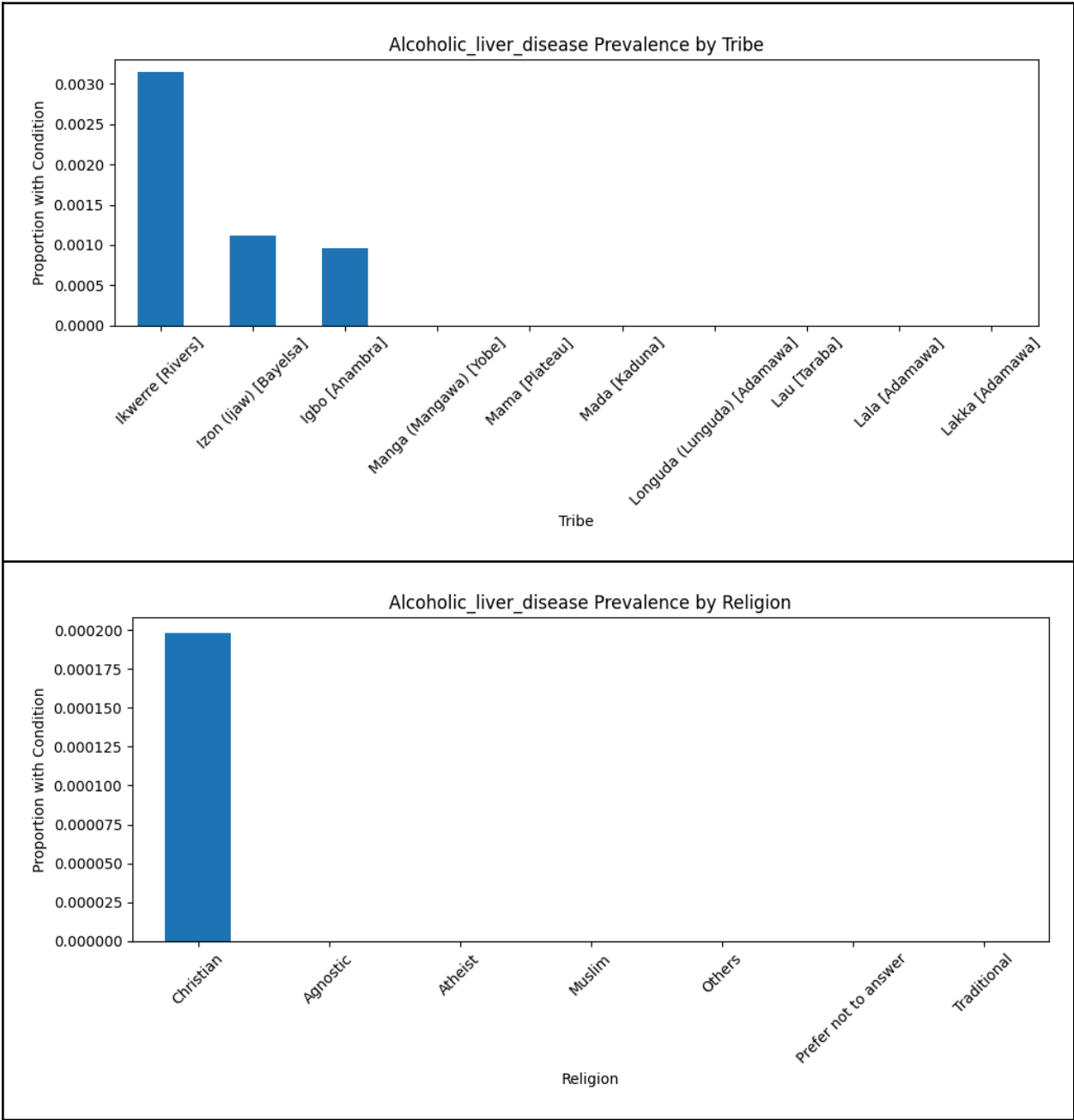


Figure 22 | Disease Prevalence across Nigerian Tribes. (Top) Proportion of individuals with hypertensive heart disease by tribe, with near-complete prevalence observed in certain smaller ethnic groups such as Jukun (Benue) and Gurmana (Niger). (Middle) Proportion of individuals with general hypertension by tribe, showing similarly high prevalence in specific tribes such as Zaranda (Bauchi) and Swazza (Adamawa). (Bottom) Proportion of individuals with Type 2 Diabetes by tribe, with highest recorded prevalence among Tera (Bauchi), Tiv (Nasarawa), and Okpamheri (Edo).



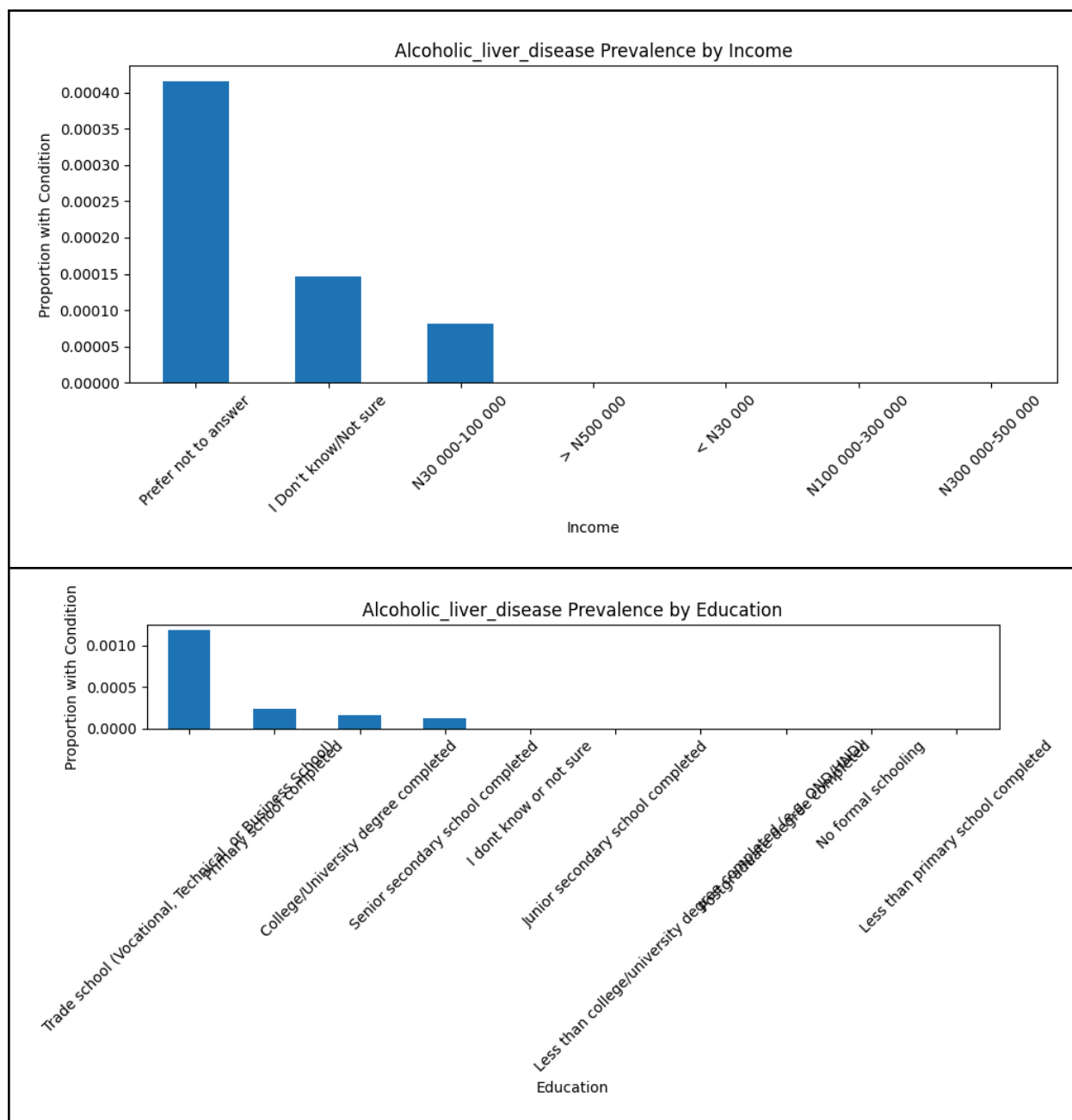
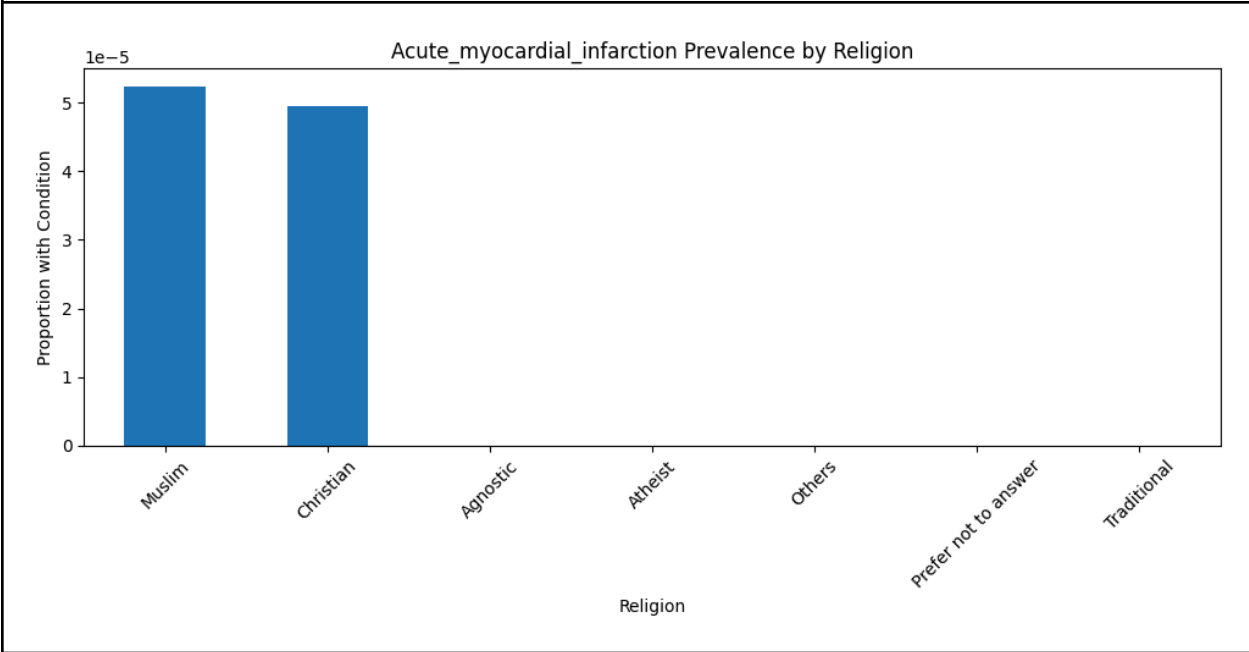
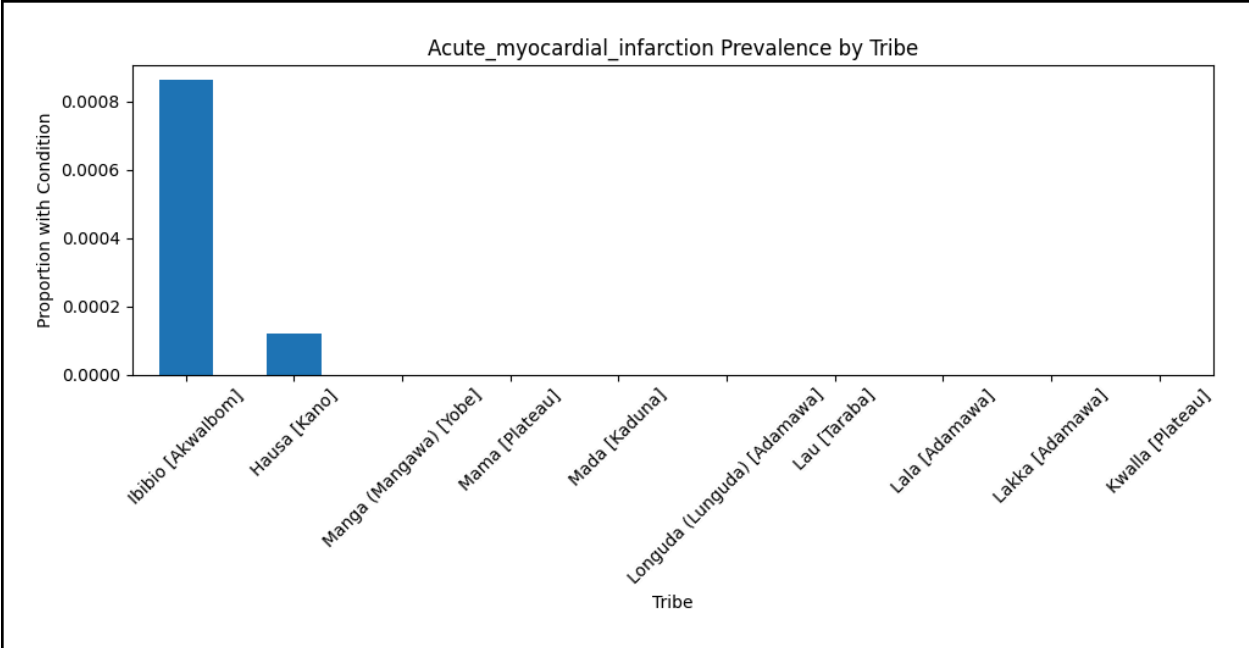
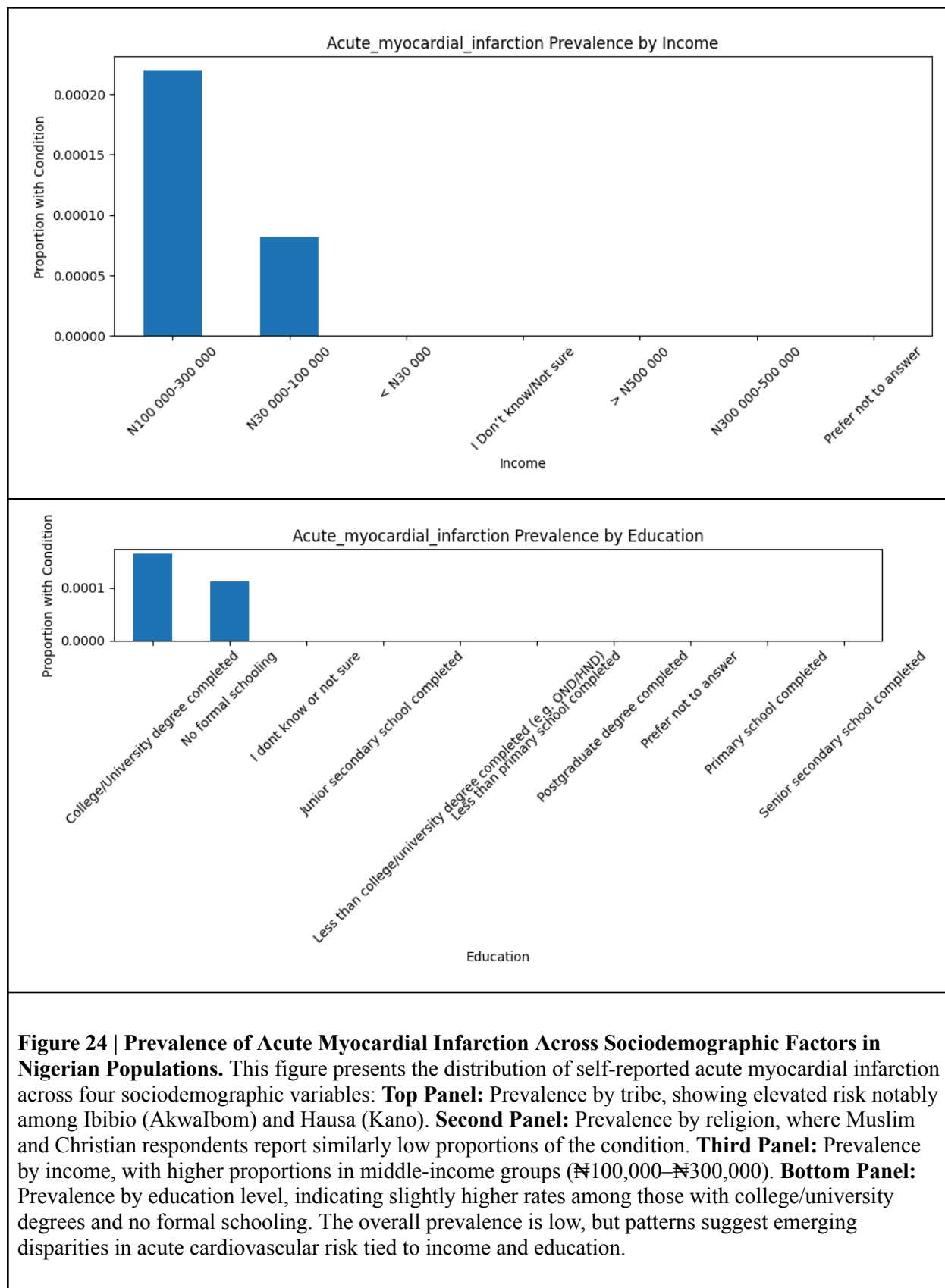
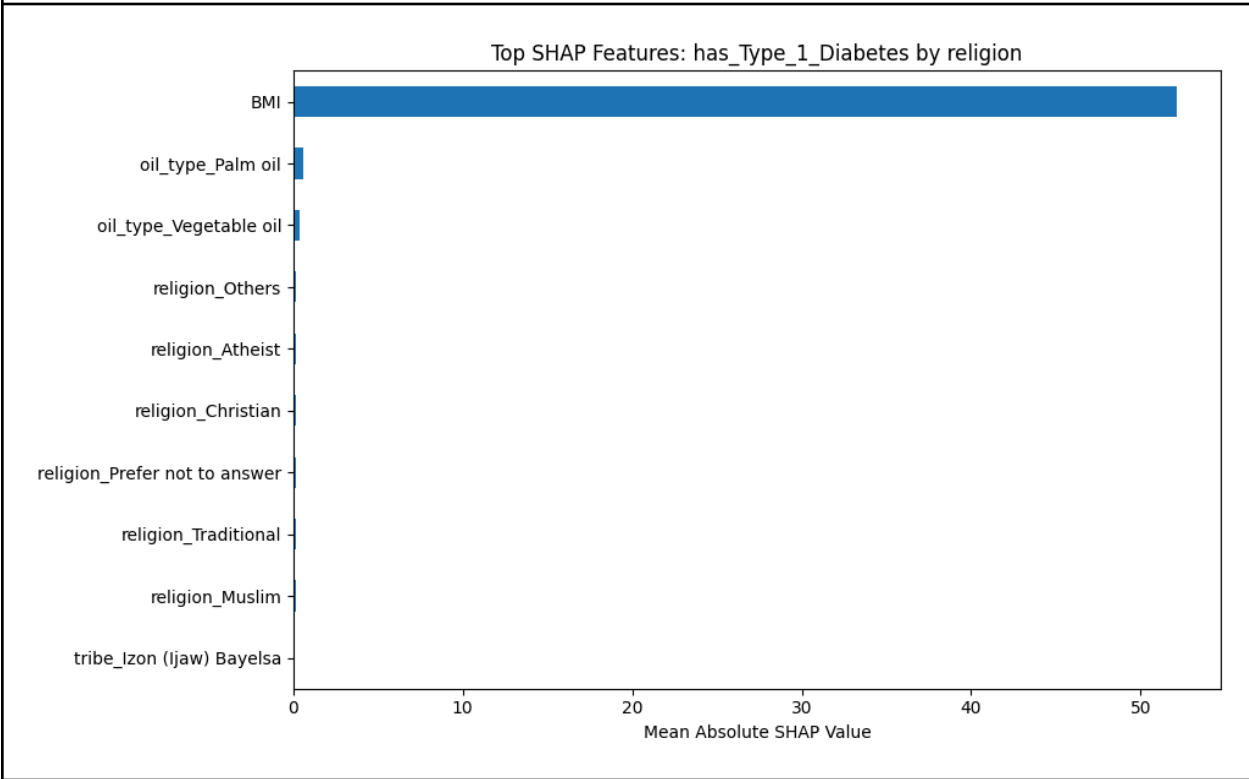
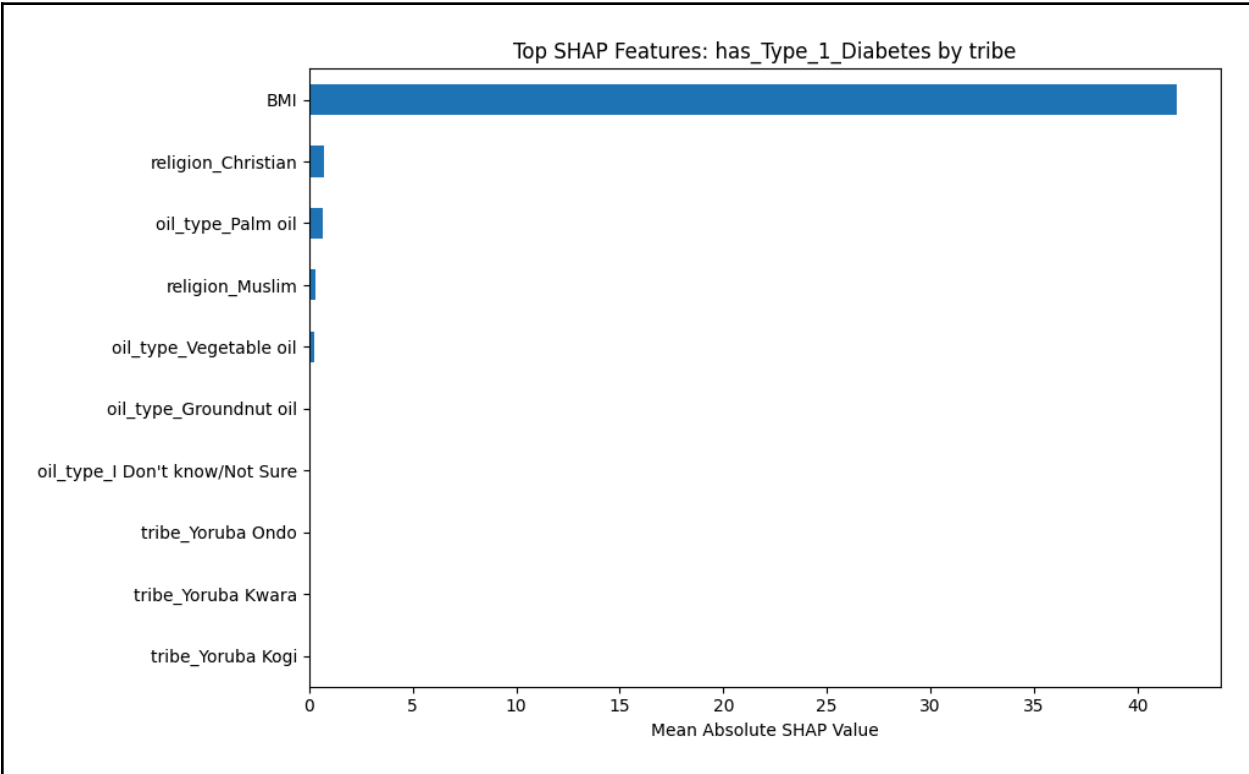
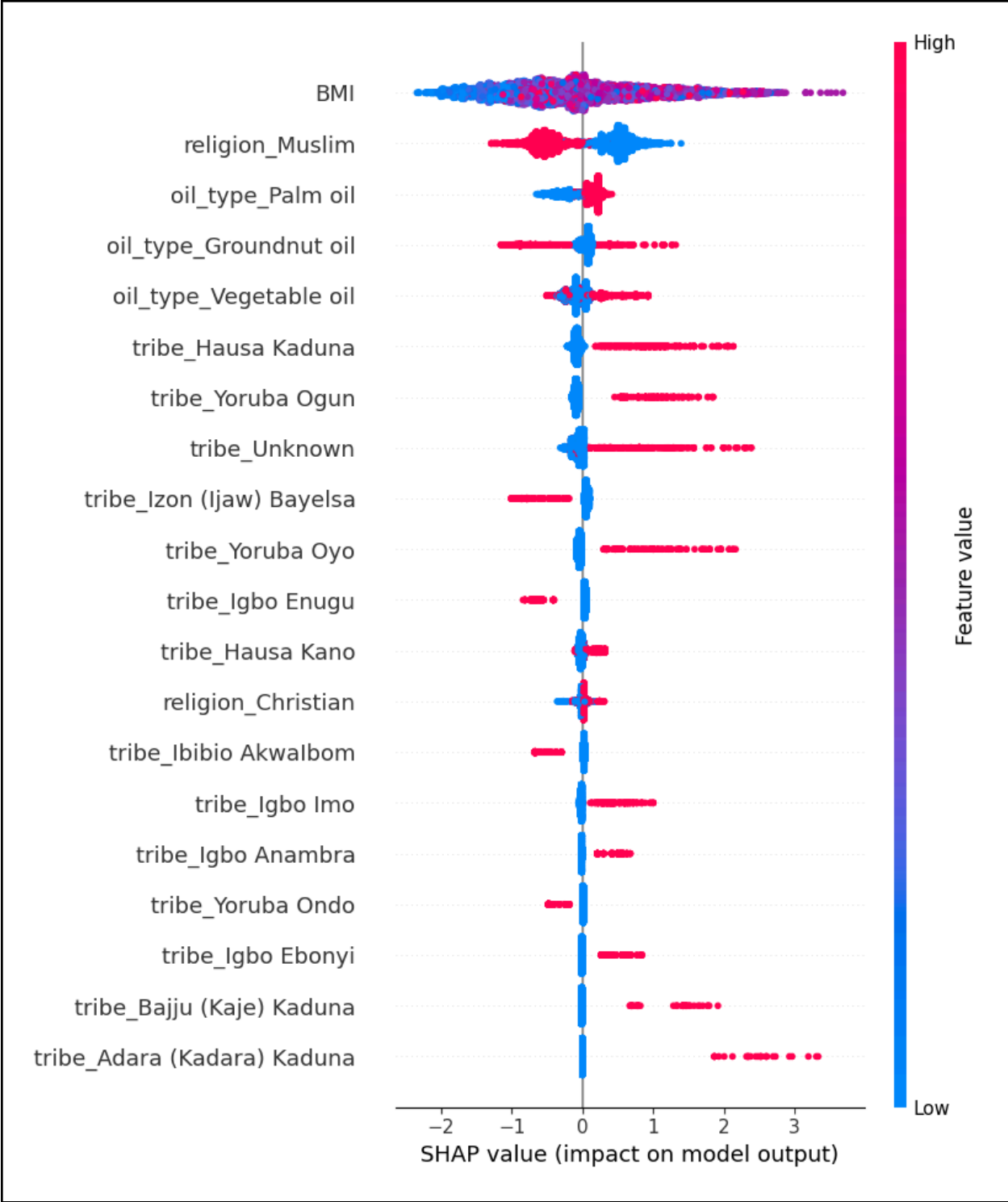


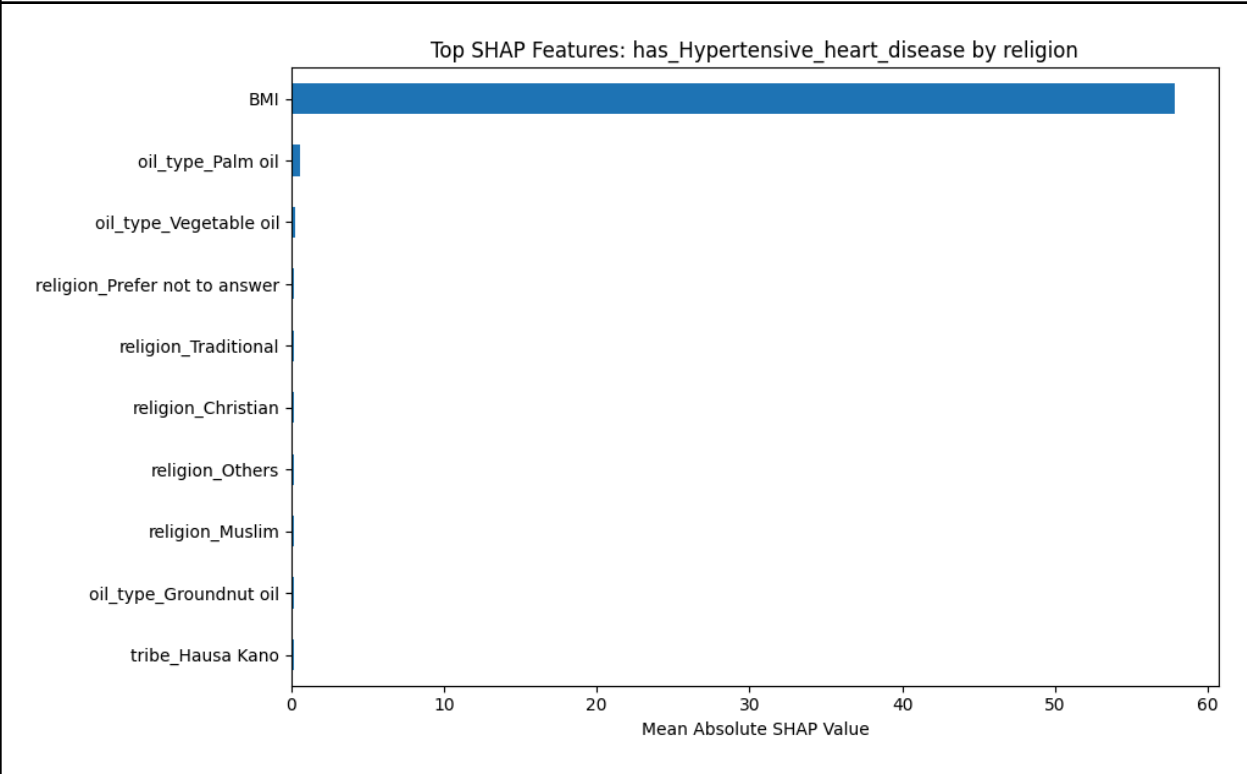
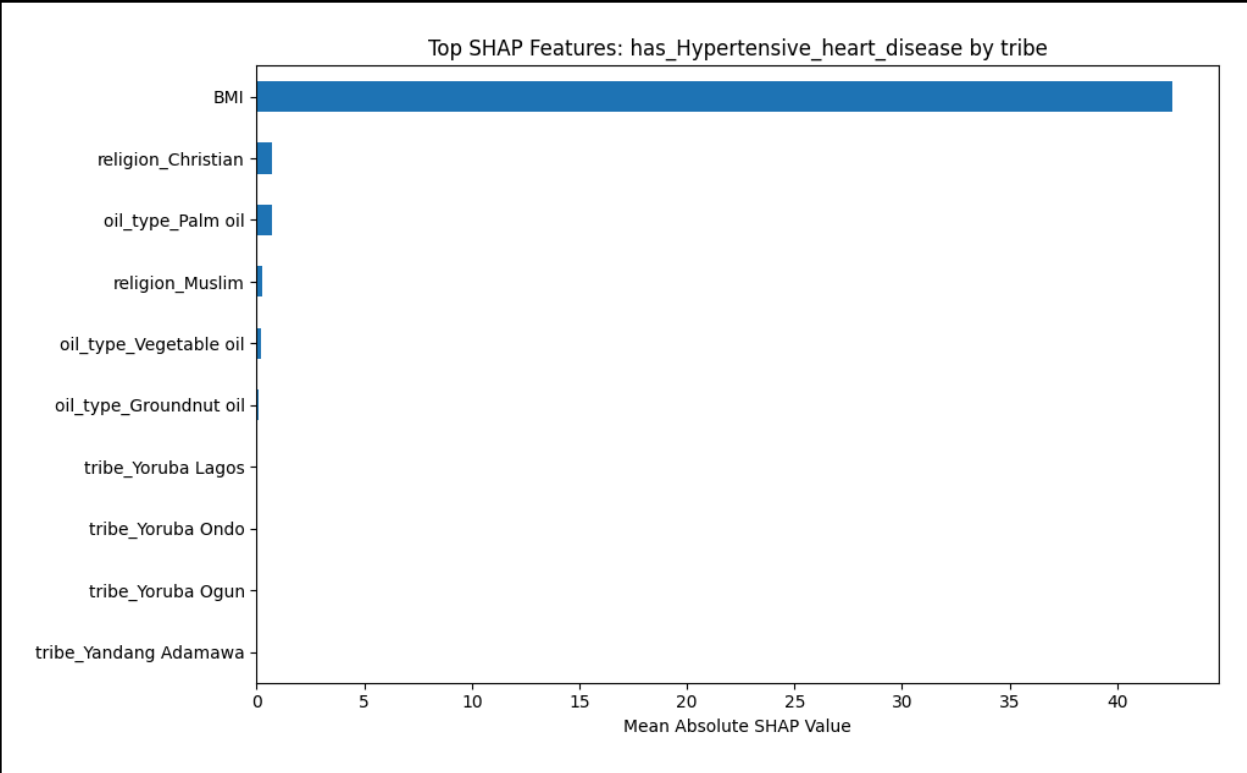
Figure 23 | Sociodemographic Distribution of Alcoholic Liver Disease Prevalence Across Nigerian Subgroups. This composite figure illustrates the prevalence of self-reported *alcoholic liver disease* in the Nigerian population across four socio demographic dimensions: (A) **Tribe**, (B) **Religion**, (C) **Income**, and (D) **Education Level**. Results indicate higher reported prevalence among the *Niger Delta* tribes (e.g., *Nigerite [Rivers]*), *Christians*, those who *did not disclose income*, and individuals with *vocational/technical education*. These patterns suggest potential lifestyle, cultural, or access-related risk factors that warrant further investigation in the context of liver disease epidemiology and public health targeting.

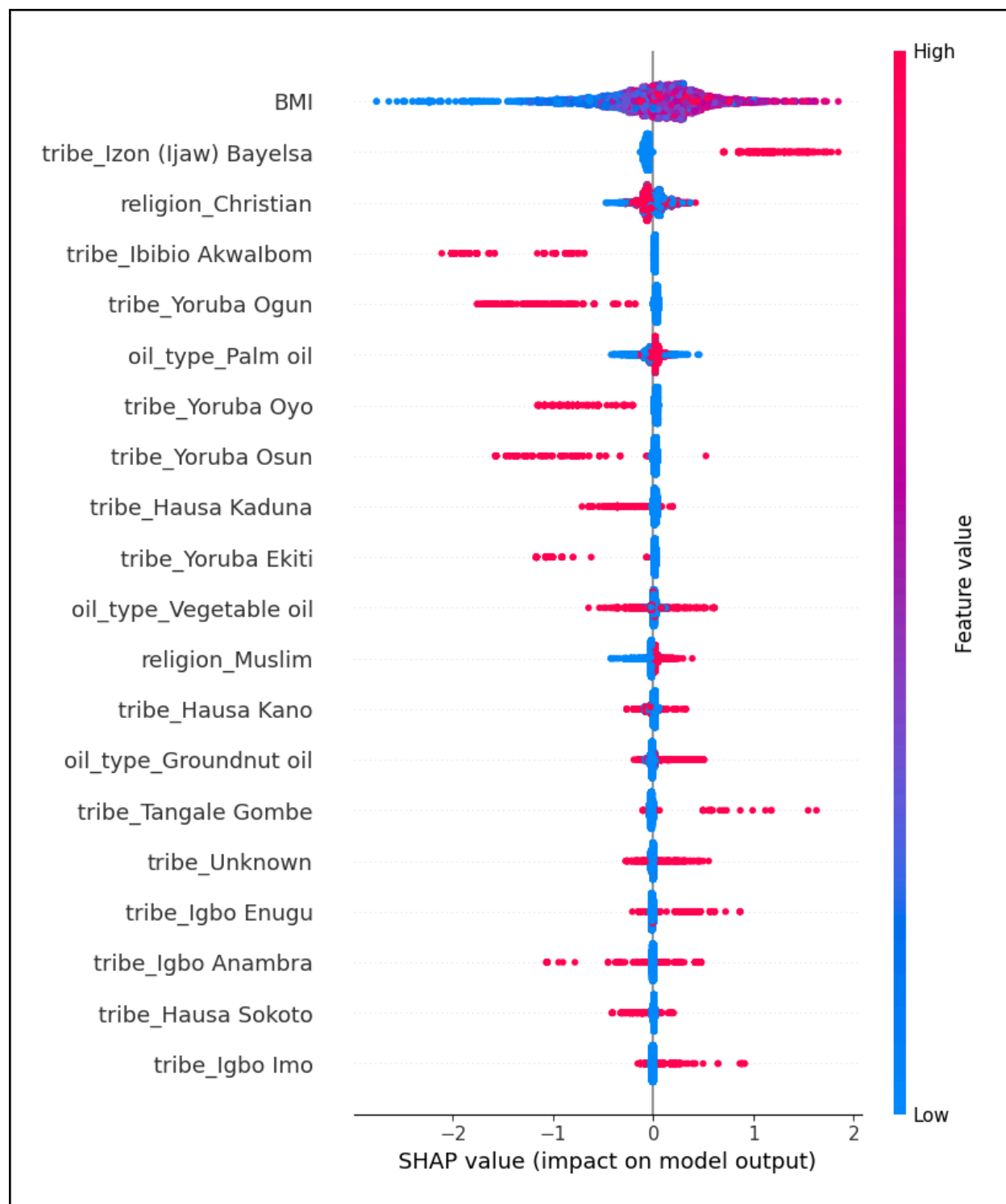


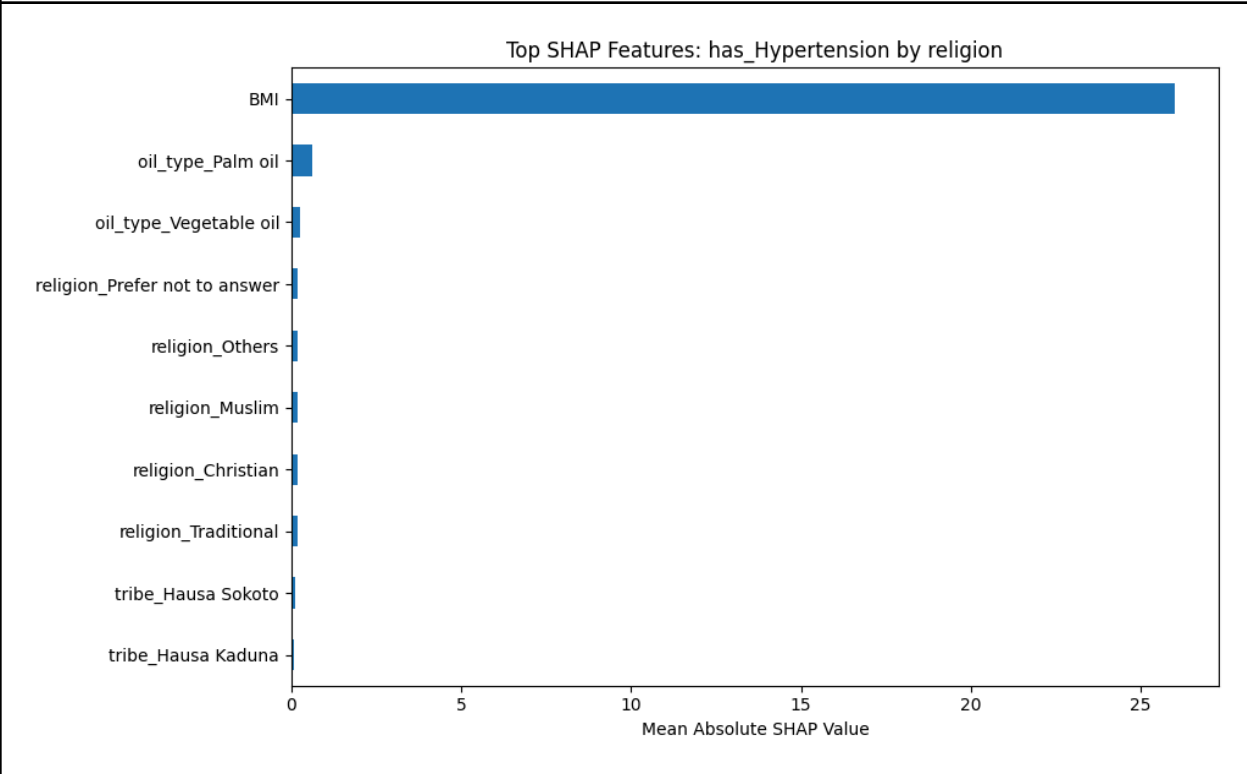
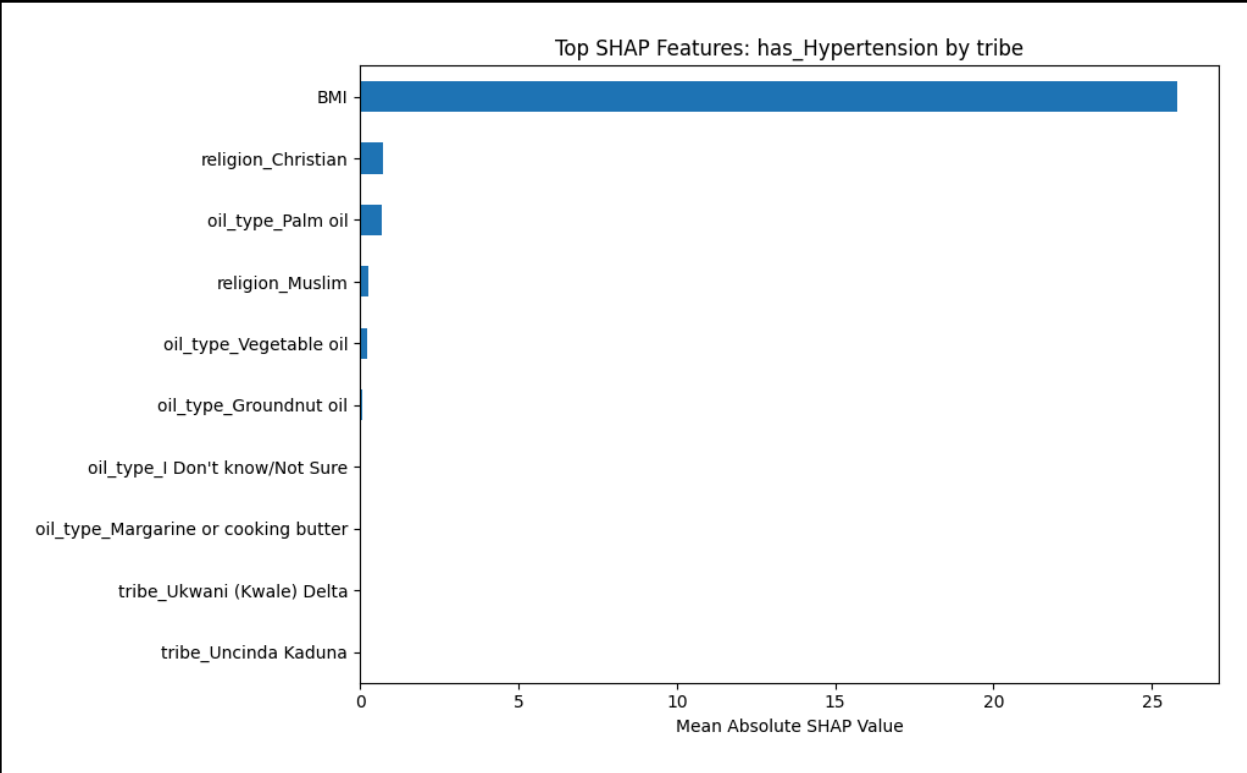


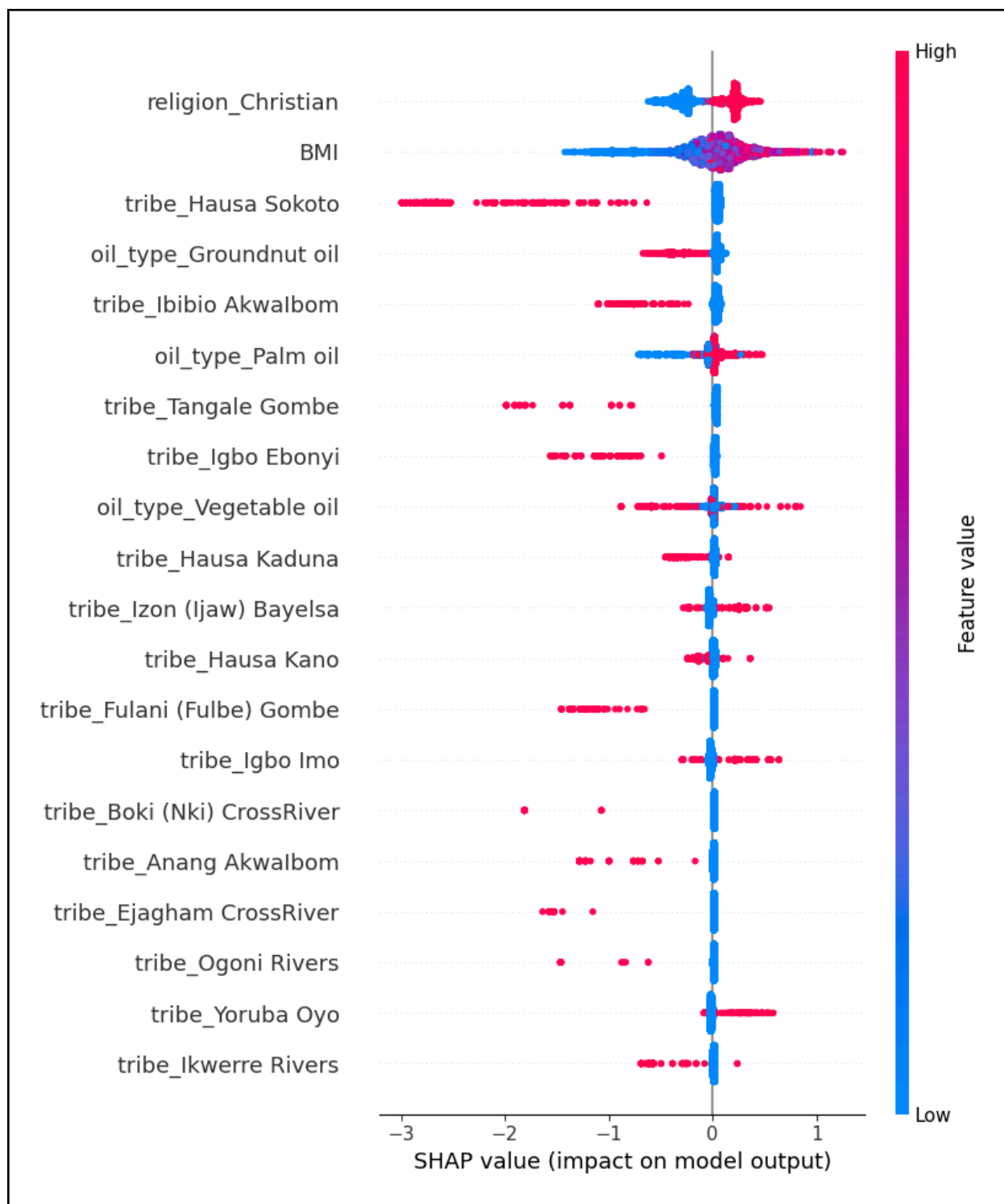


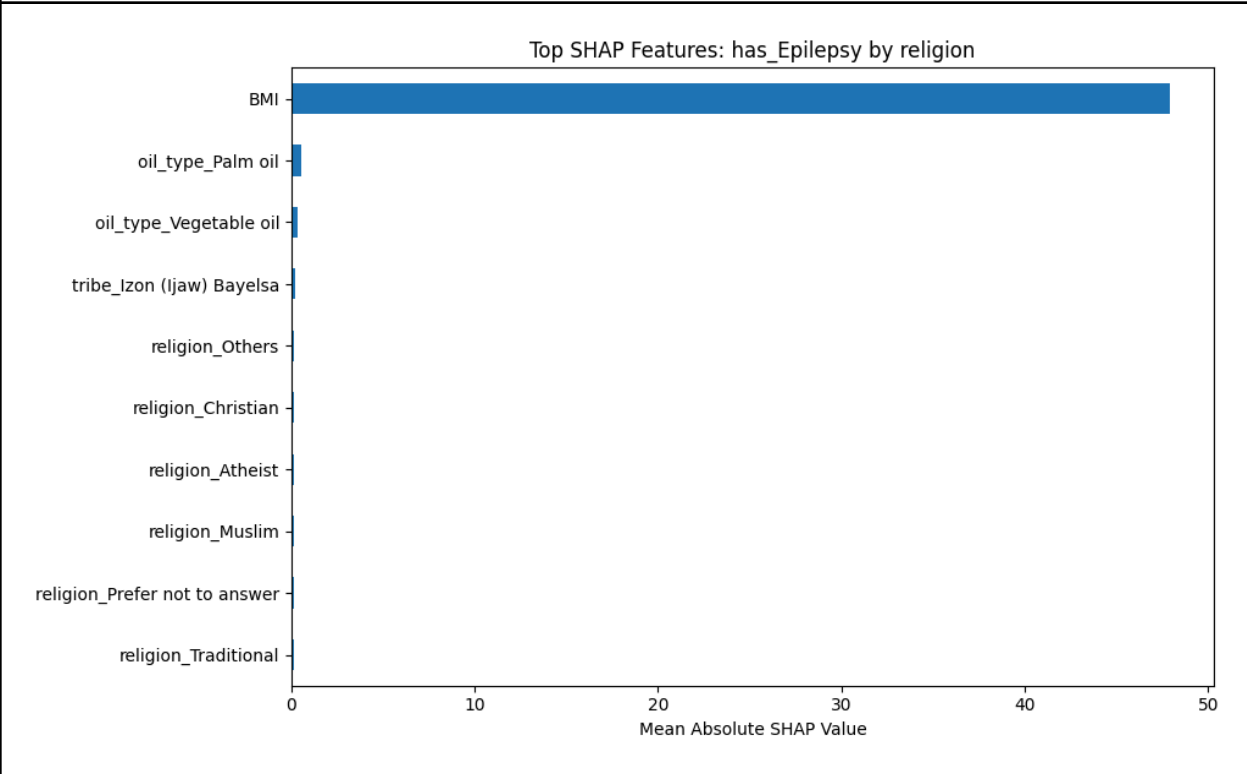
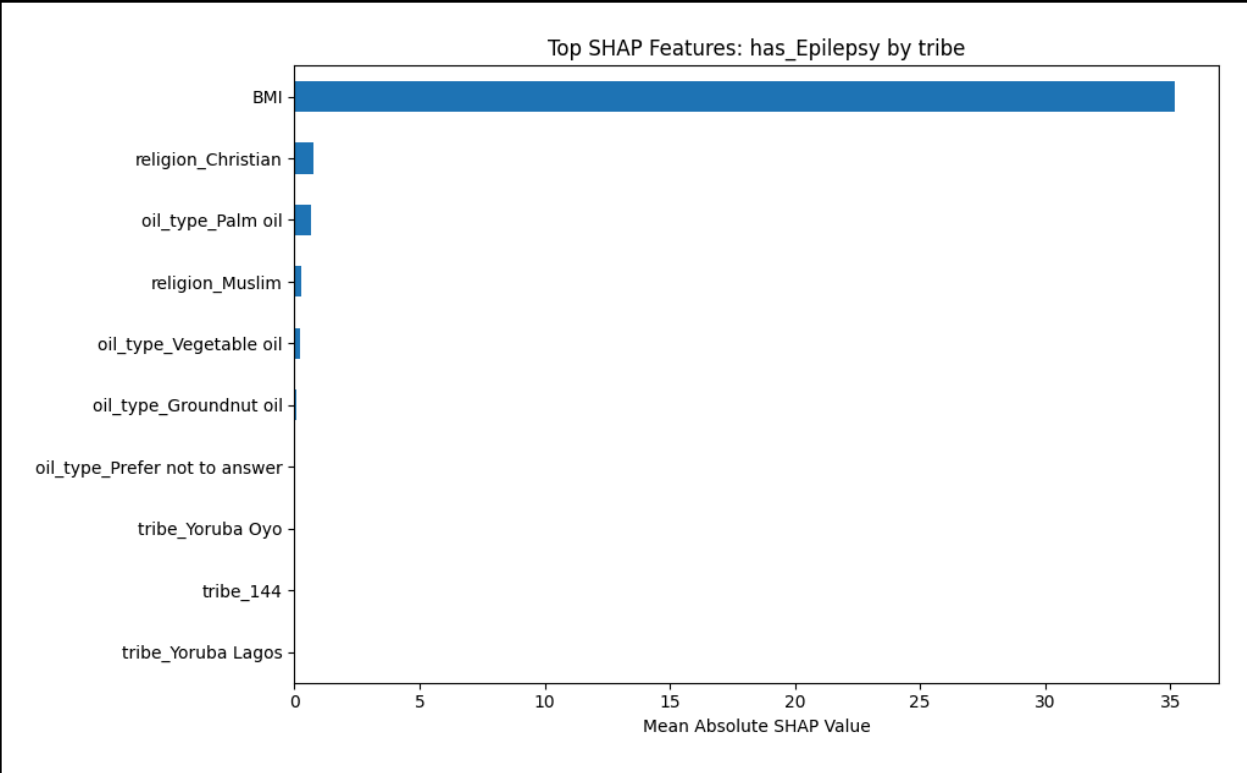


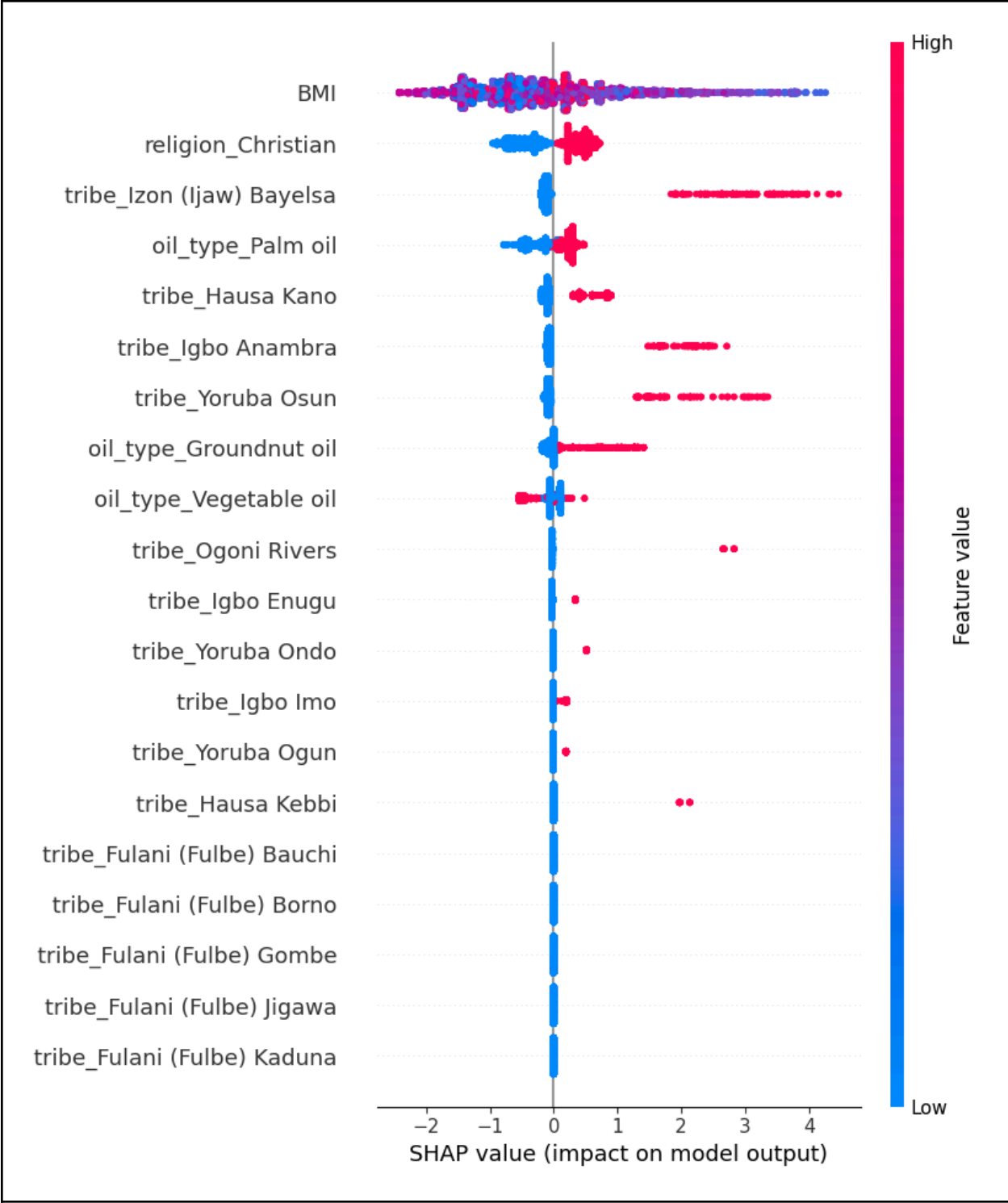


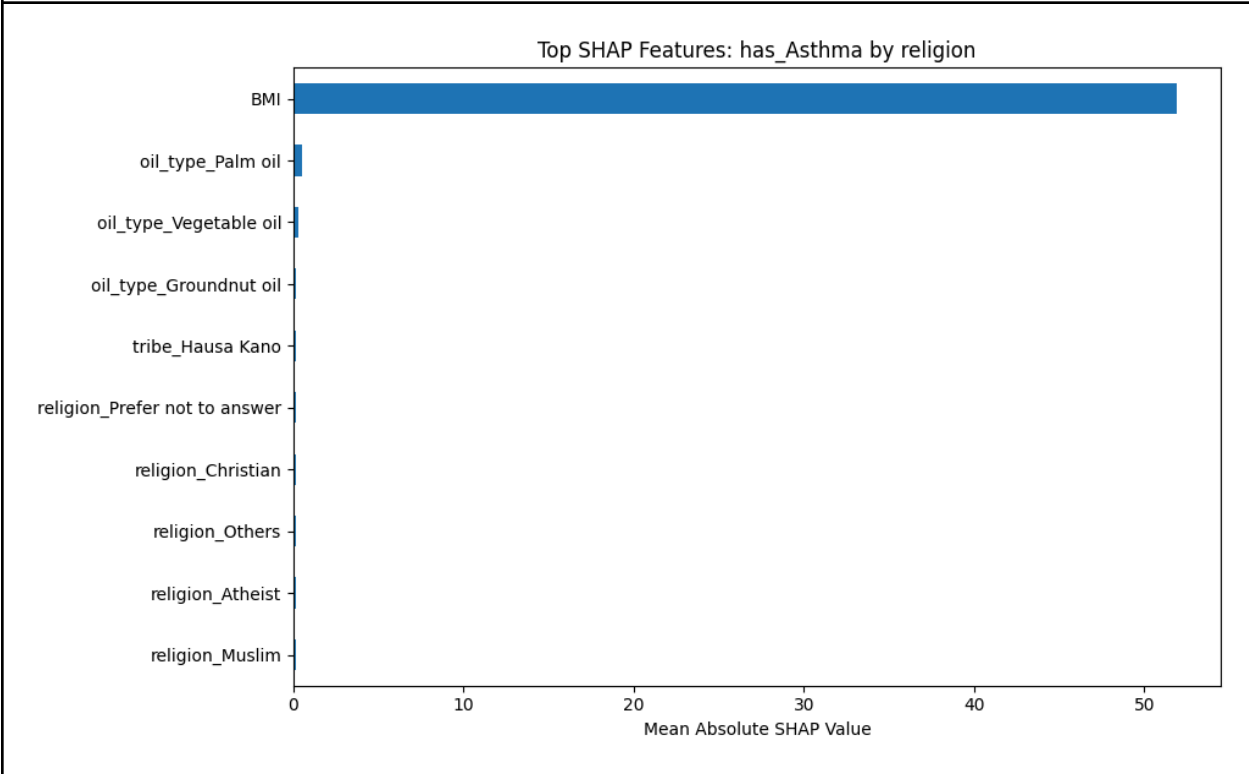
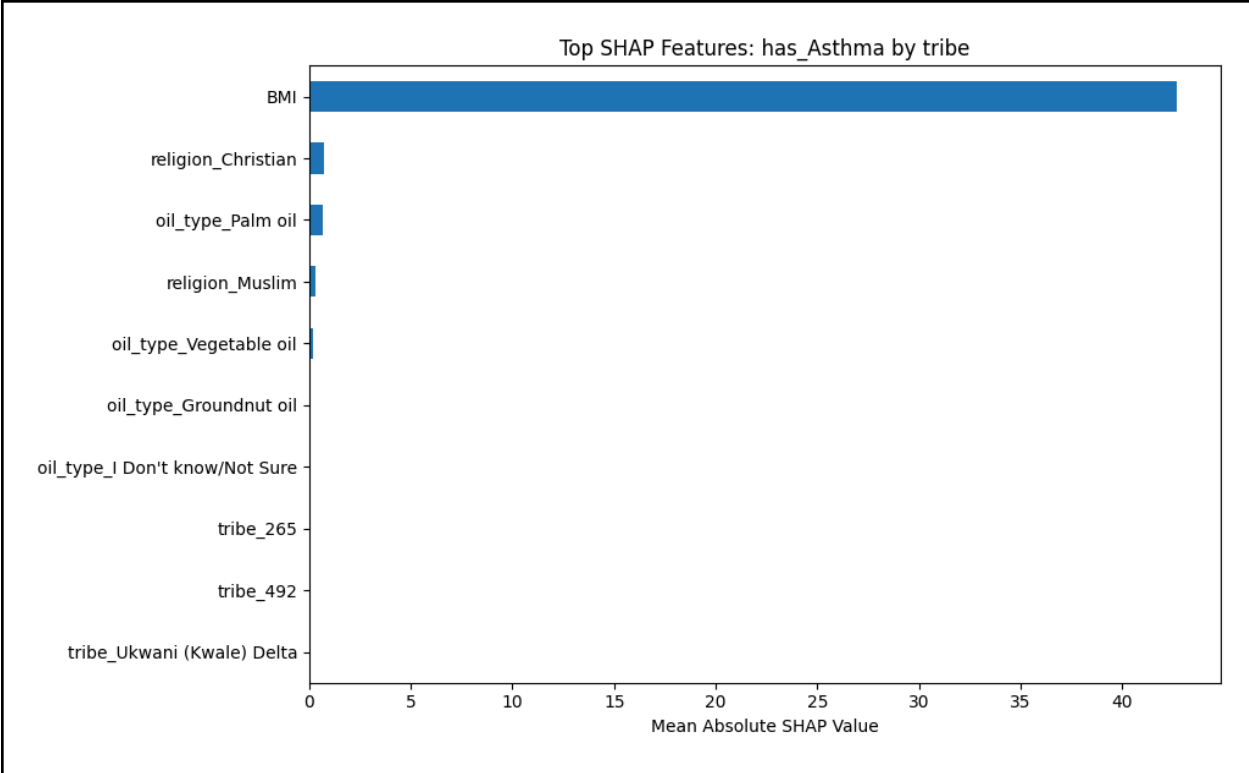


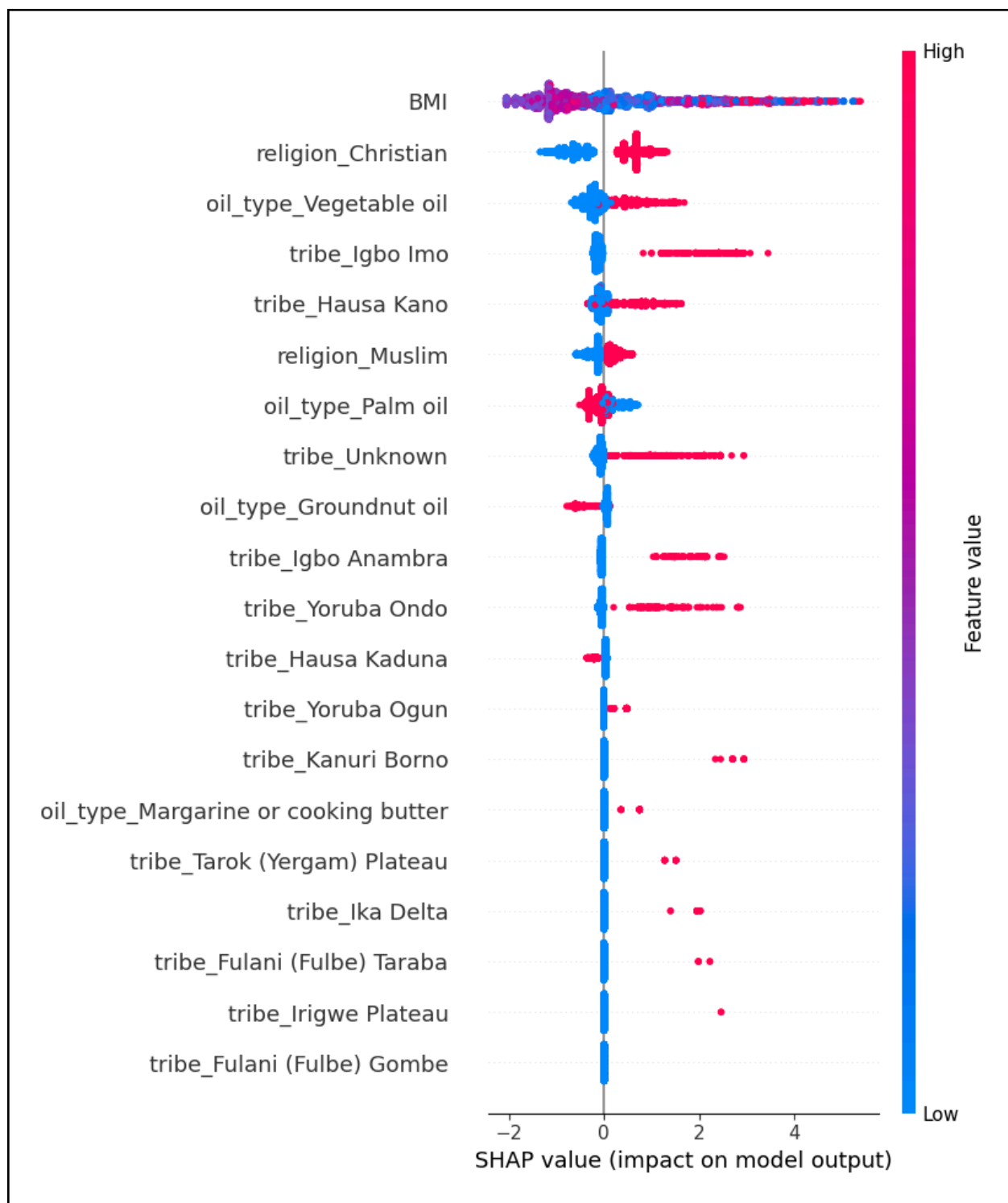


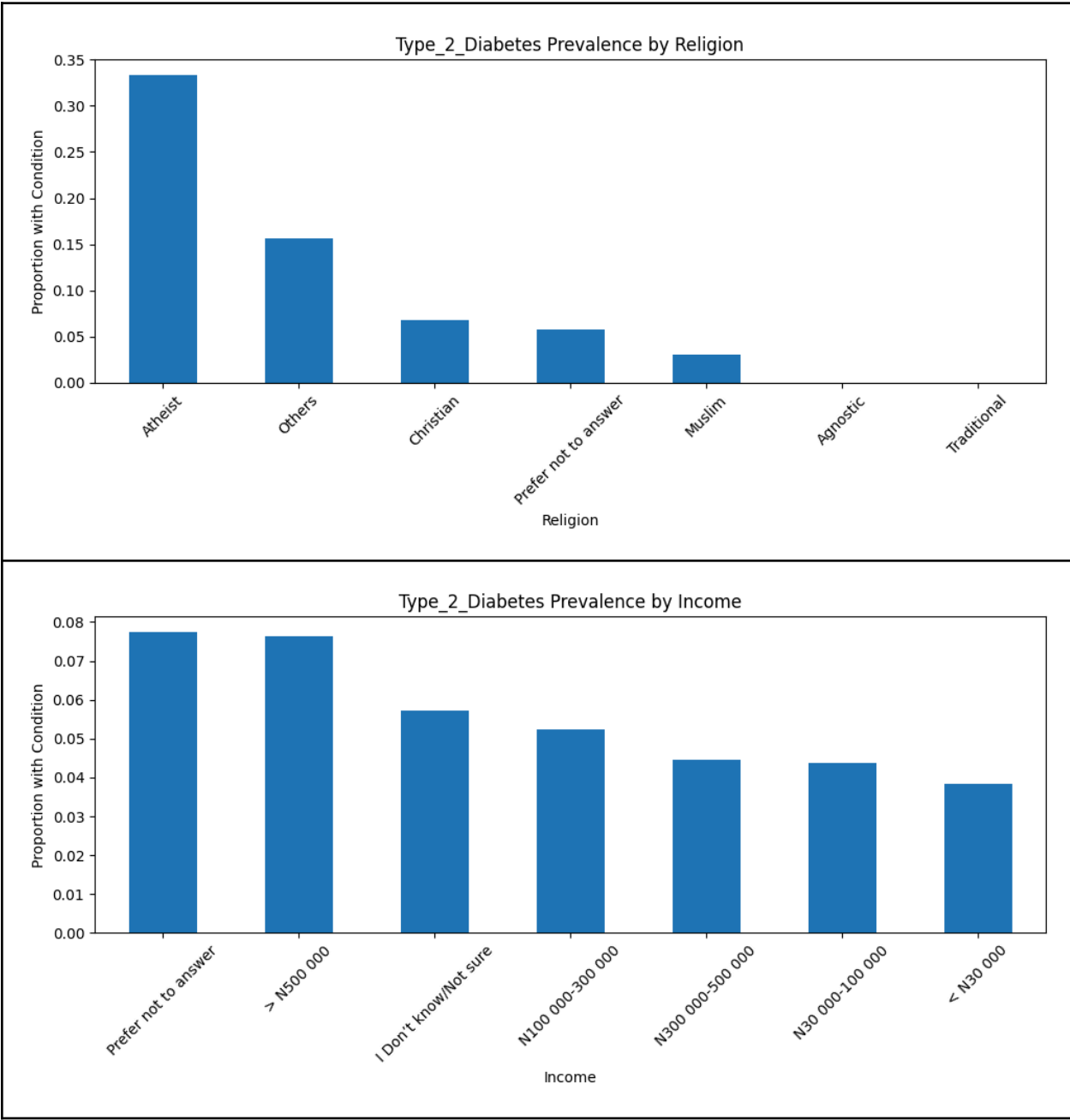


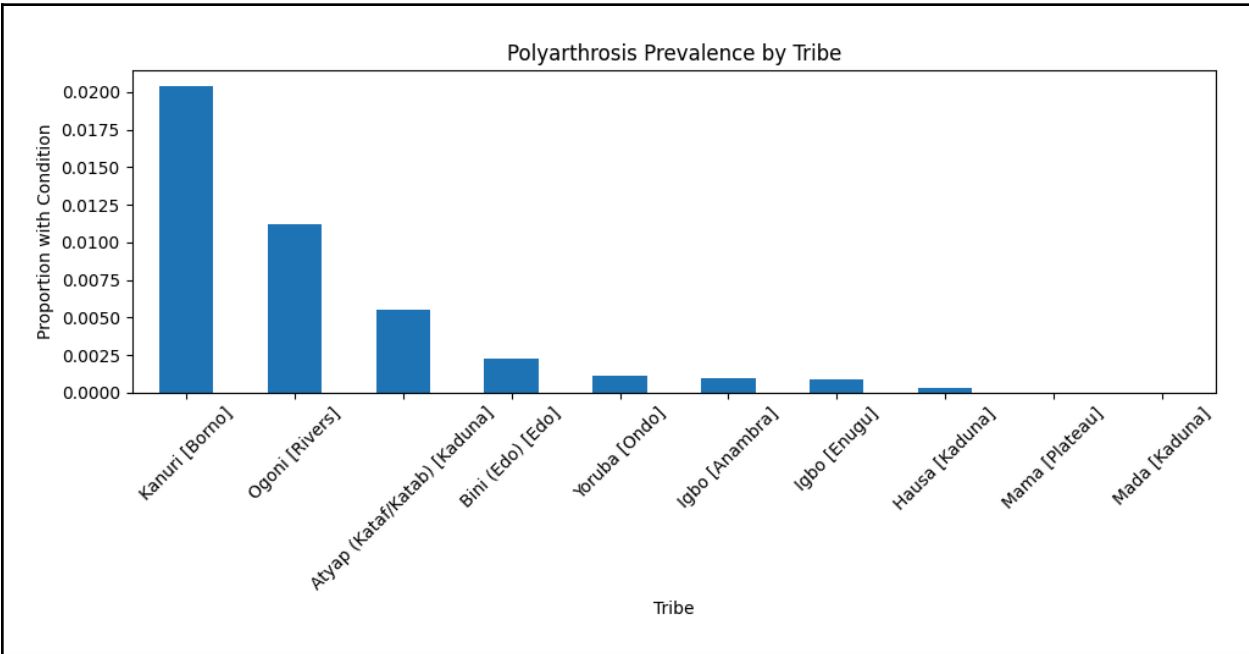
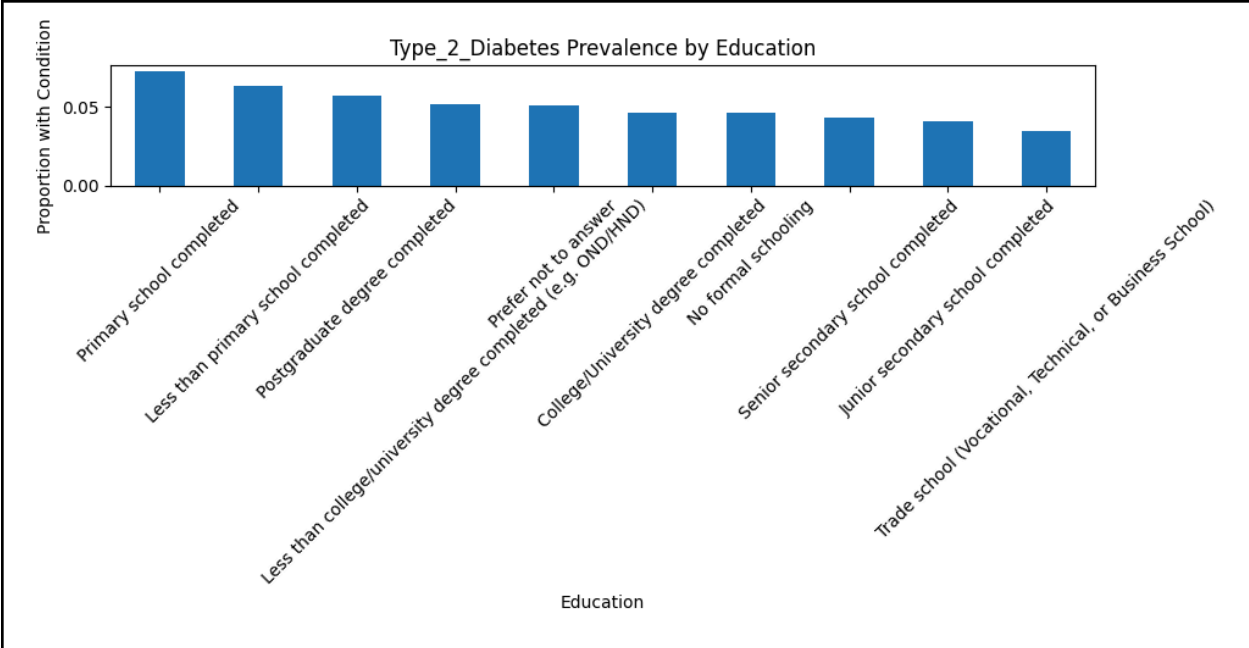


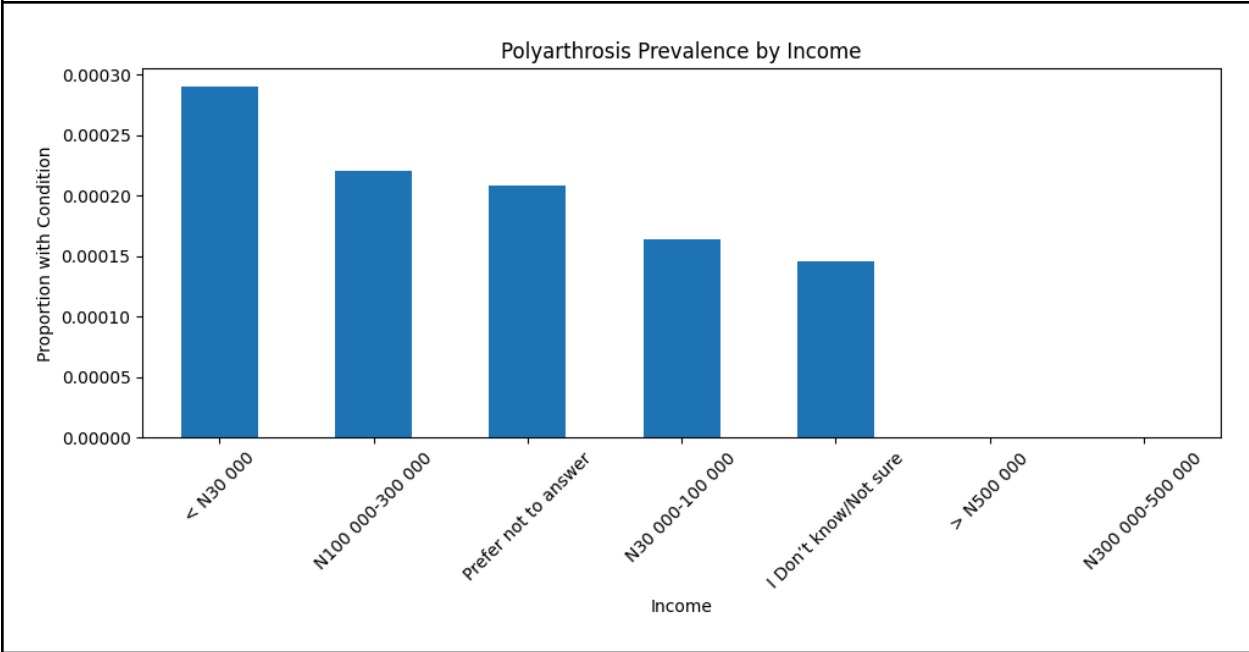
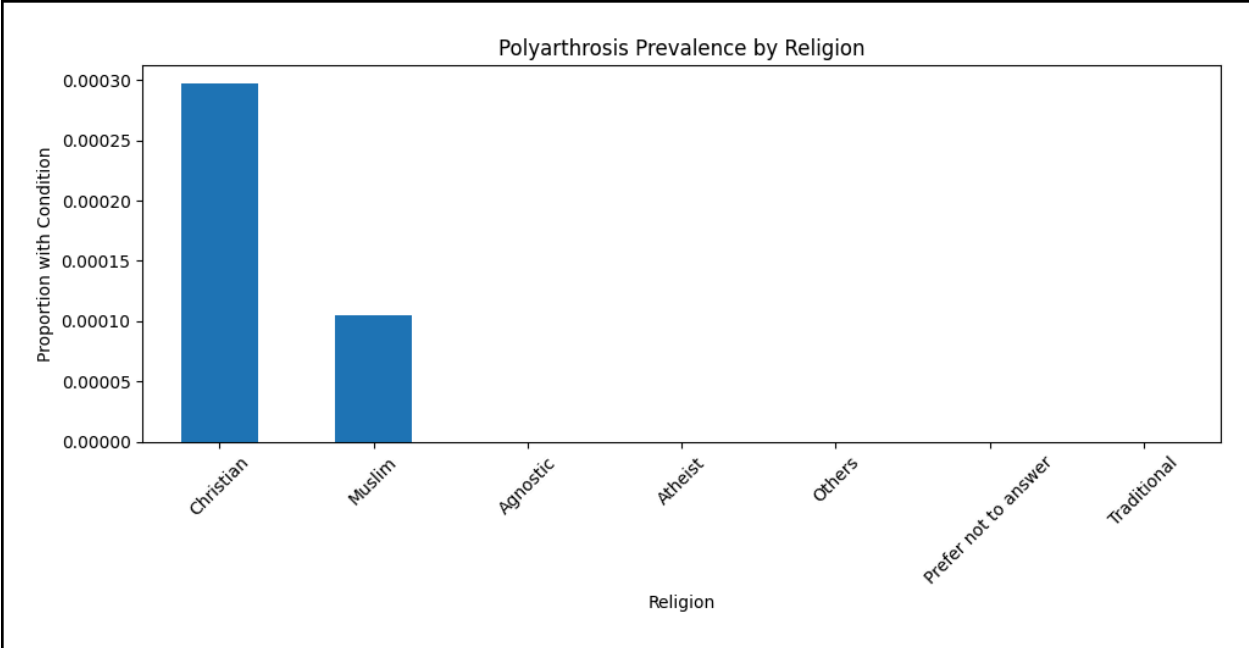


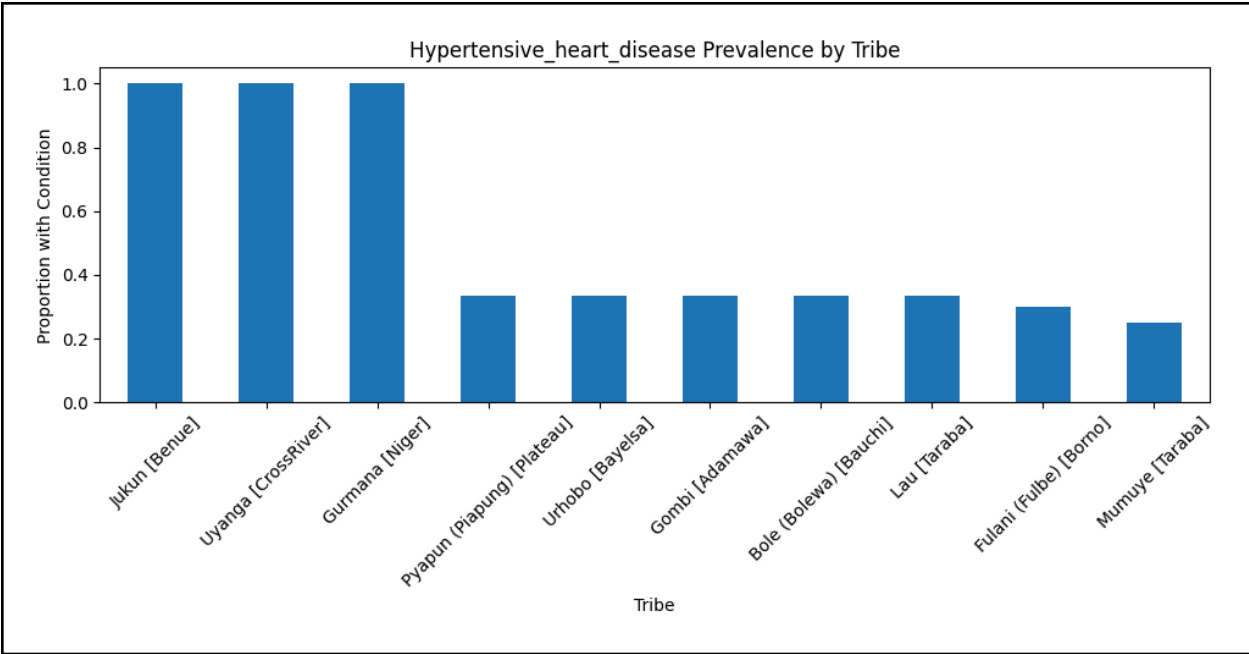
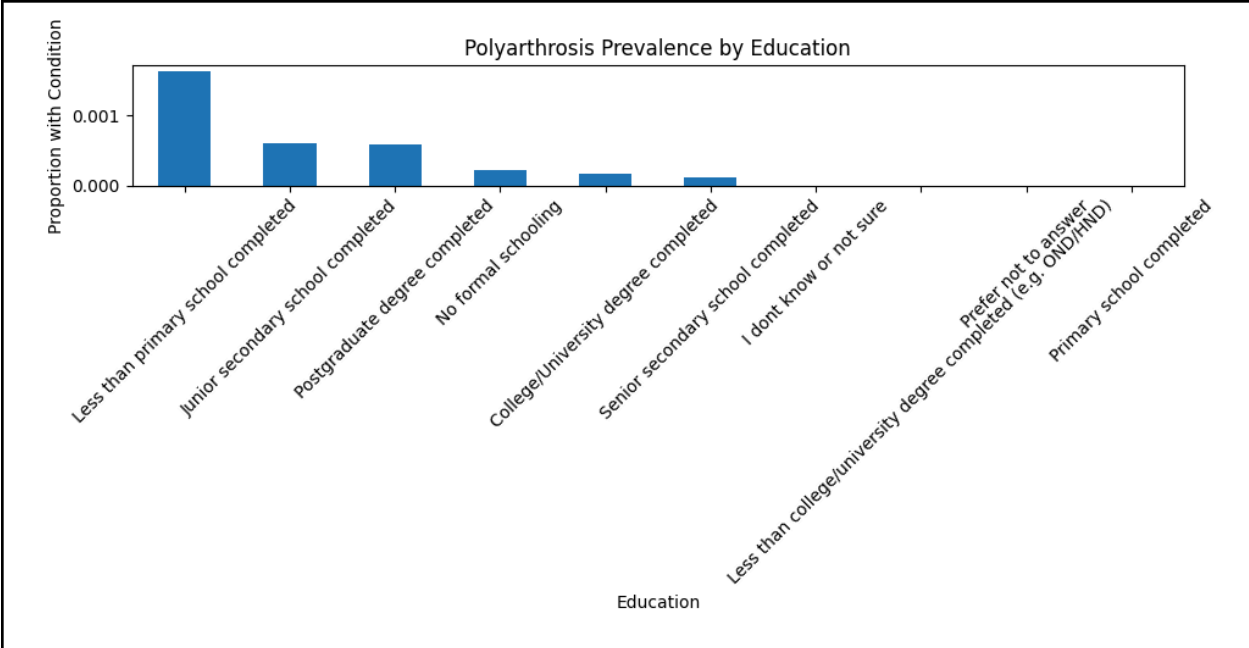


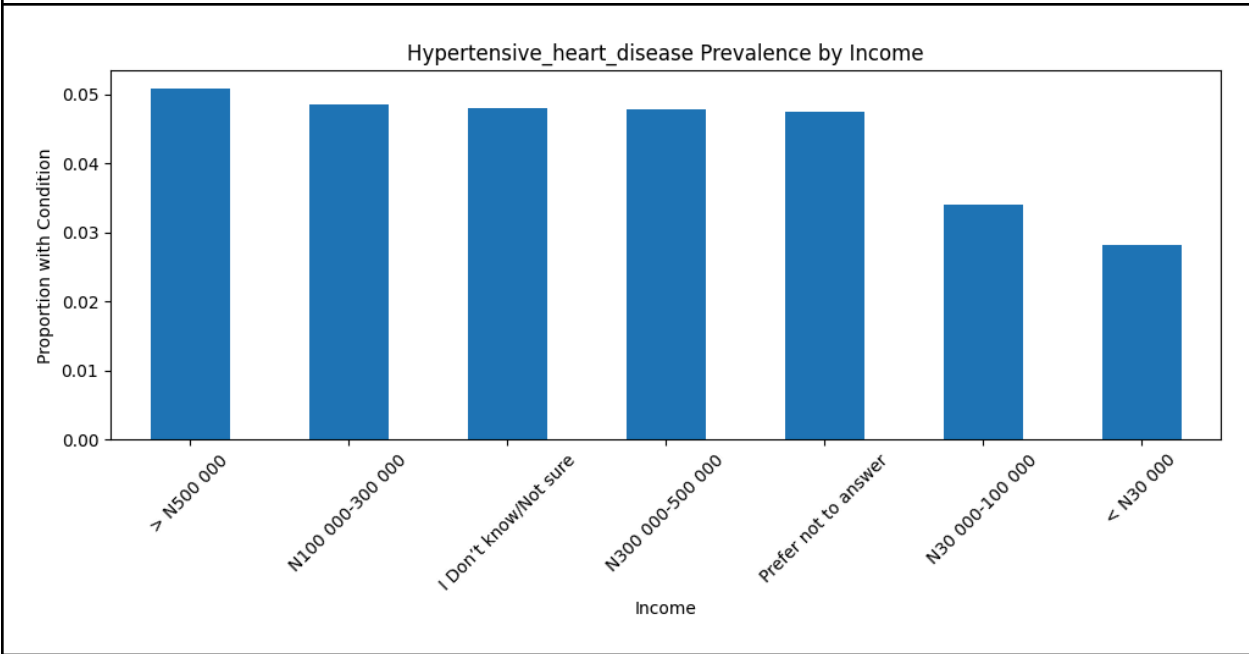
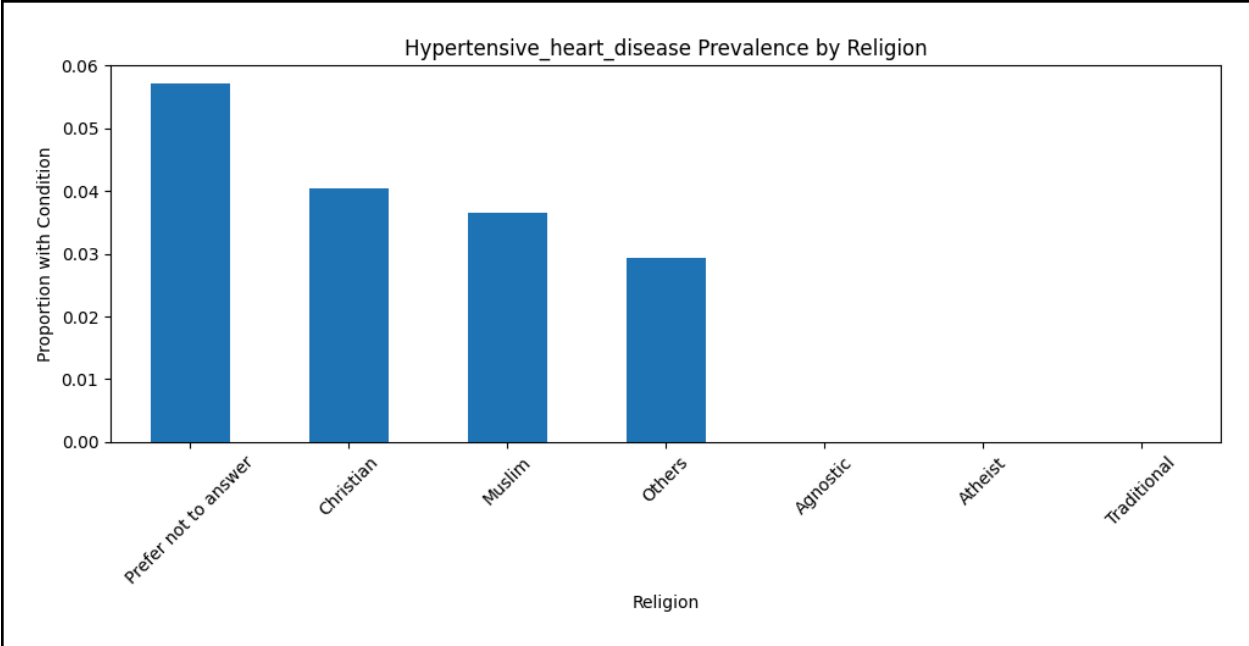


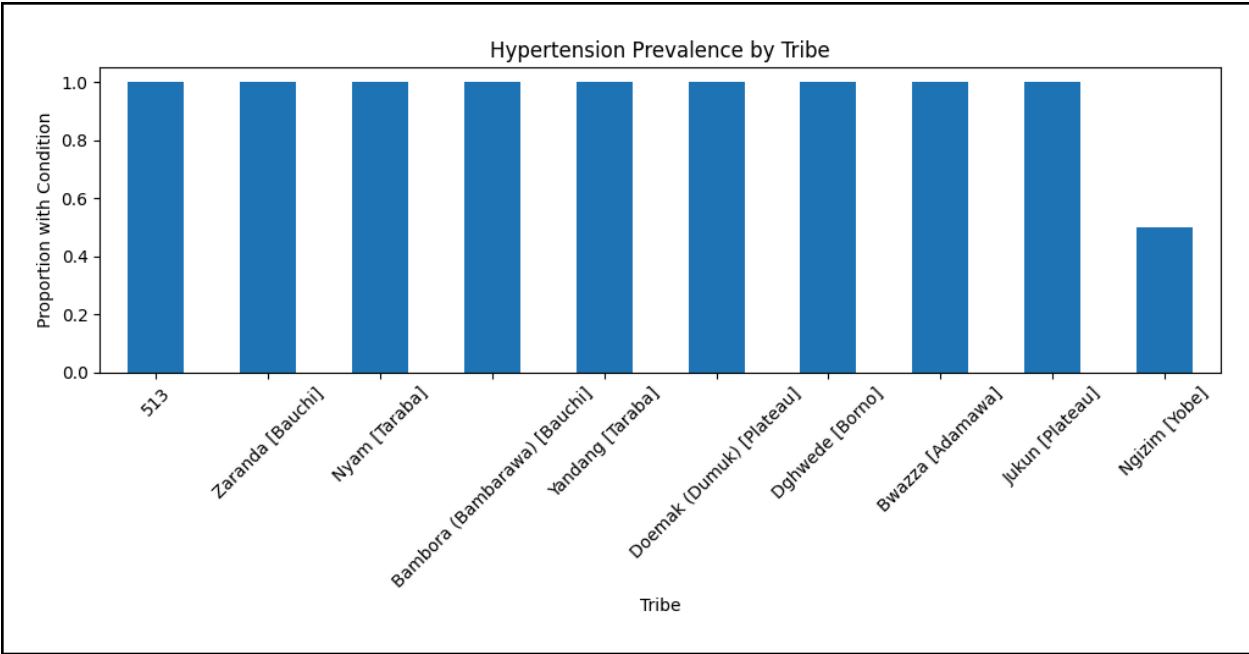
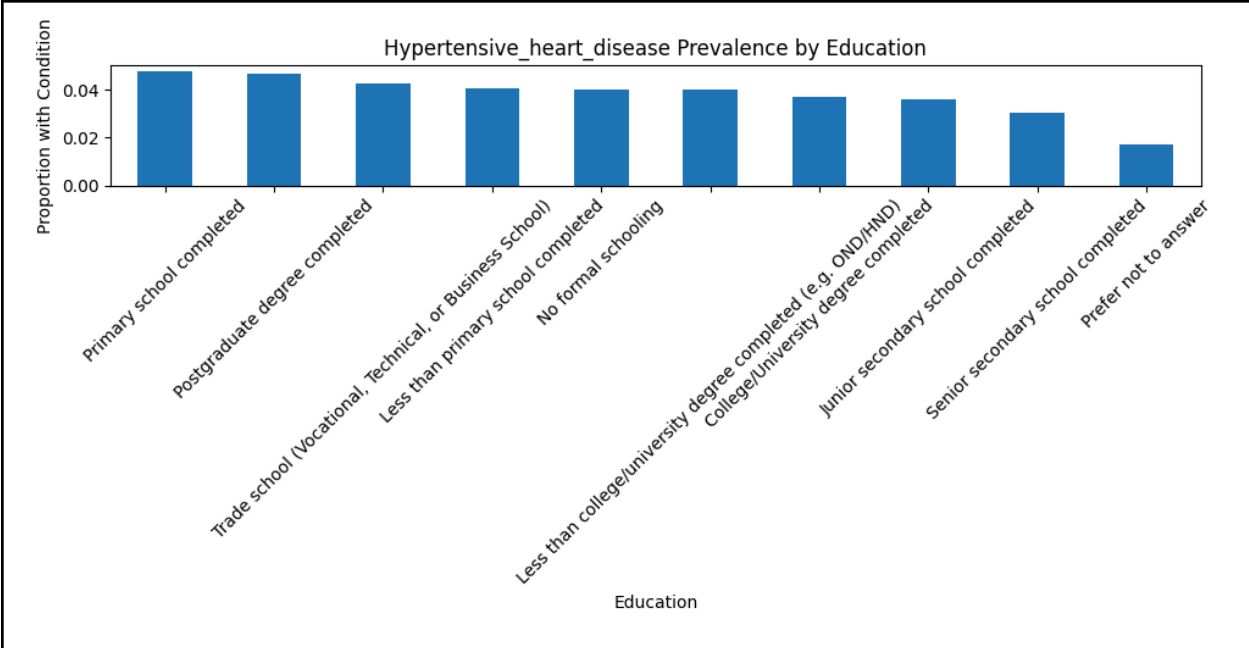


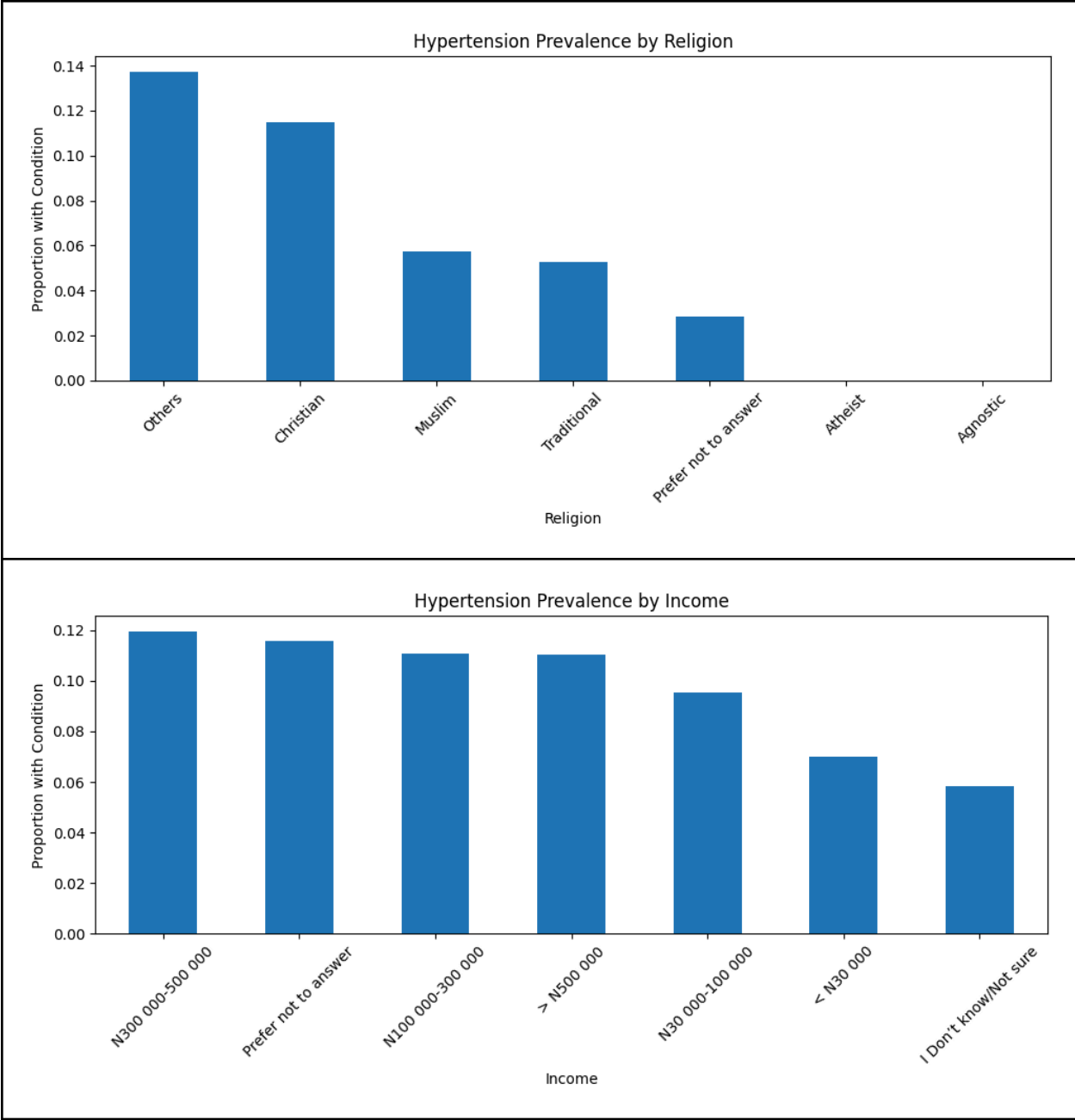


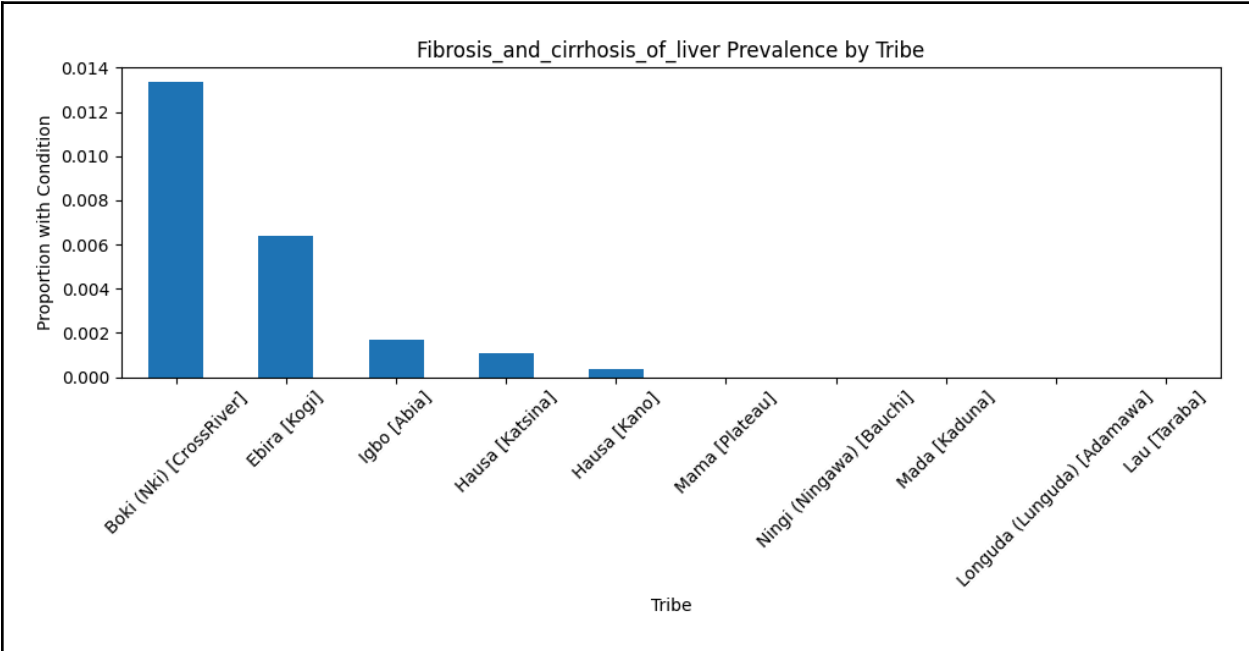
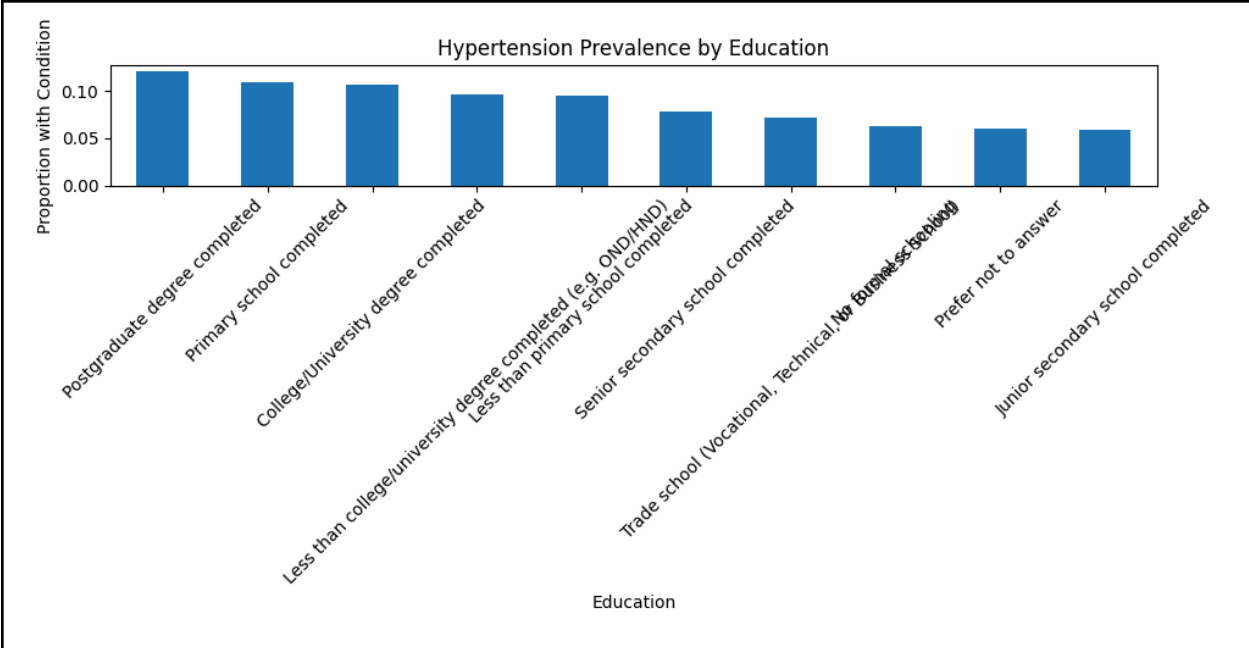


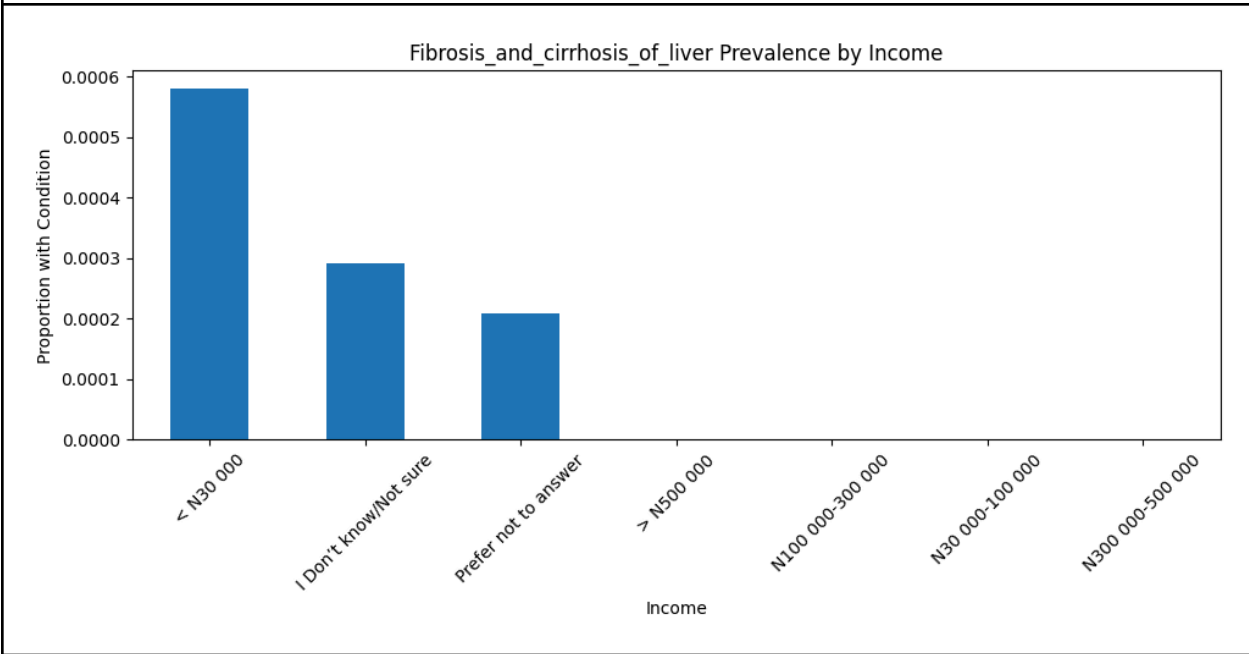
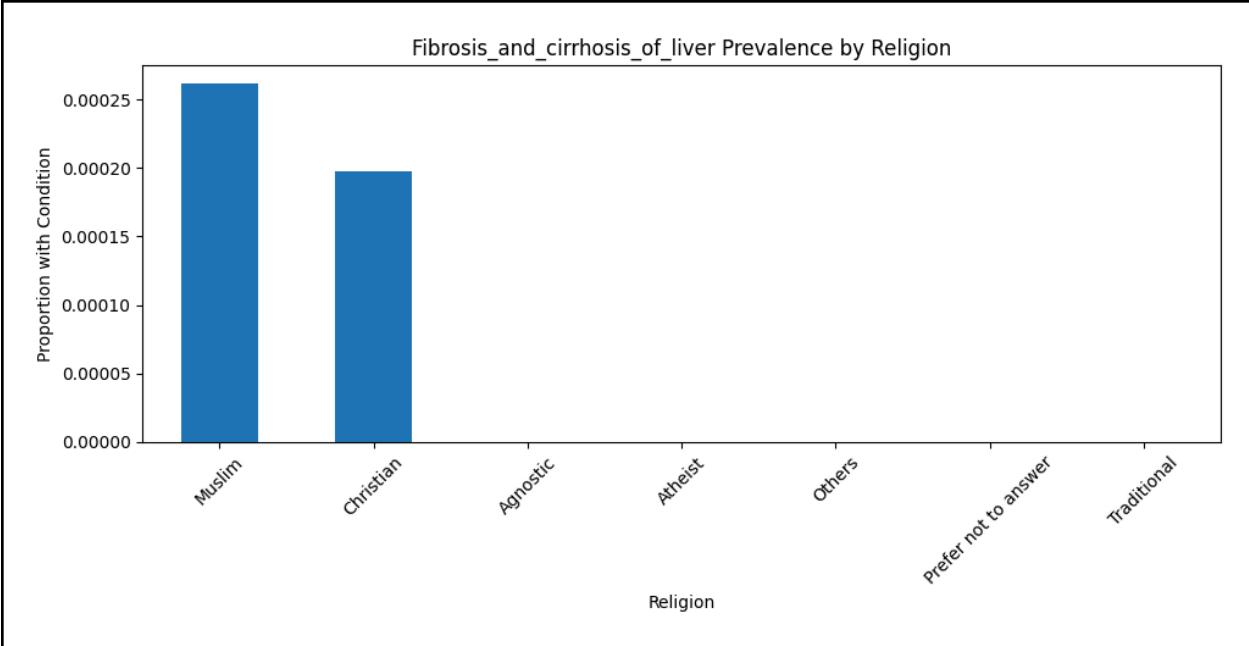


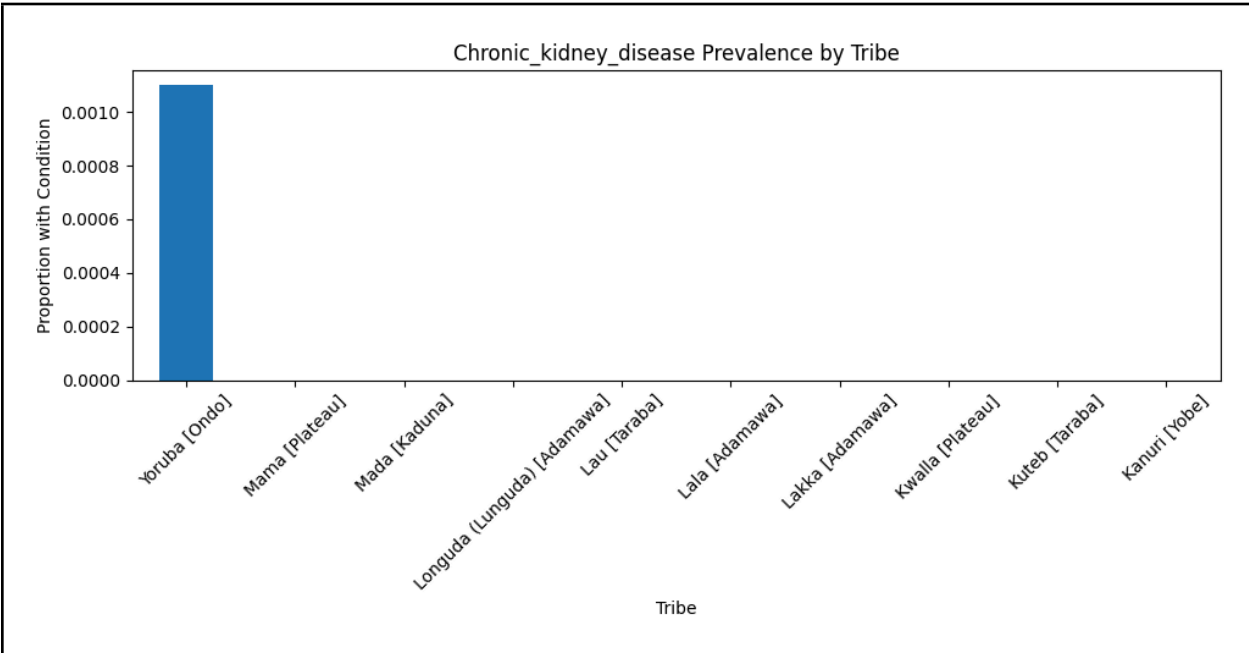
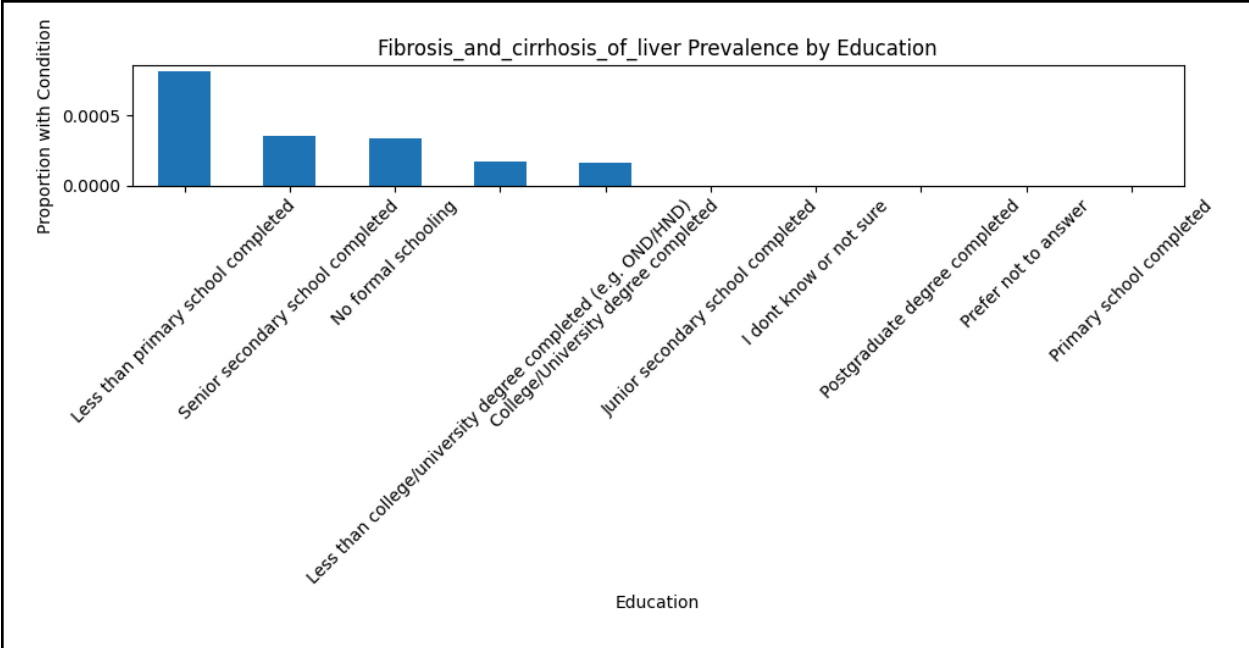


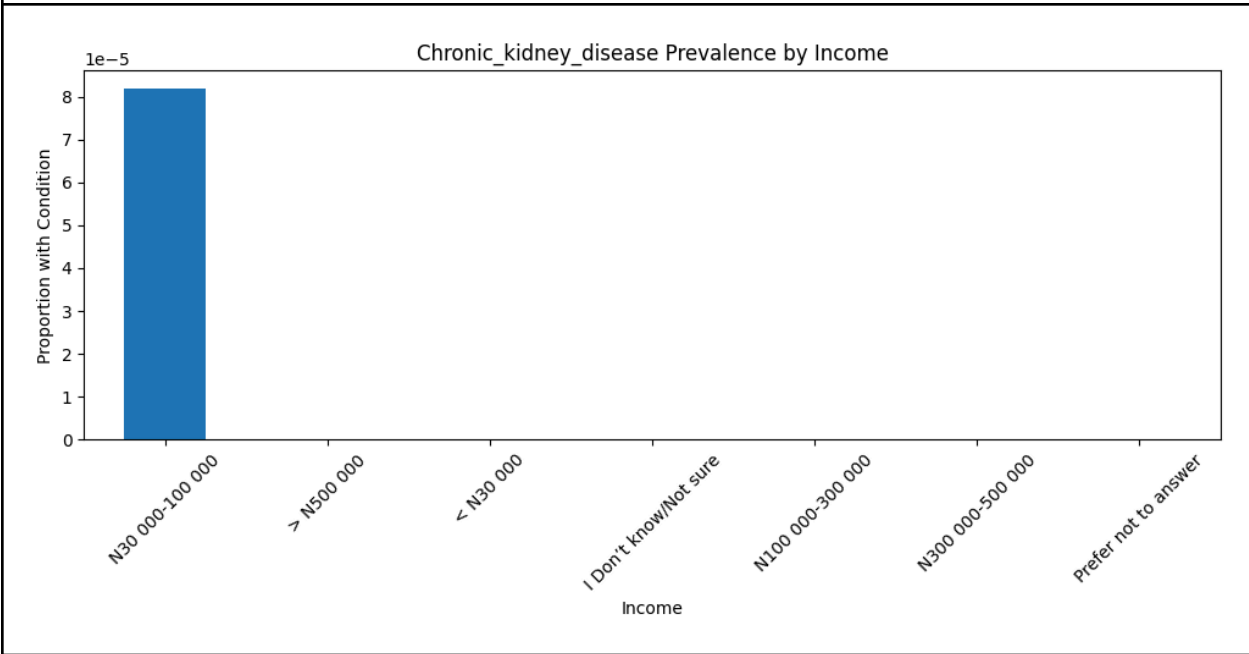
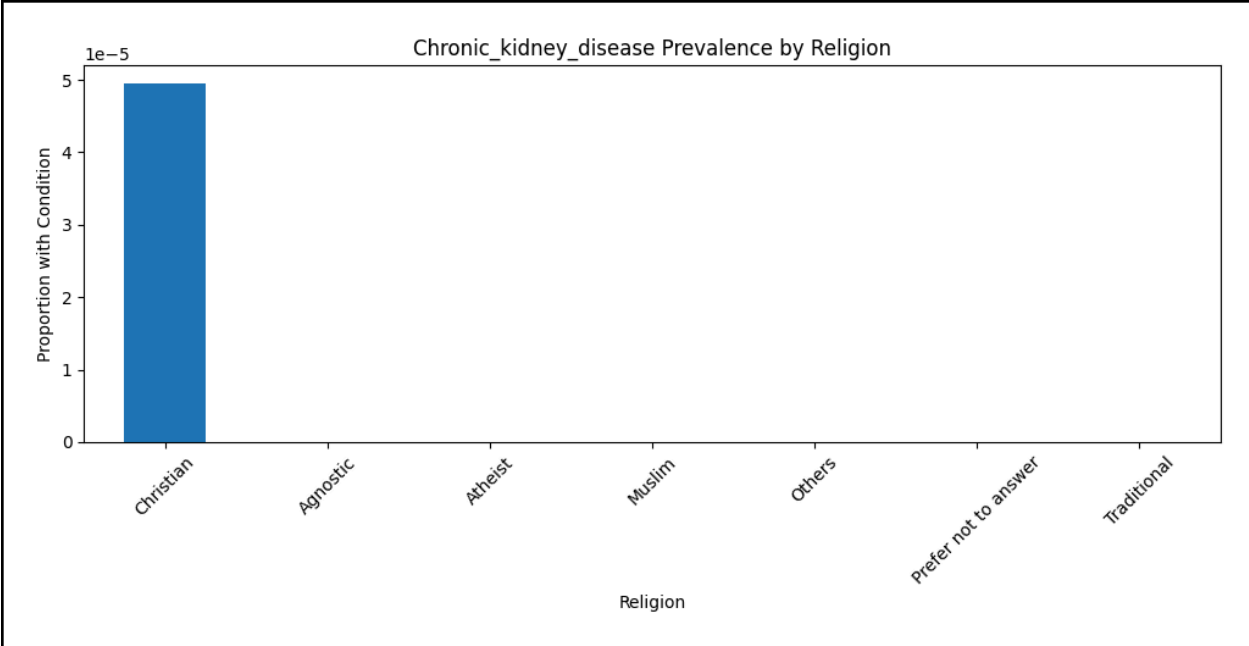


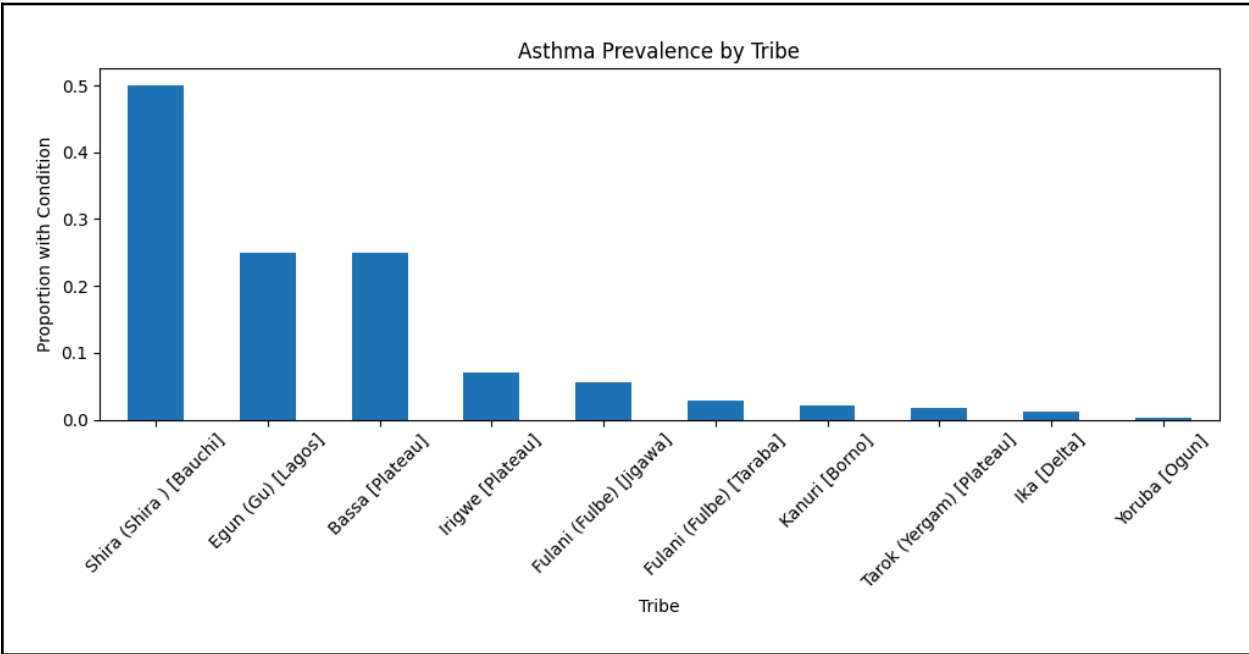
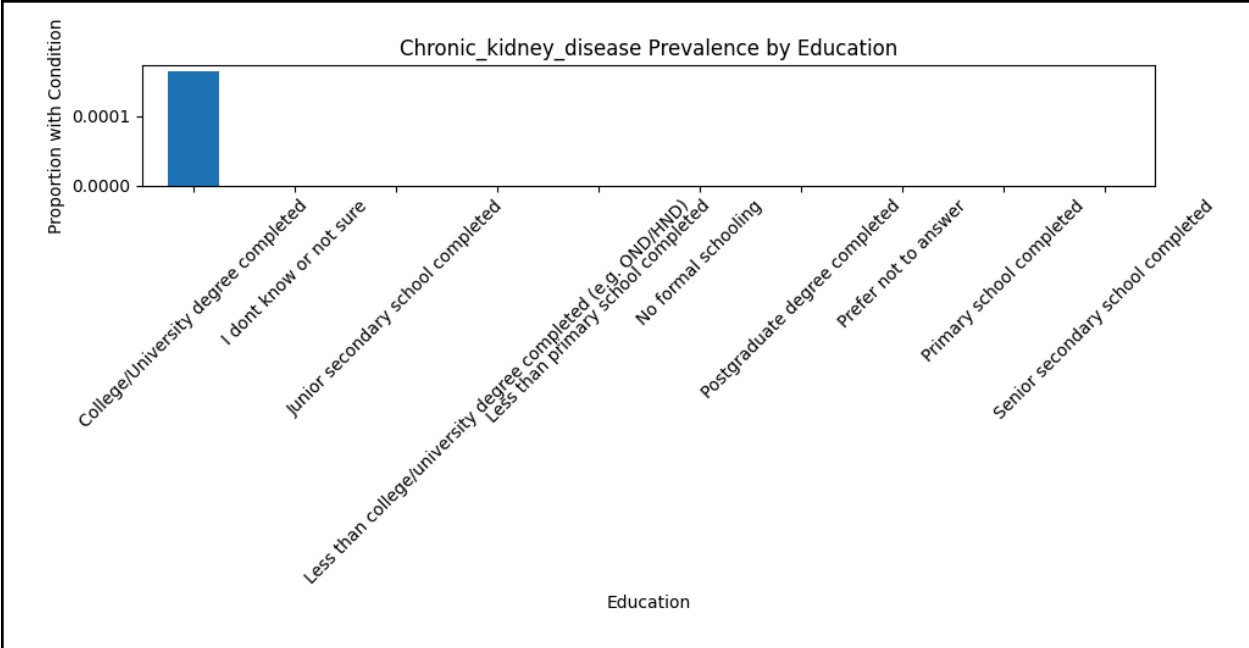


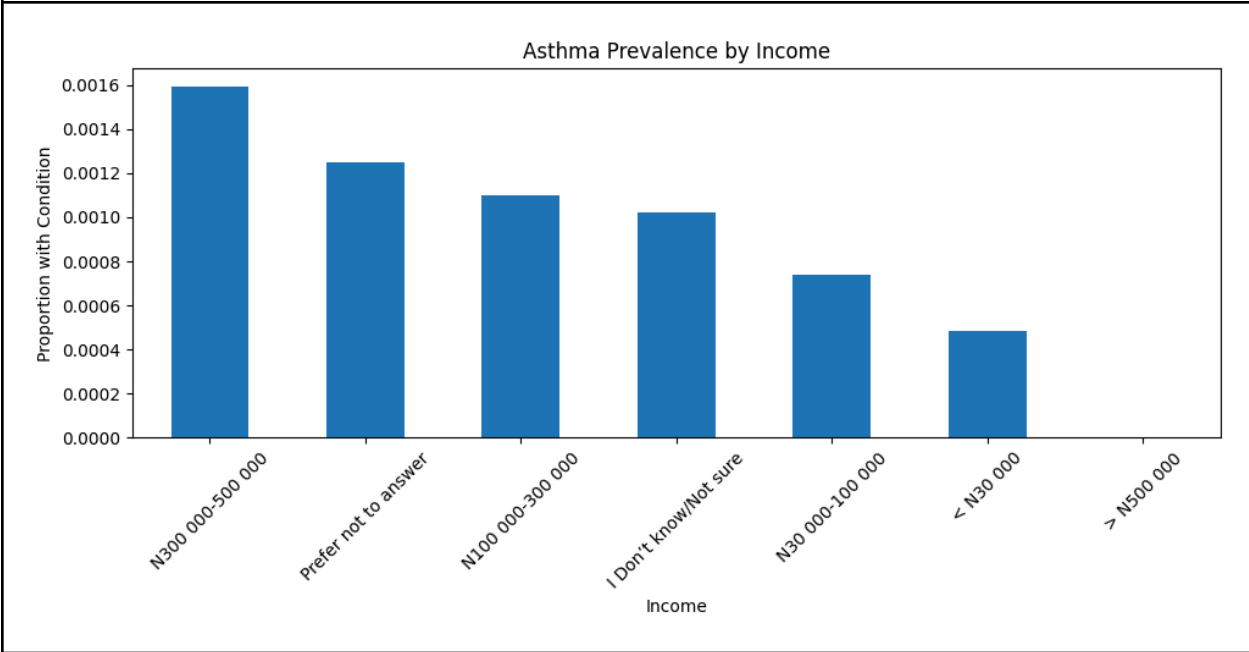
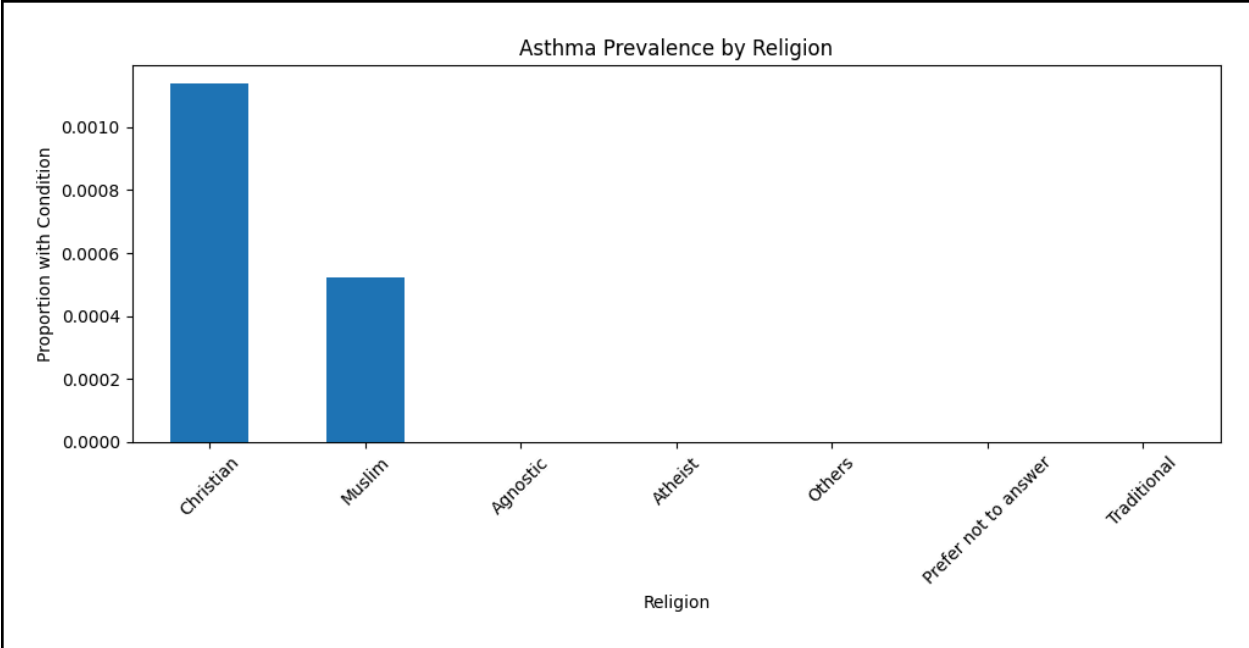


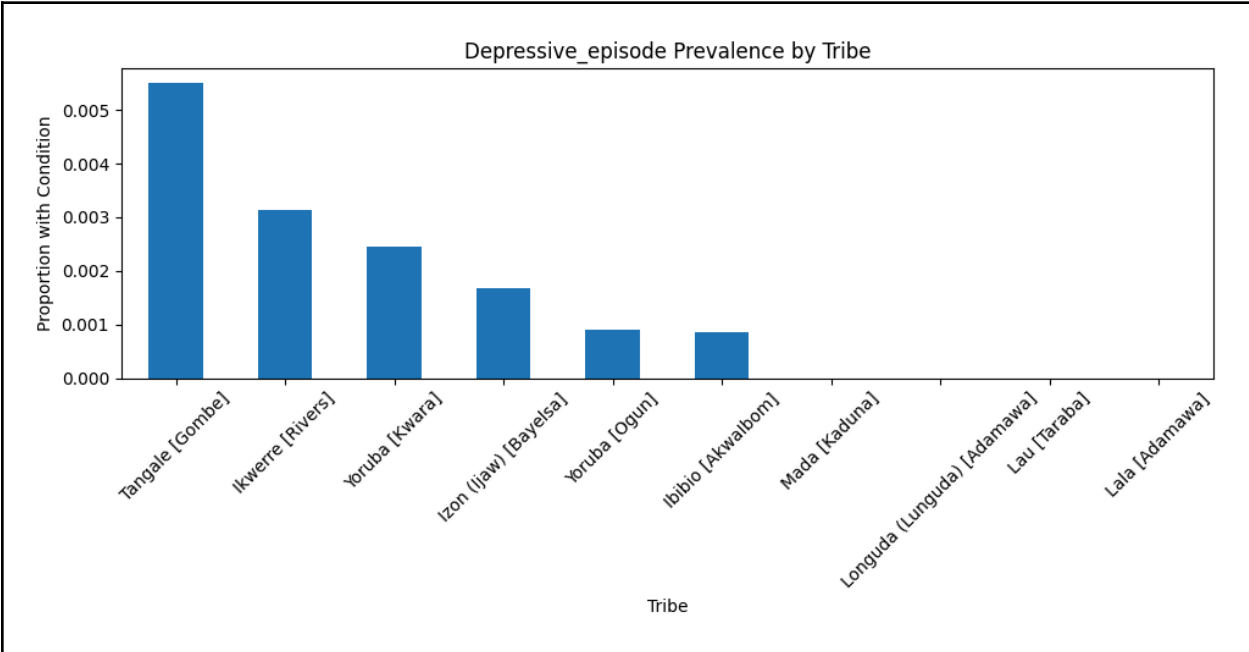
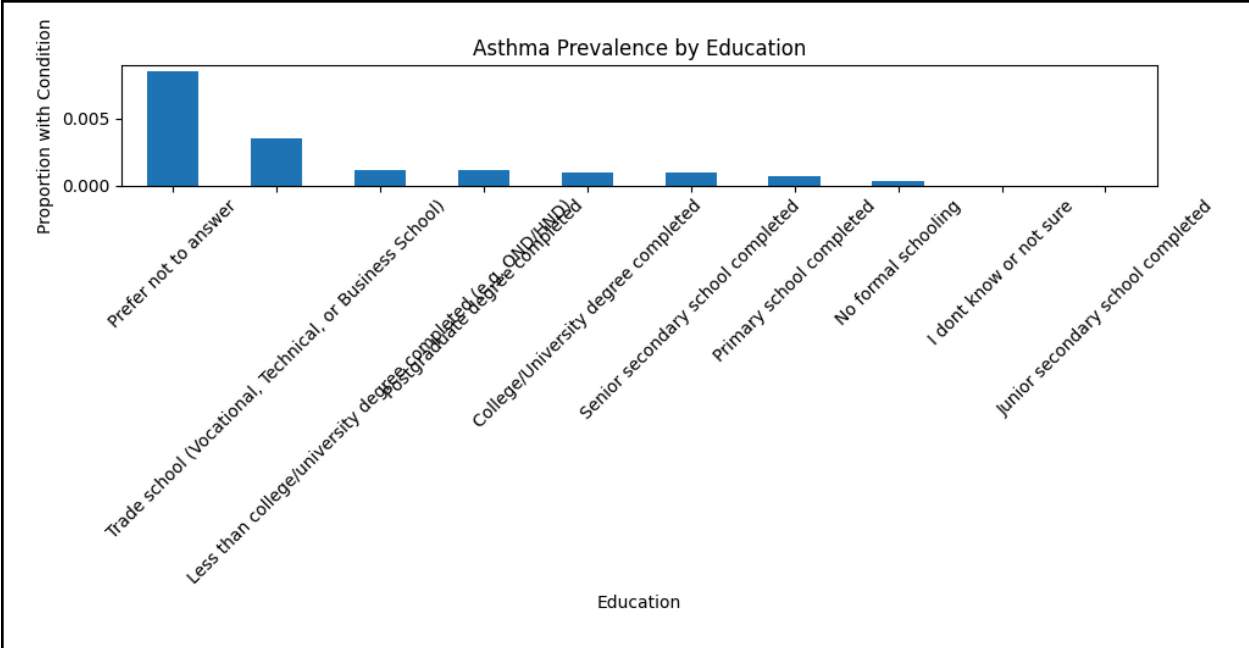


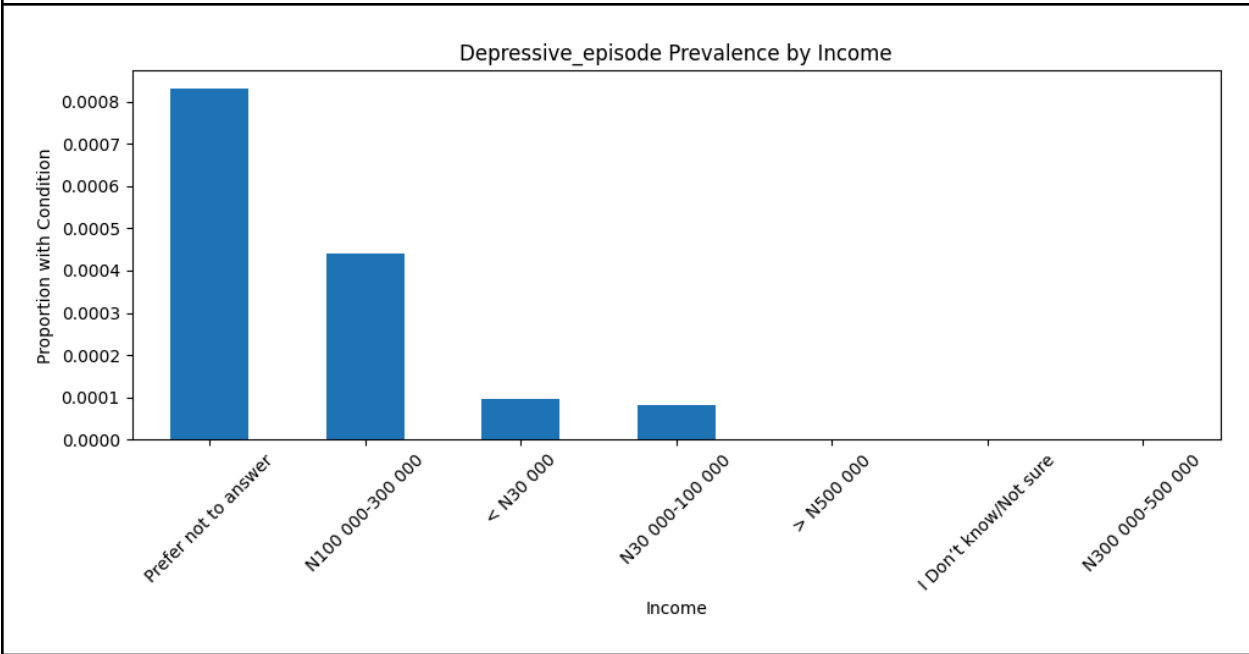
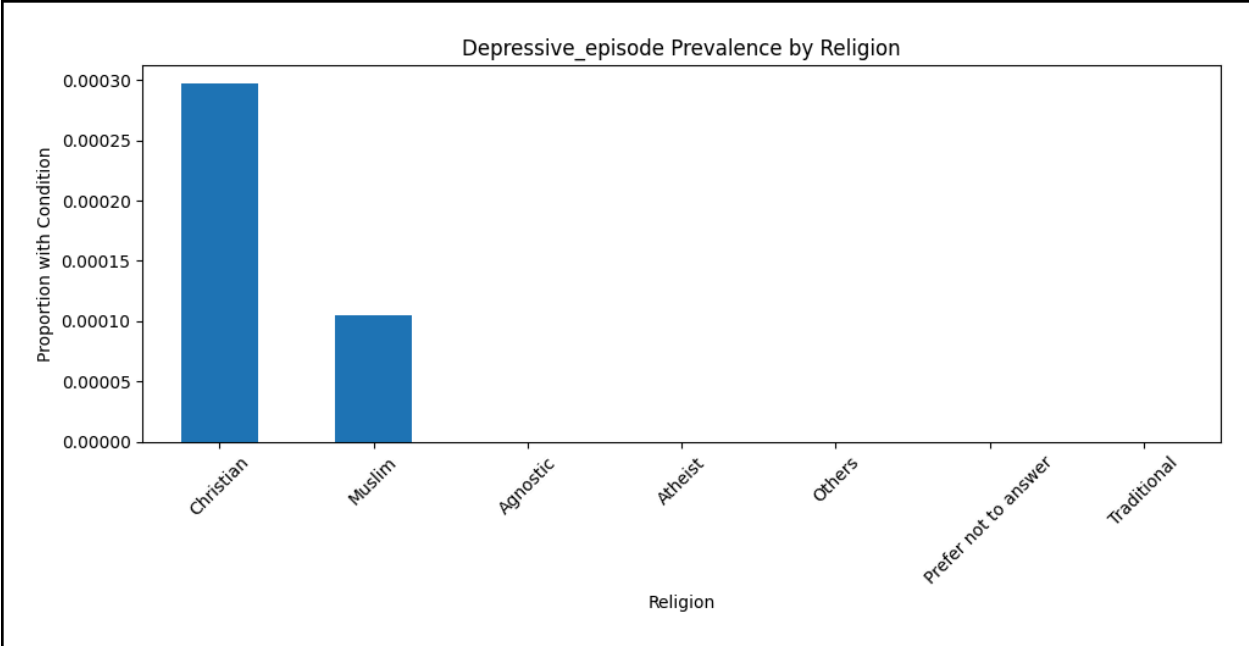


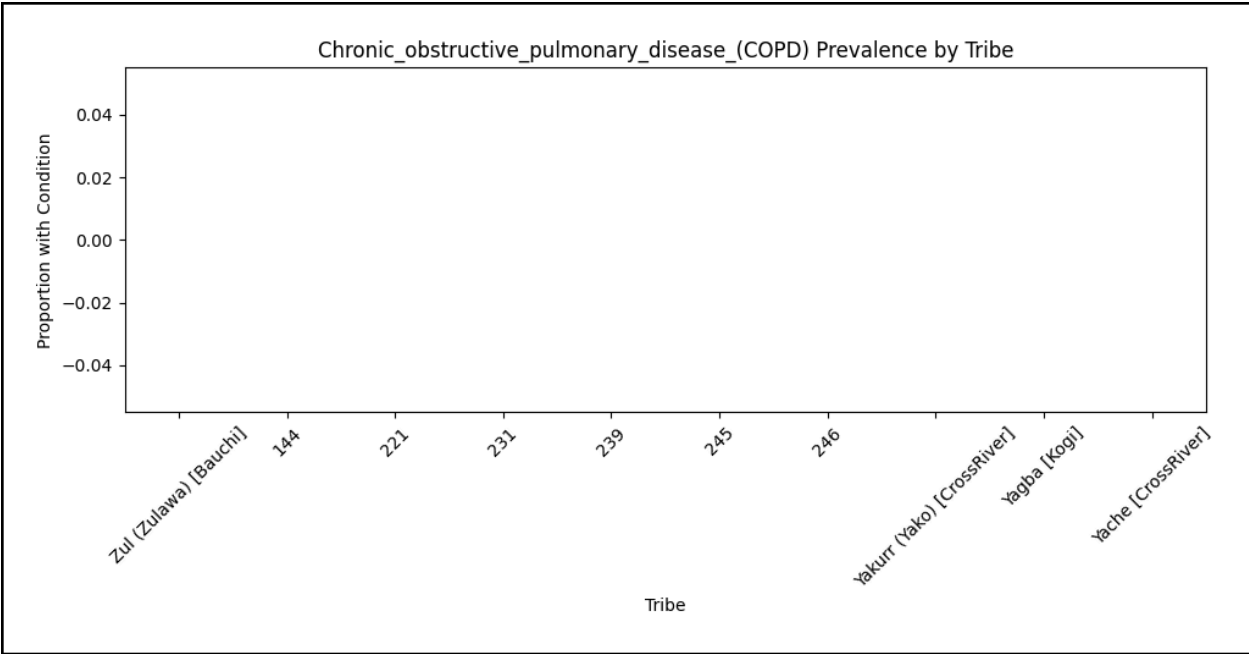
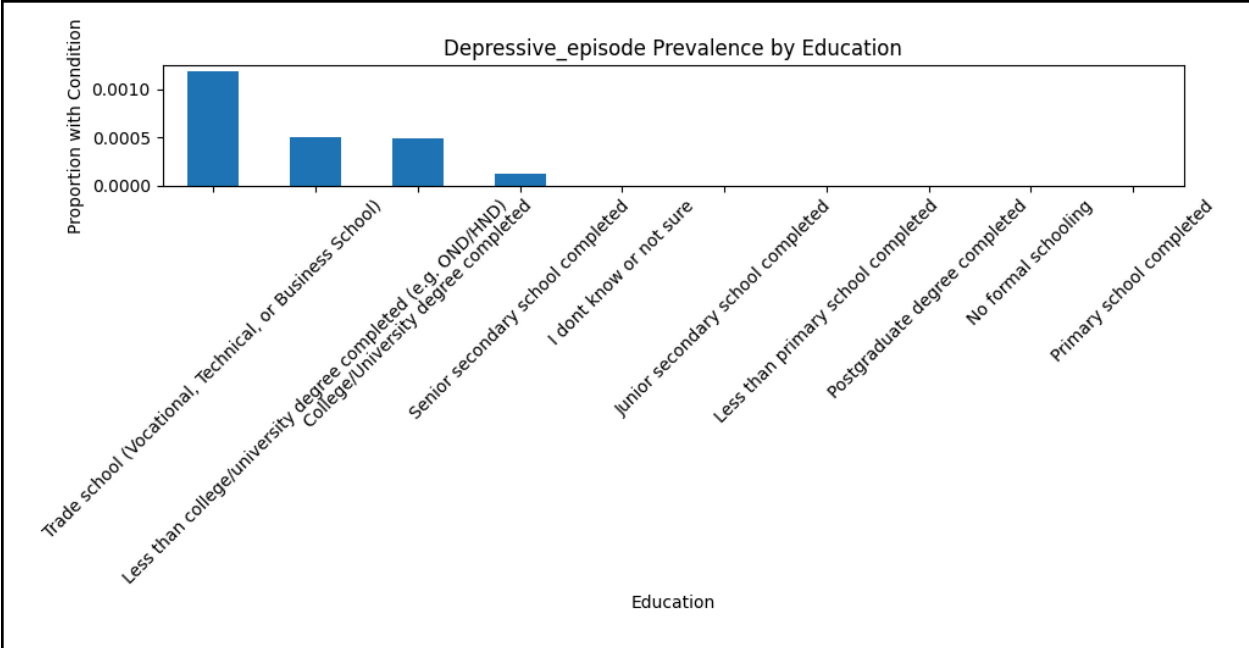


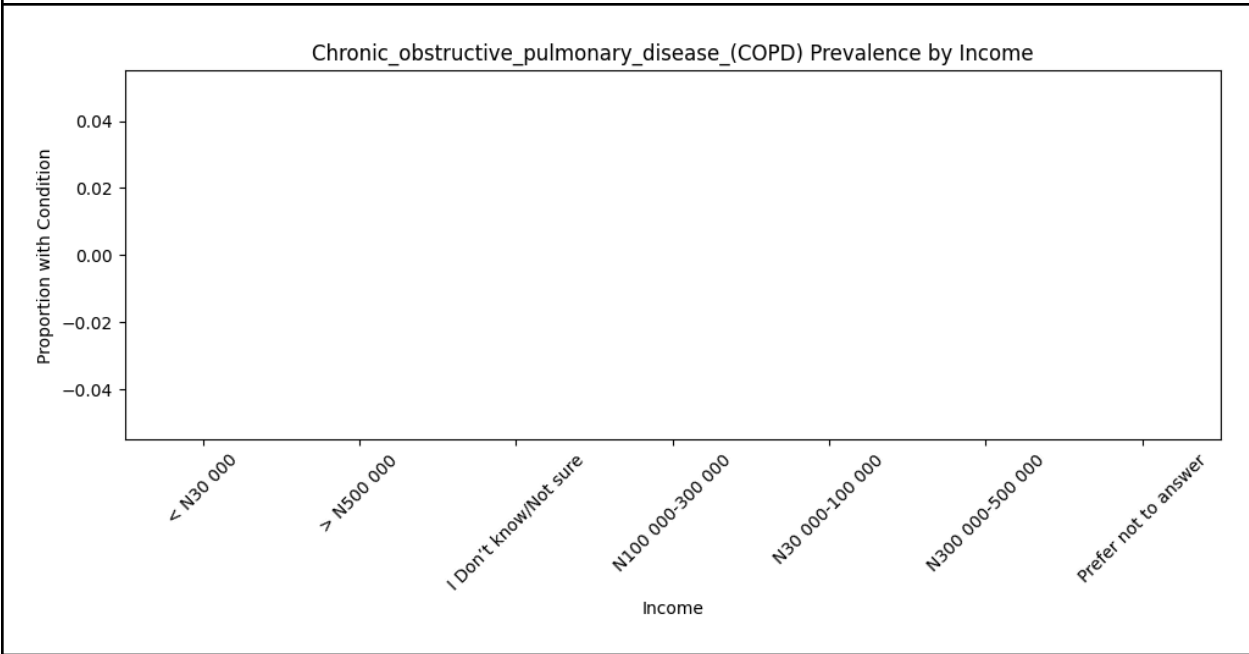
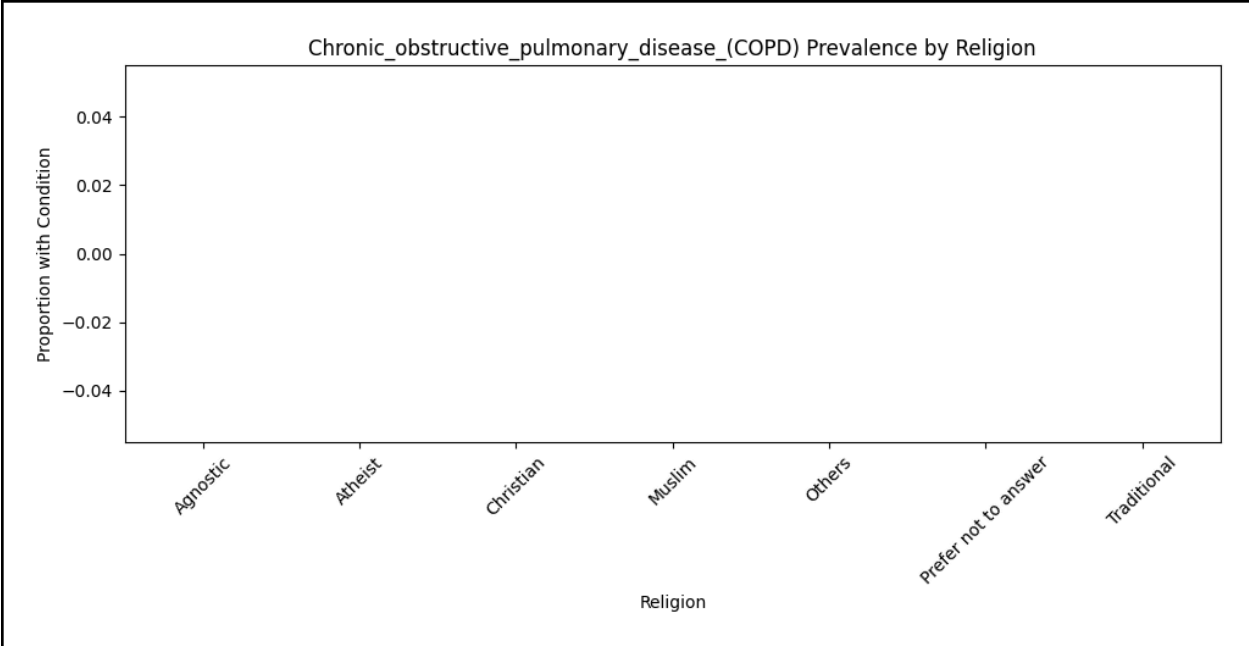












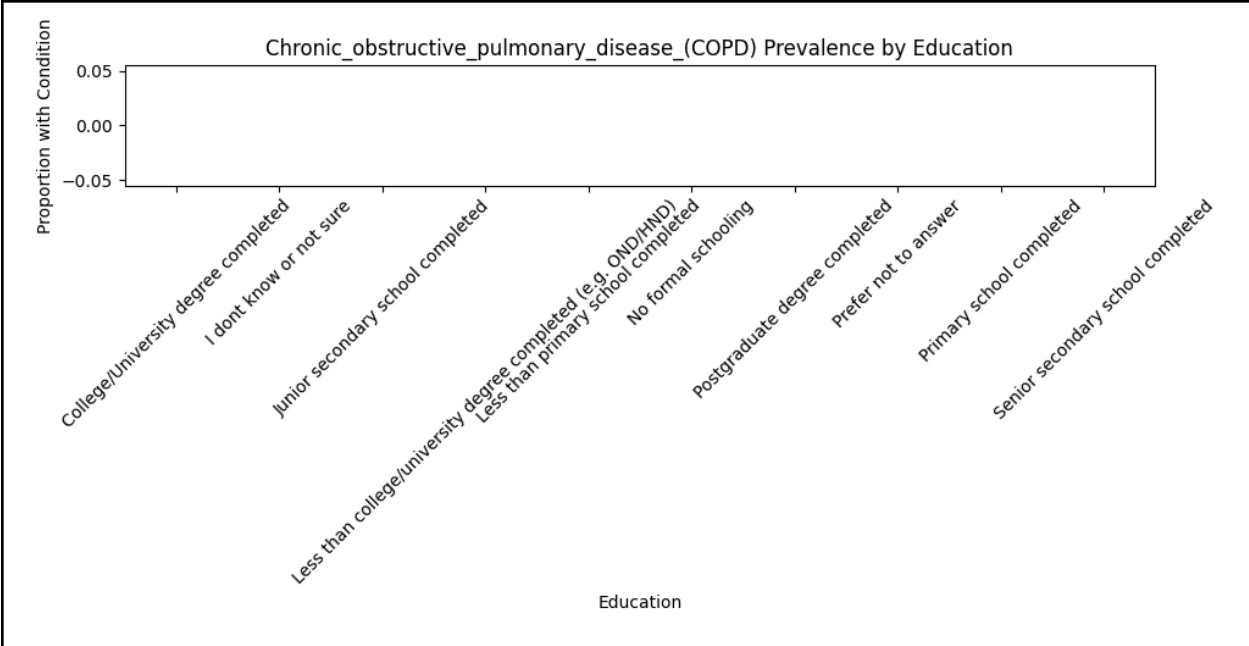
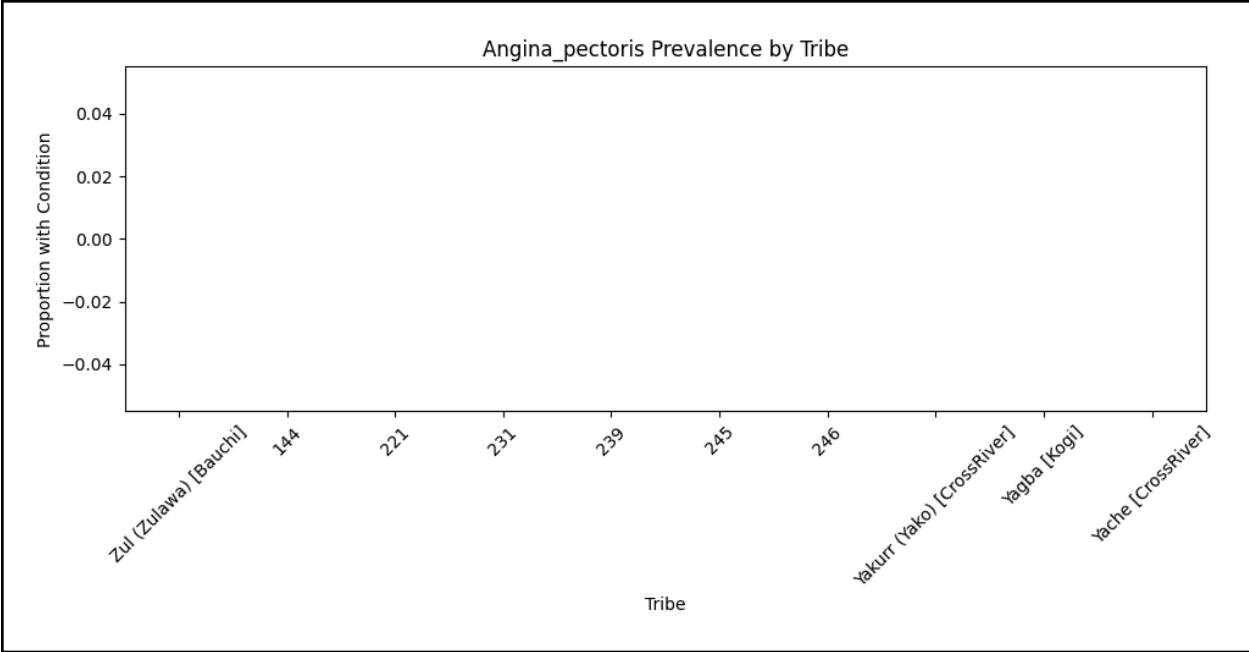
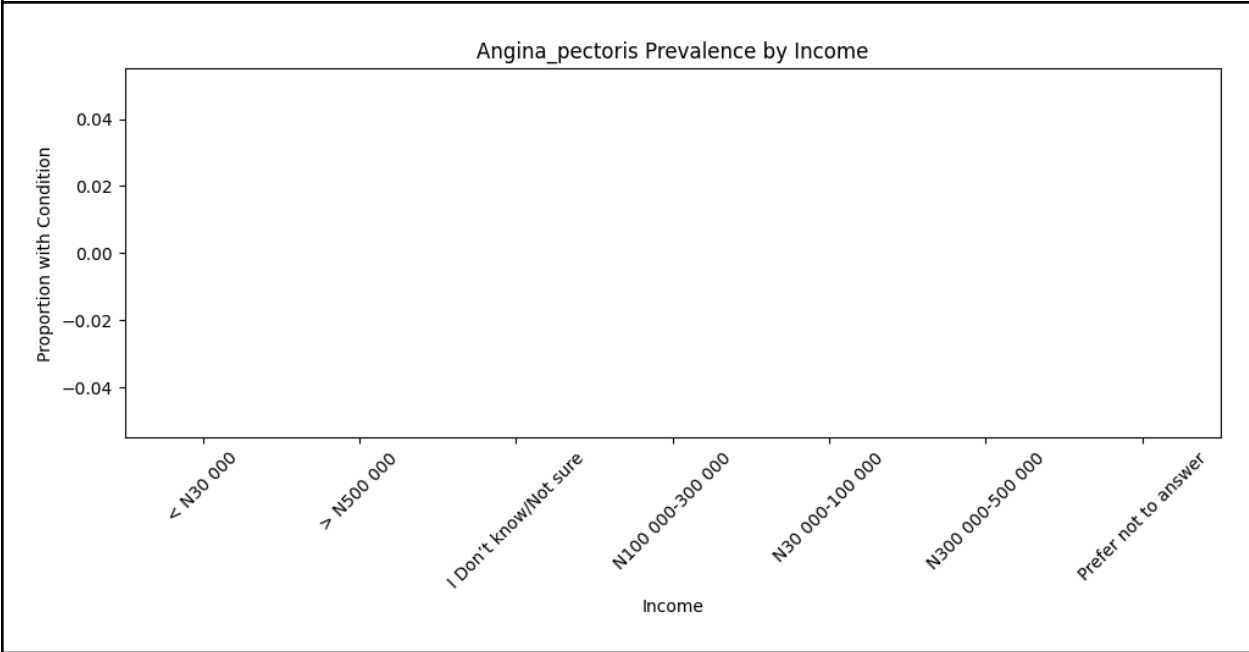
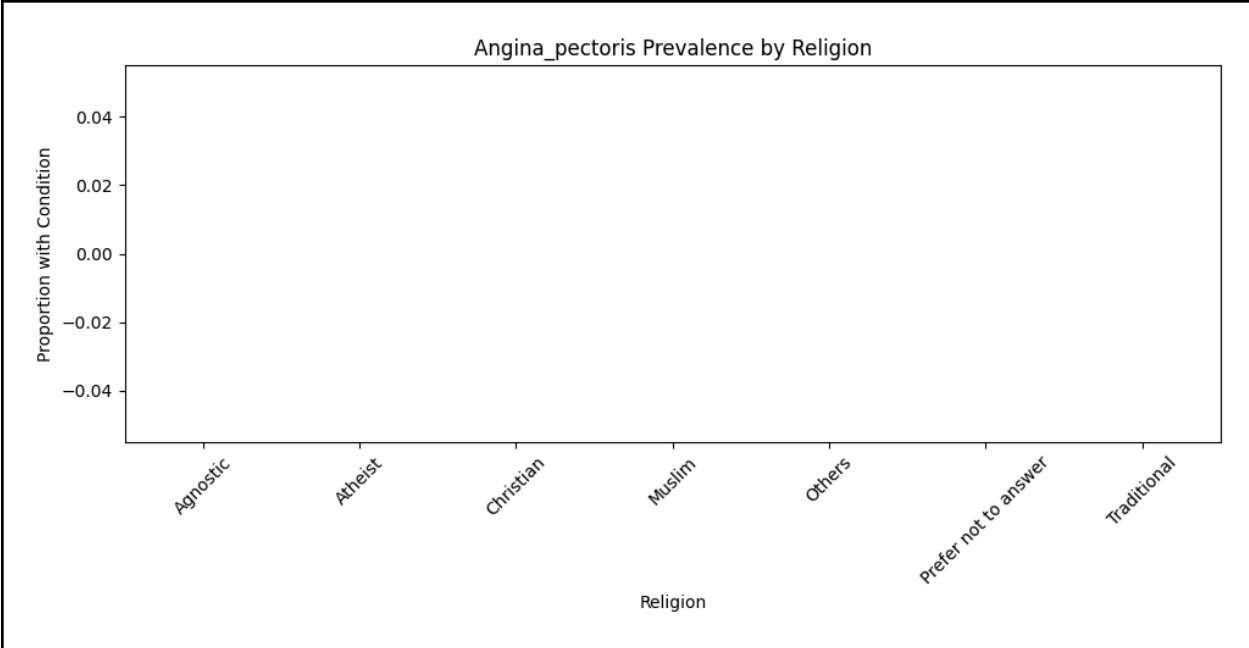


Figure 17 | Sociodemographic Distribution of Chronic Obstructive Pulmonary Disease Prevalence in Nigeria. Bar plots displaying the prevalence of Chronic Obstructive Pulmonary across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





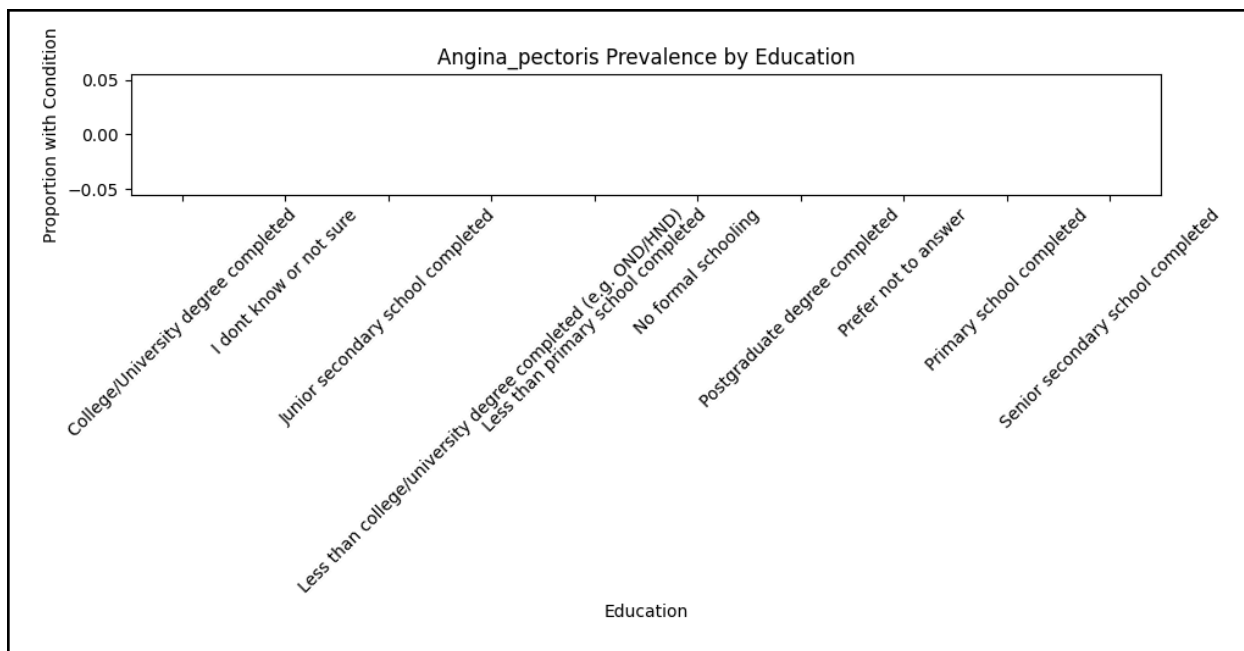
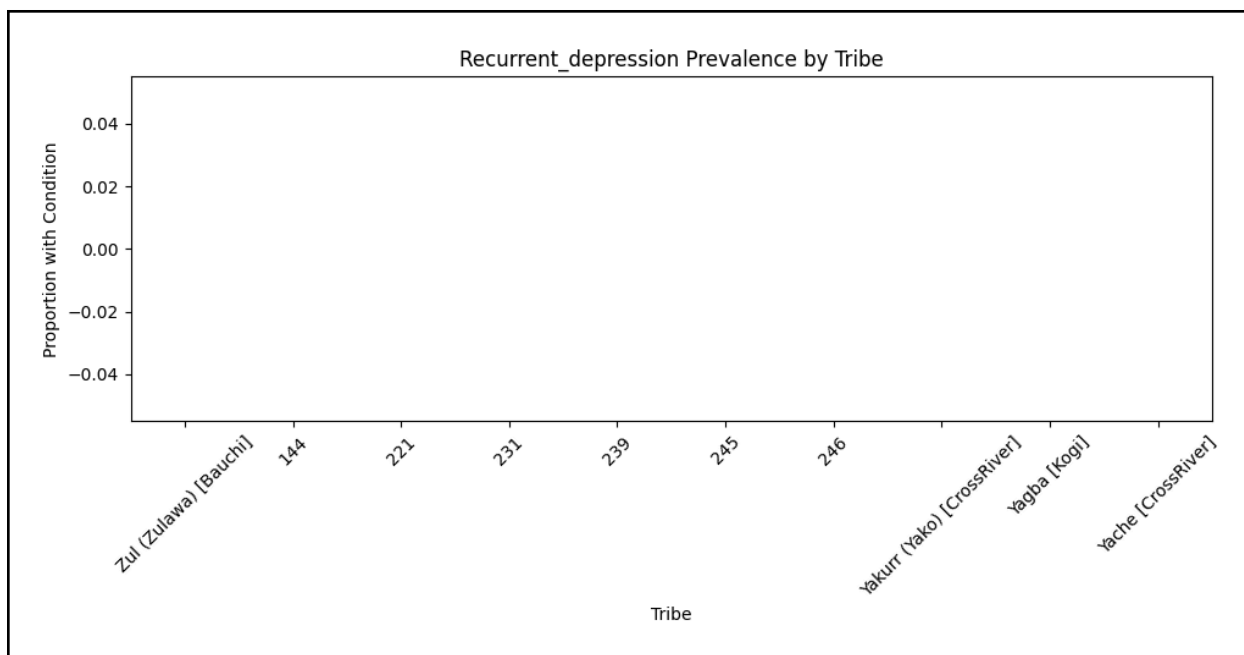
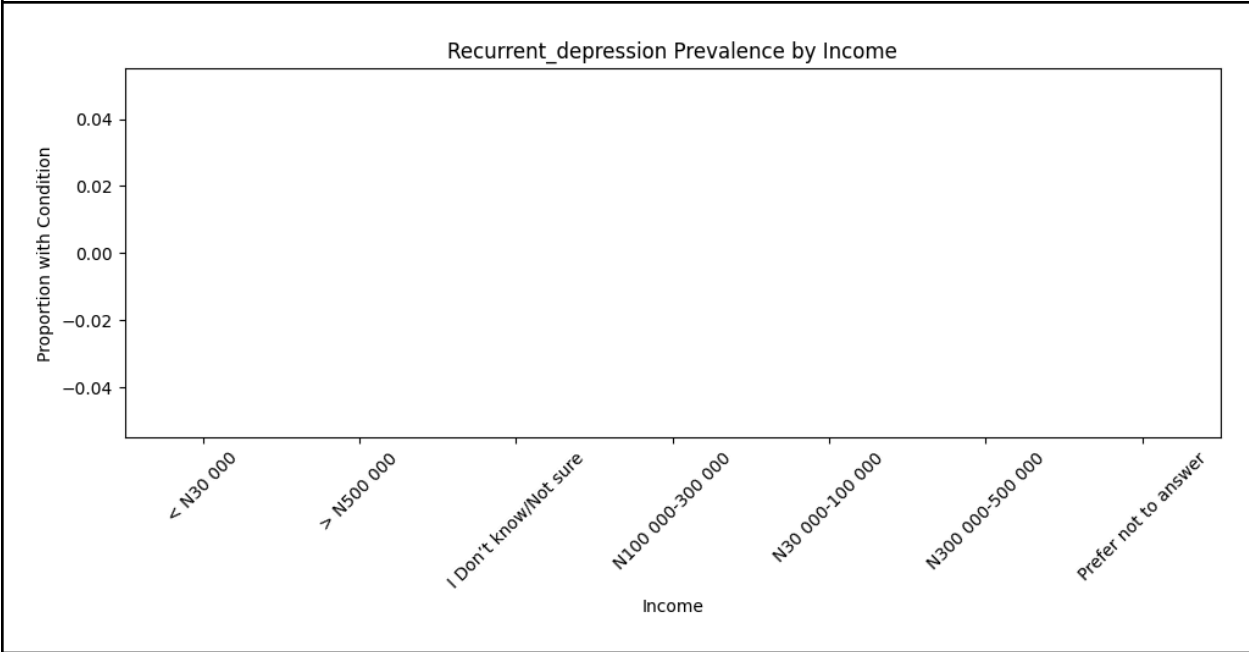
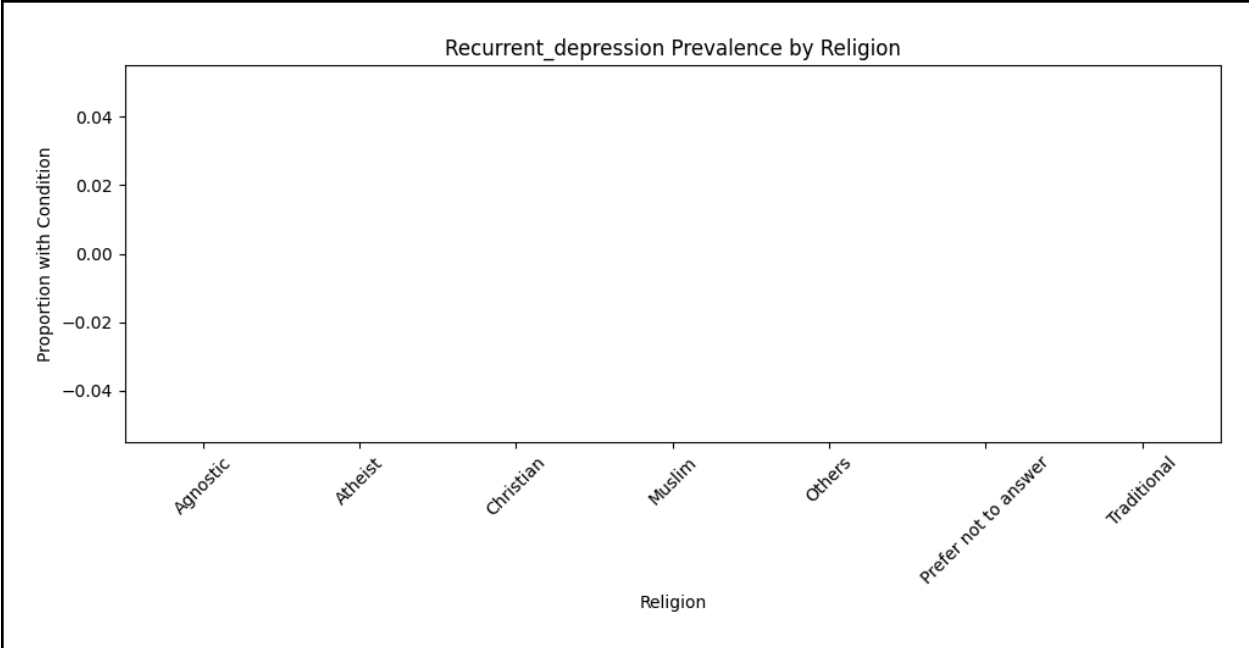


Figure 25 | Sociodemographic Distribution of Angina Pectoris Prevalence in Nigeria. Bar plots displaying the prevalence of *Angina pectoris* across different socio demographic groups: **Top-left:** By **Tribe**; **Top-right:** By **Religion**; **Bottom-left:** By **Income level**; **Bottom-right:** By **Education level**. No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





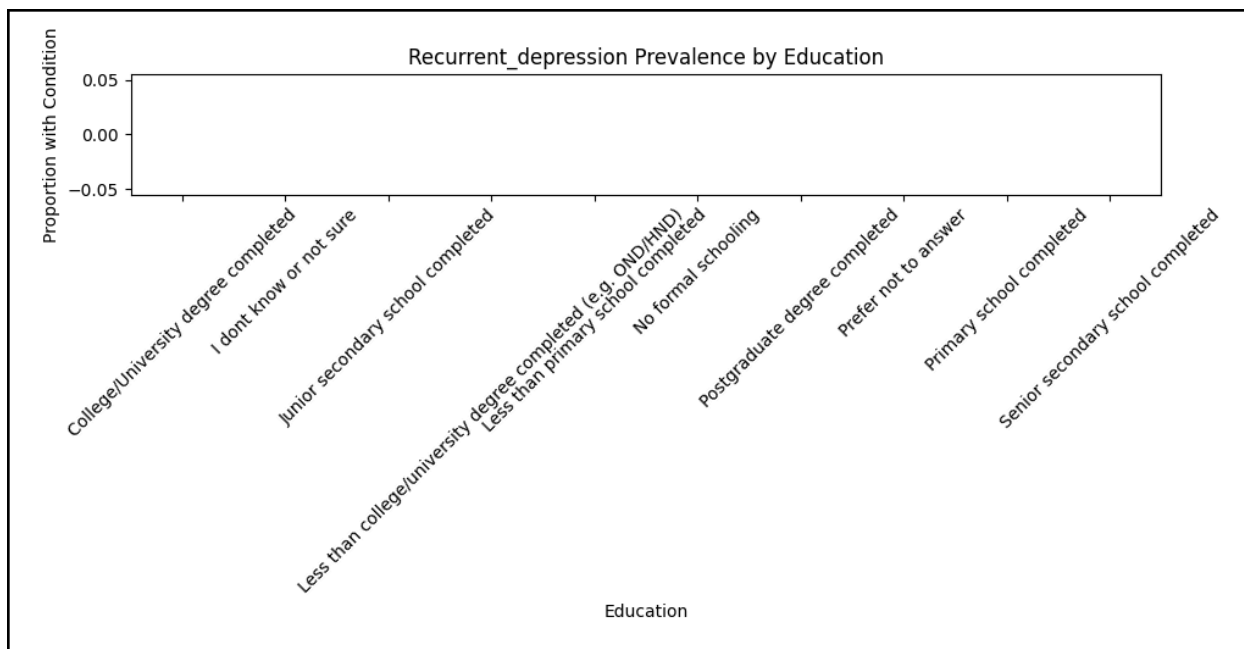
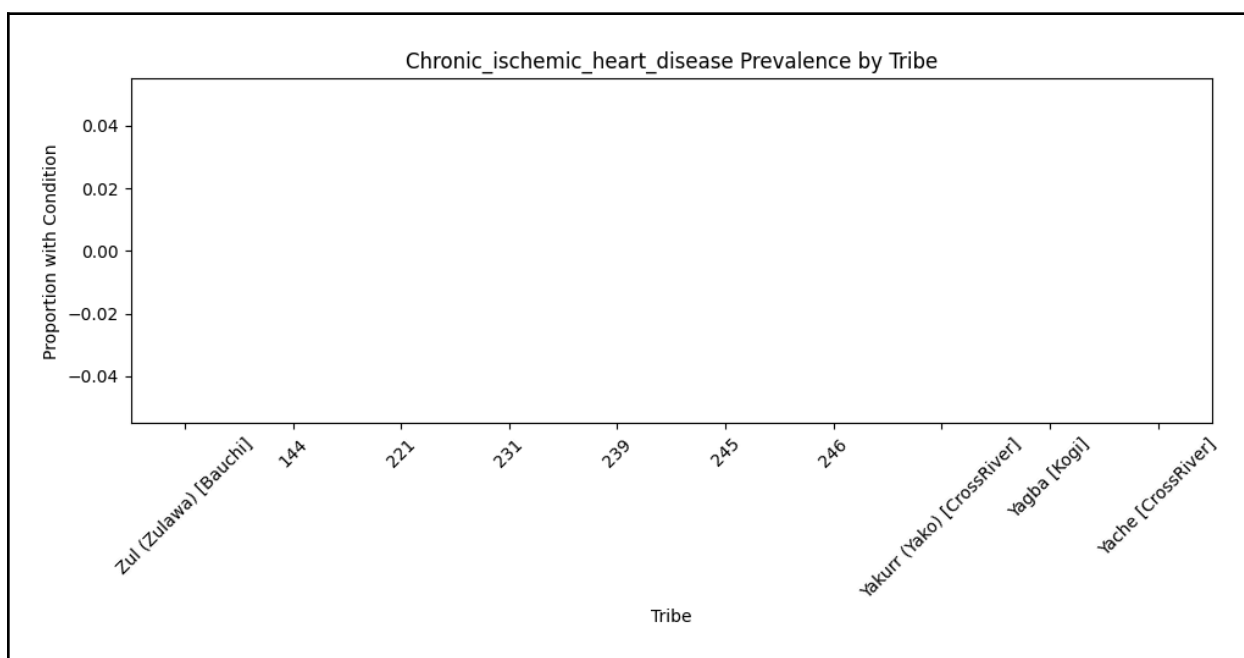
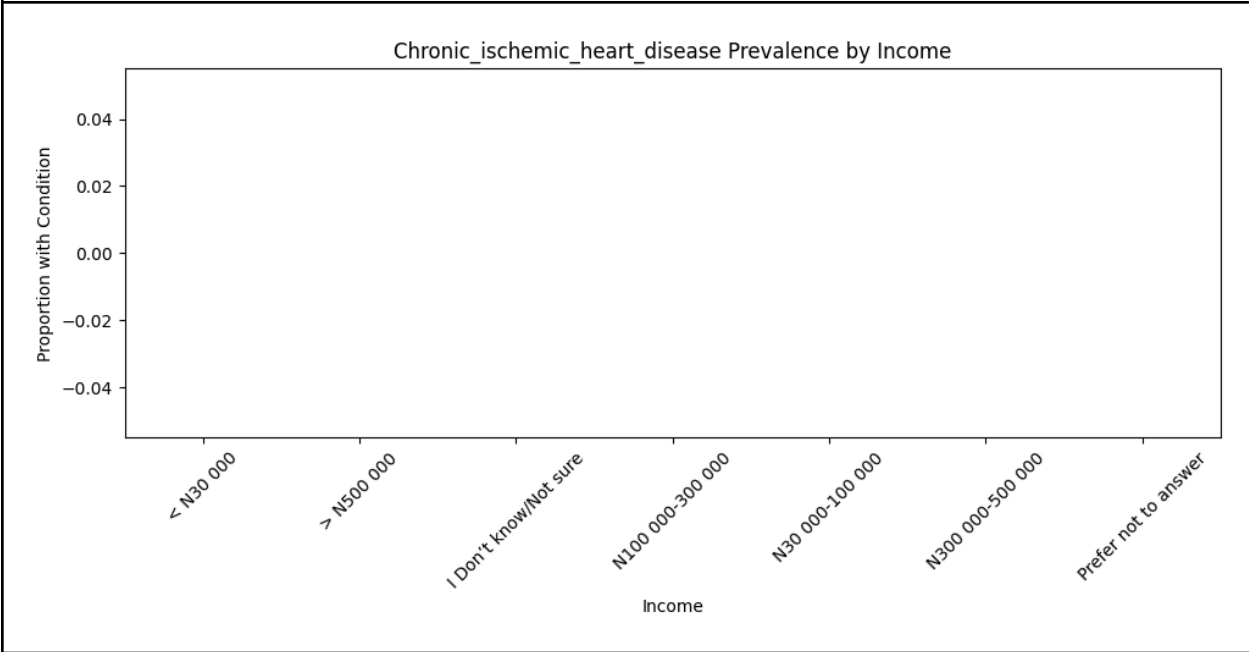
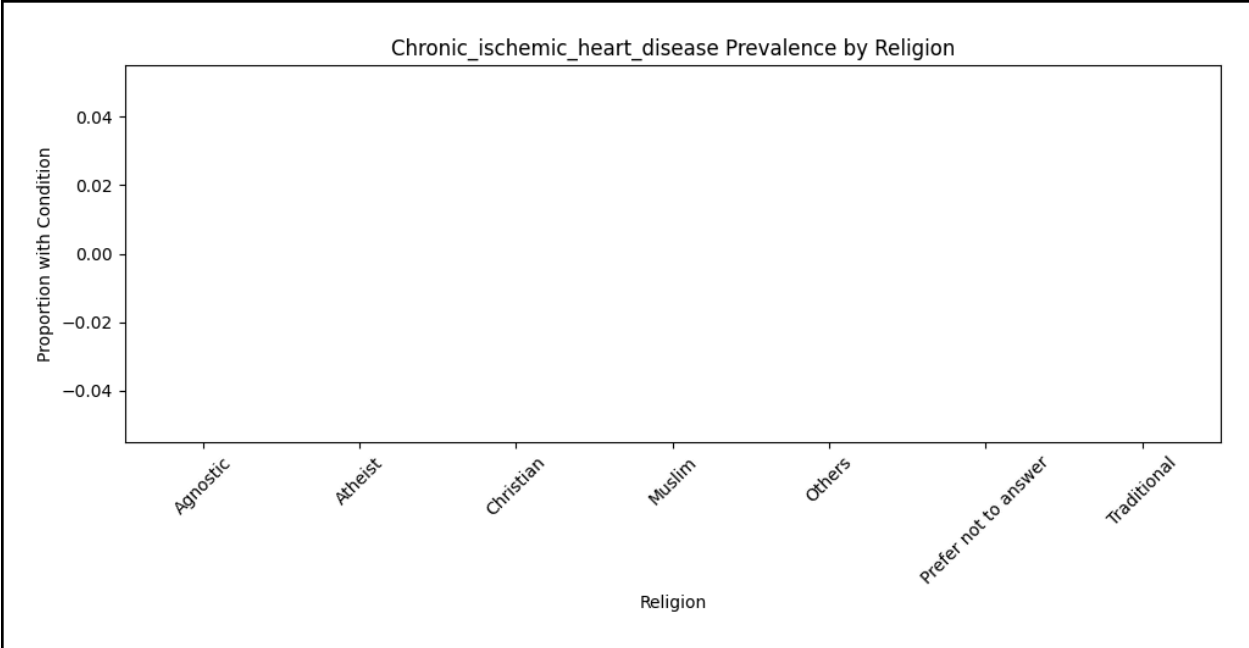


Figure 26 | Sociodemographic Distribution of Recurrent Depression Disease Prevalence in Nigeria. Bar plots displaying the prevalence of Recurrent Depression across different socio demographic groups: **Top-left: By Tribe;** **Top-right: By Religion;** **Bottom-left: By Income level;** **Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





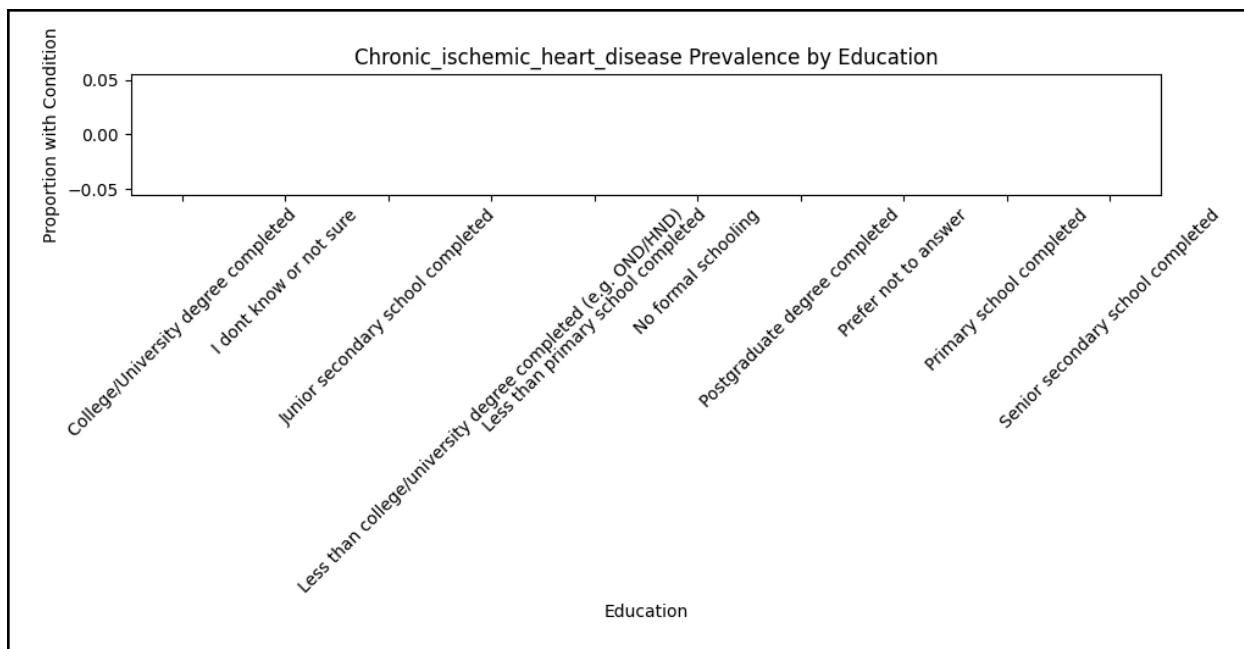
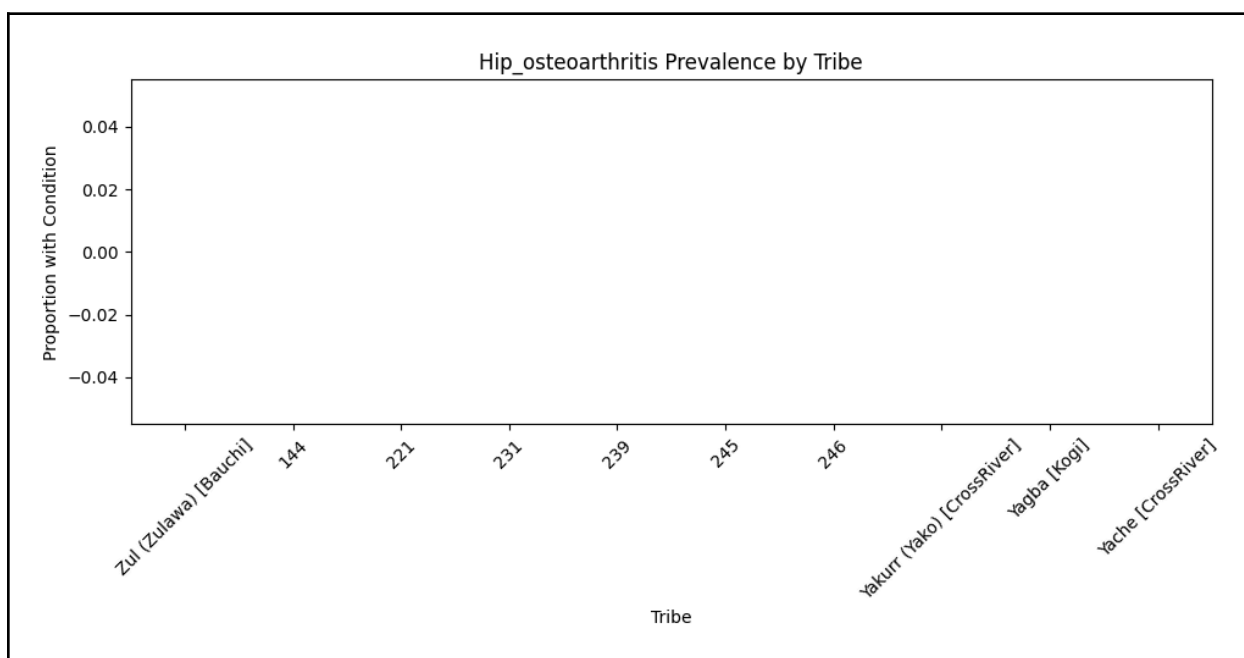
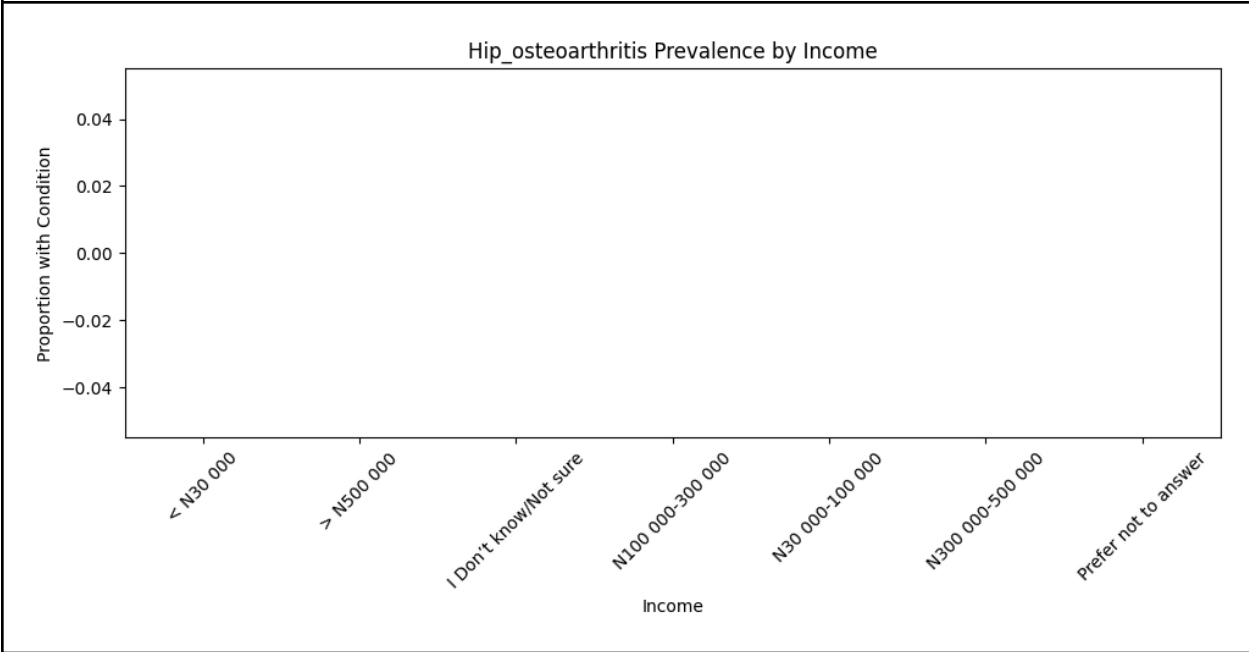
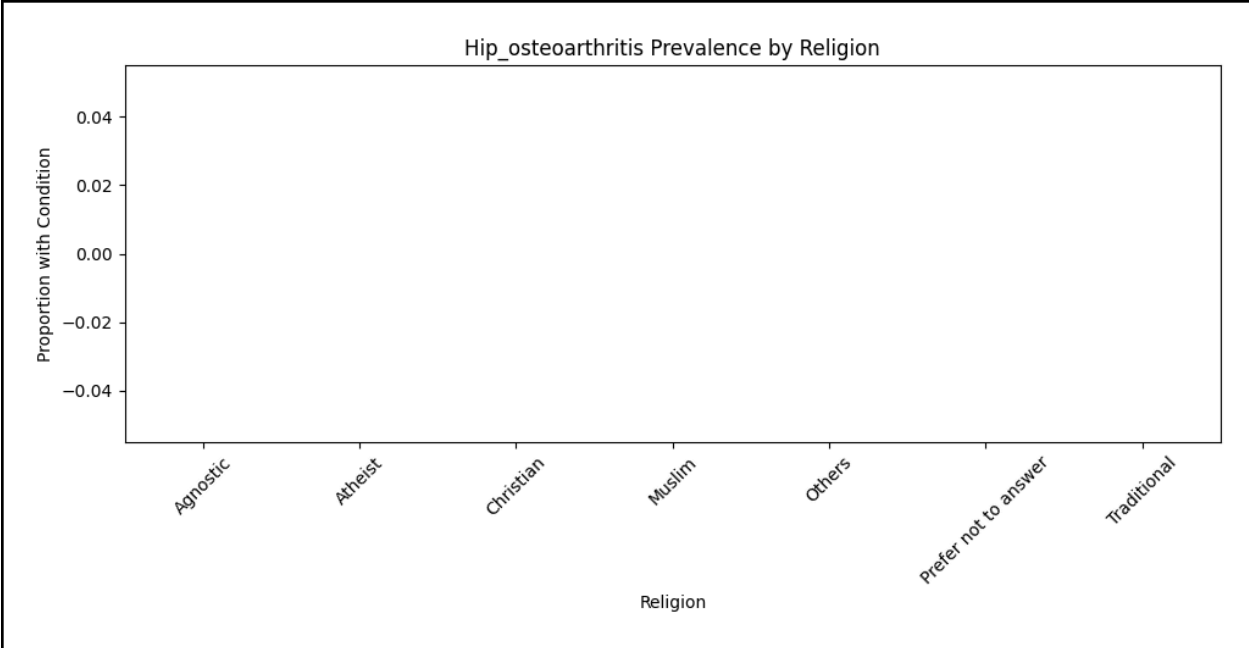


Figure 27 | Sociodemographic Distribution of Chronic Ishemic Disease Prevalence in Nigeria. Bar plots displaying the prevalence of Chronic Ishemic across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





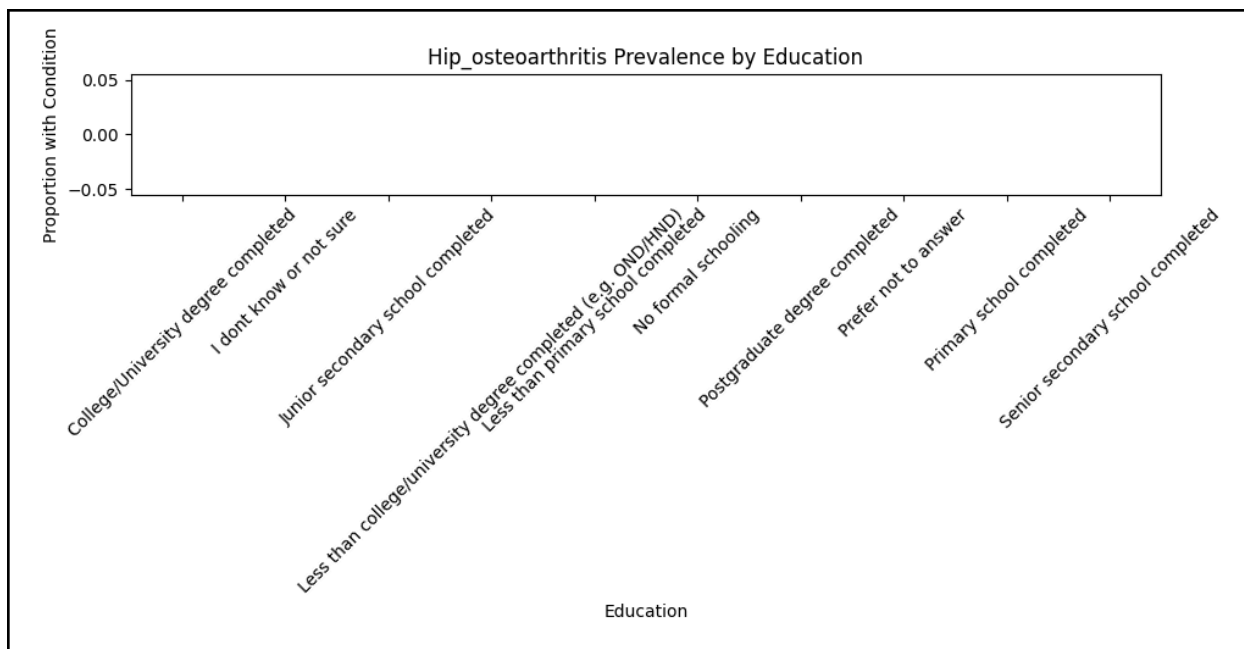
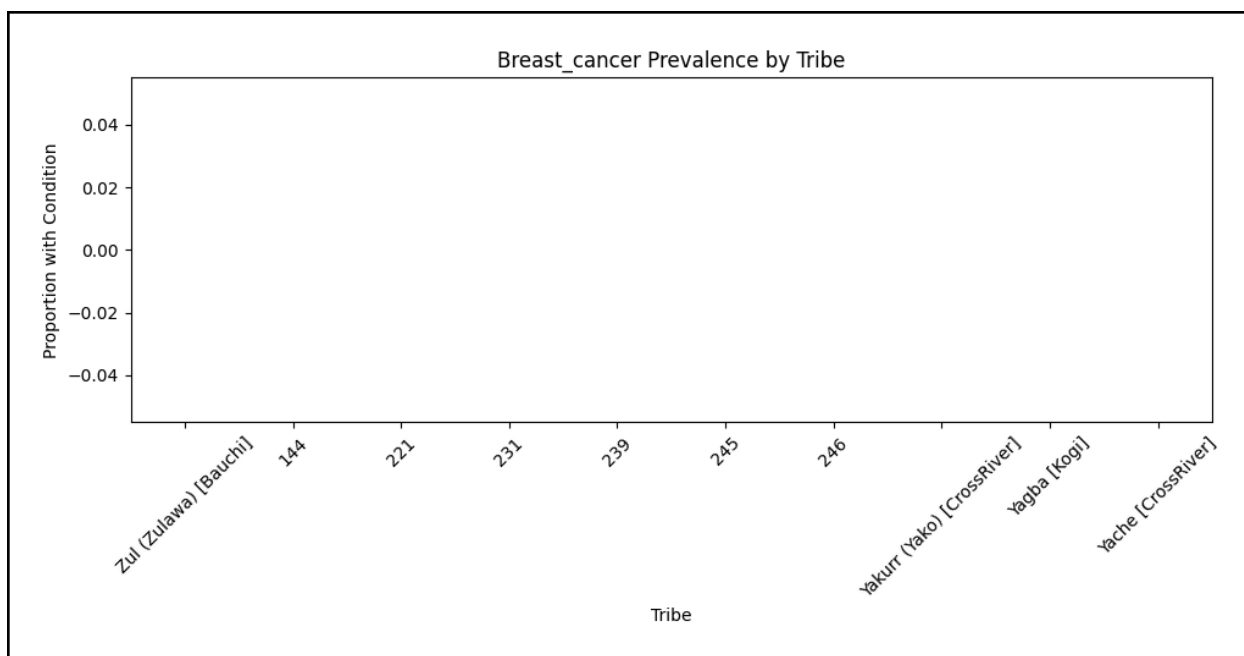
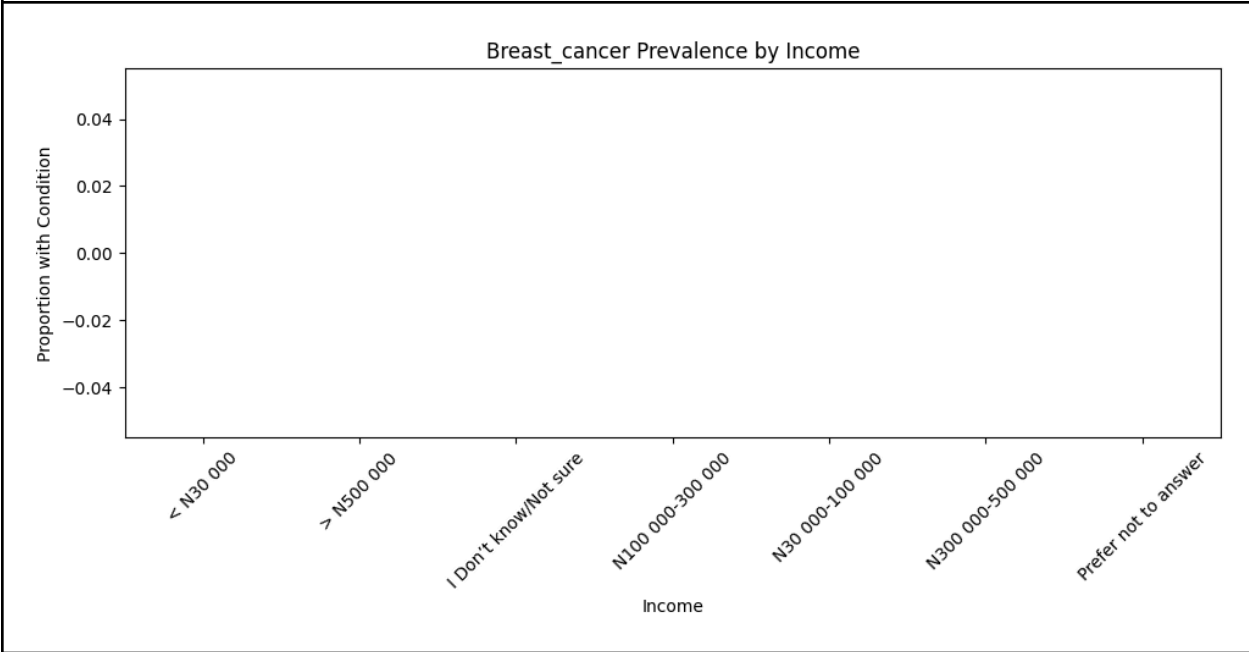
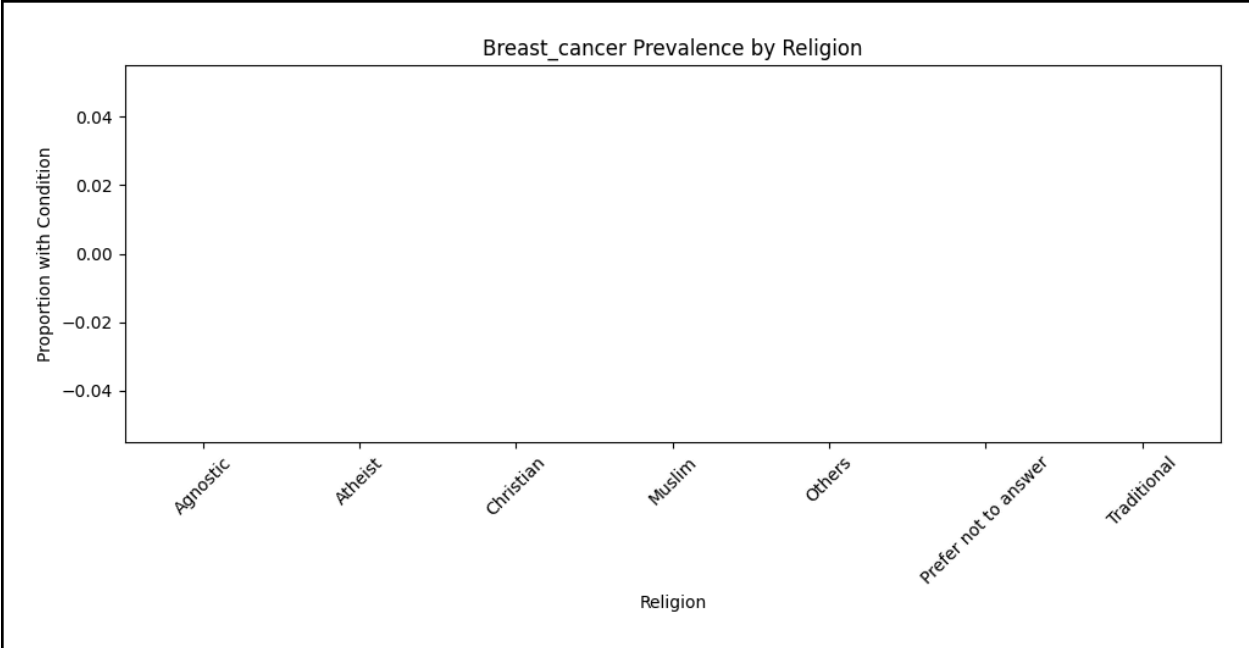


Figure 28 | Sociodemographic Distribution of Hip Osteoarthritis Disease Prevalence in Nigeria. Bar plots displaying the prevalence of **Hip Osteoarthritis** across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





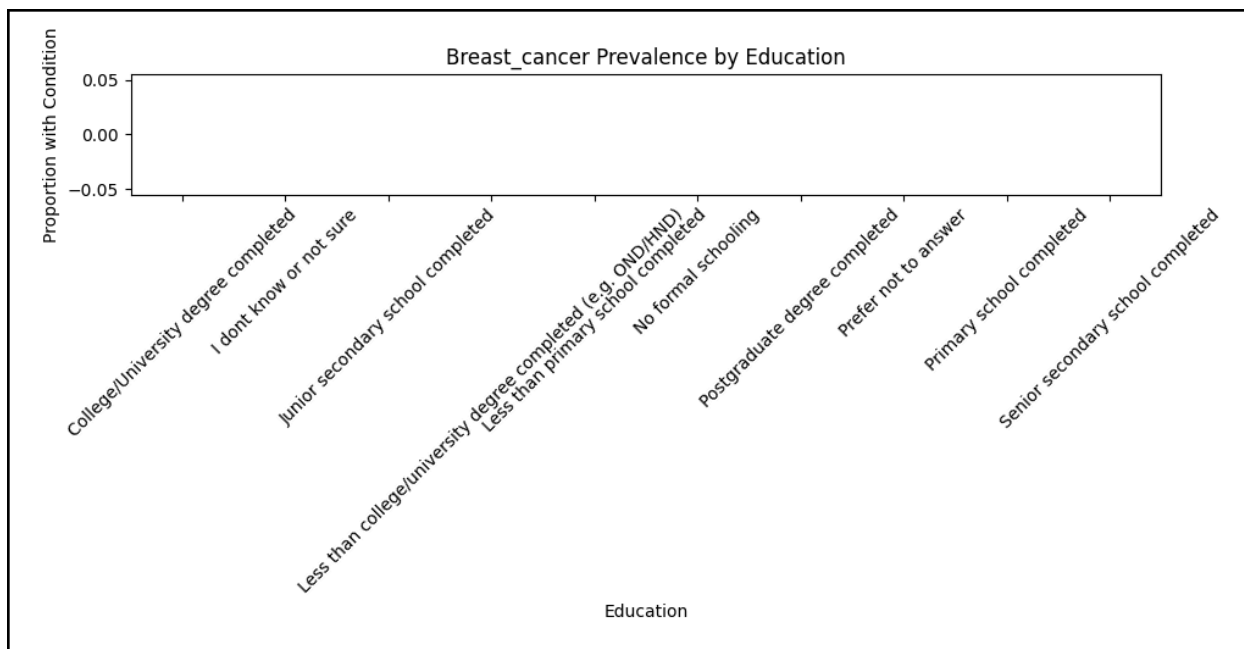
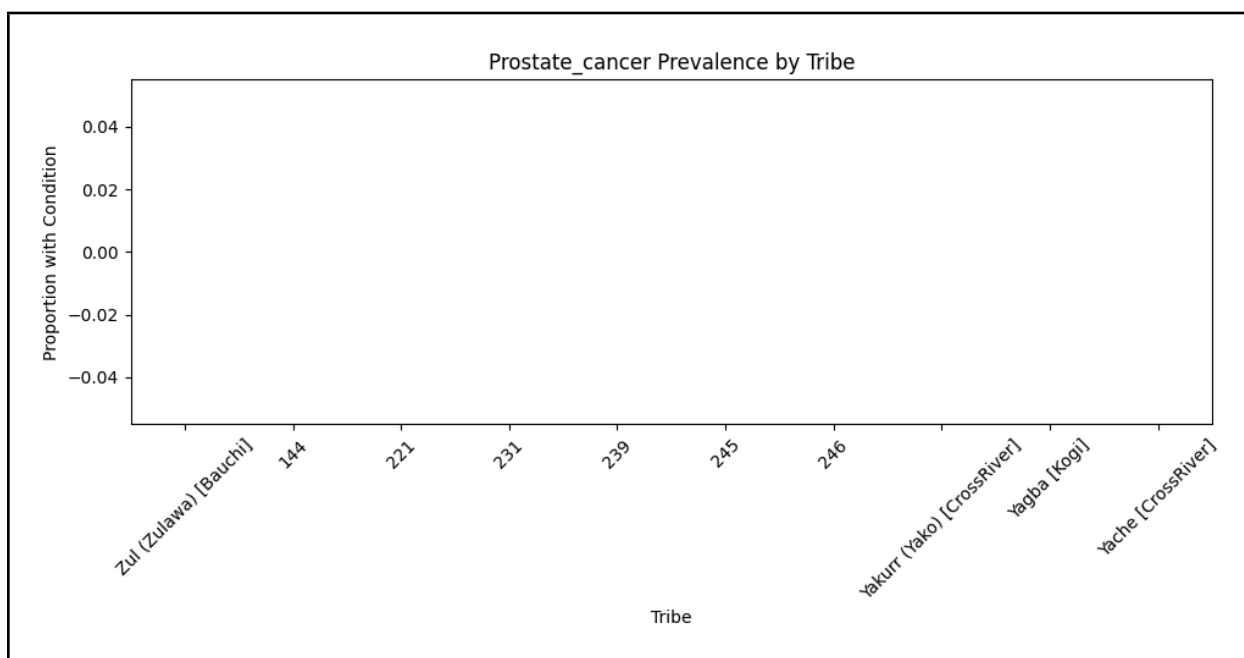
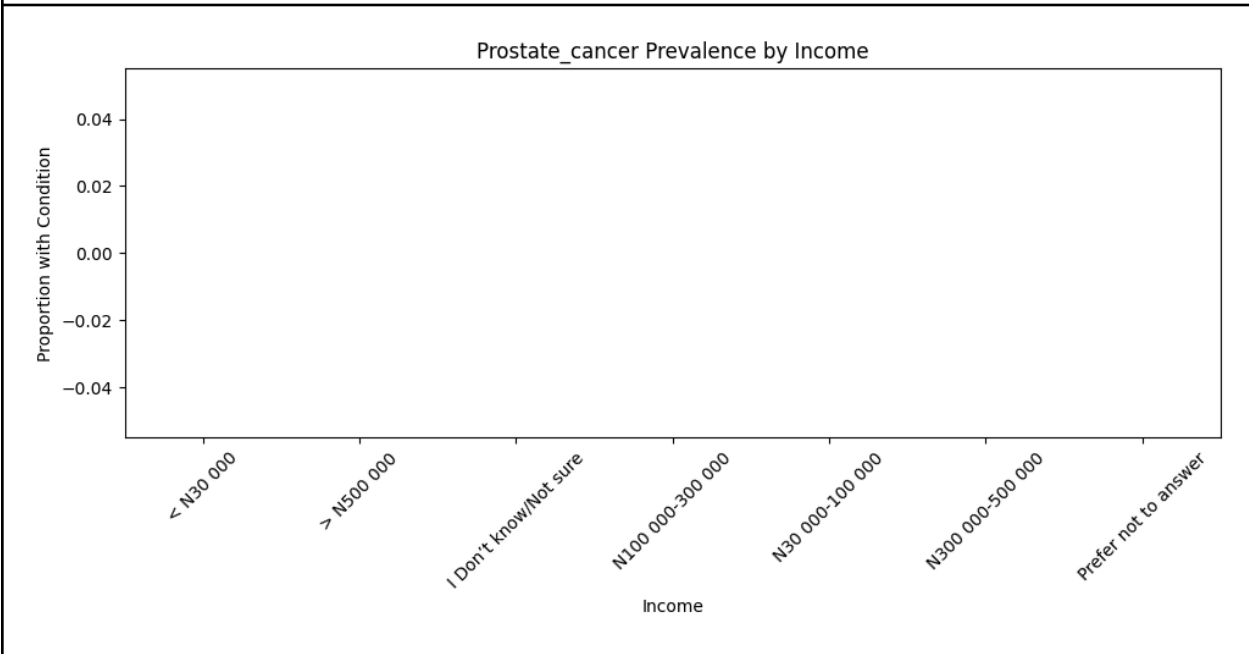
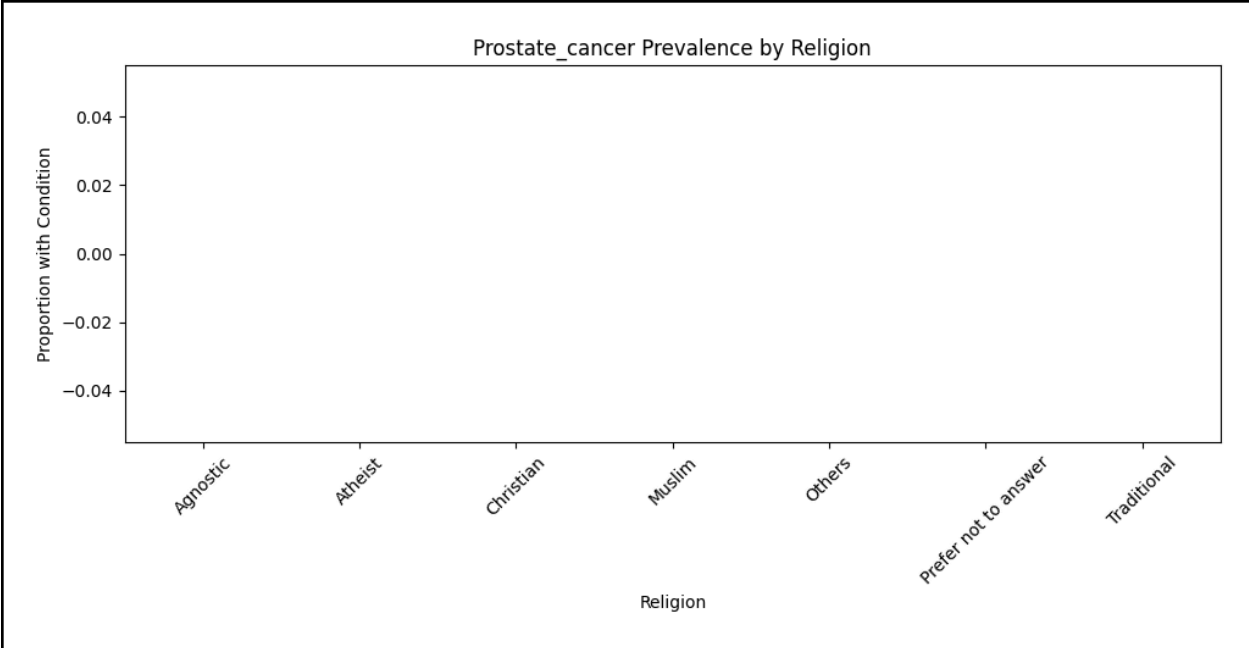


Figure 29 | Sociodemographic Distribution of Breast Cancer Prevalence in Nigeria. Bar plots displaying the prevalence of Breast Cancer across different socio demographic groups: **Top-left:** By **Tribe**; **Top-right:** By **Religion**; **Bottom-left:** By **Income level**; **Bottom-right:** By **Education level**. No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





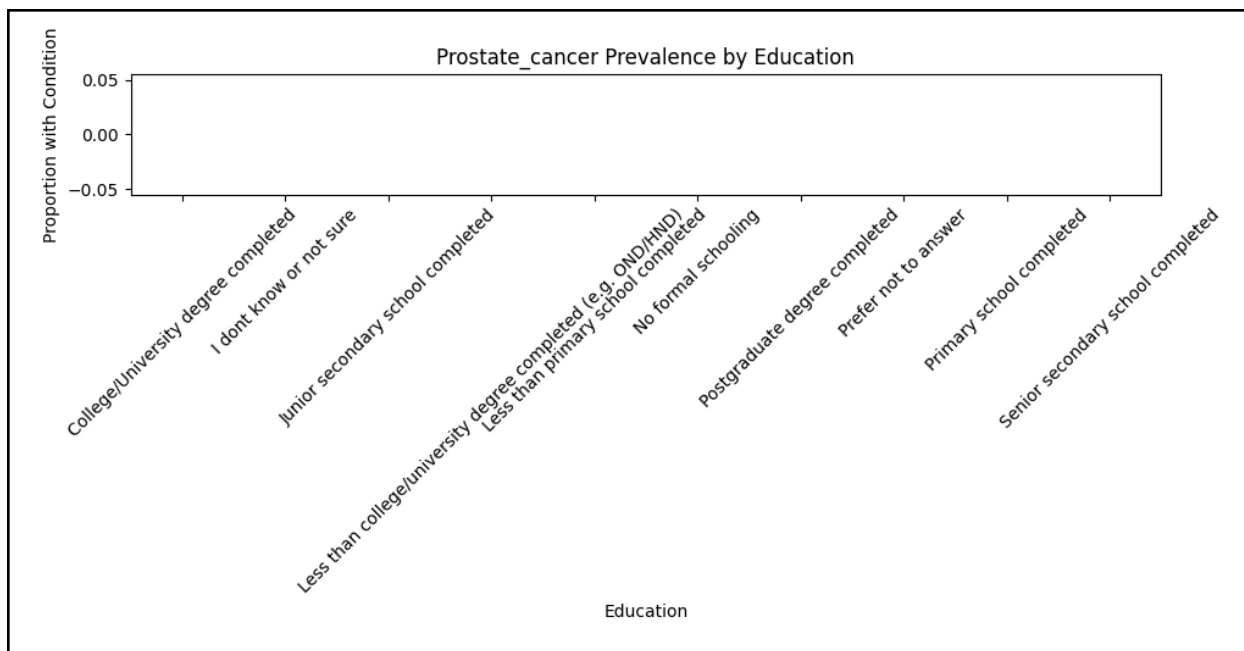
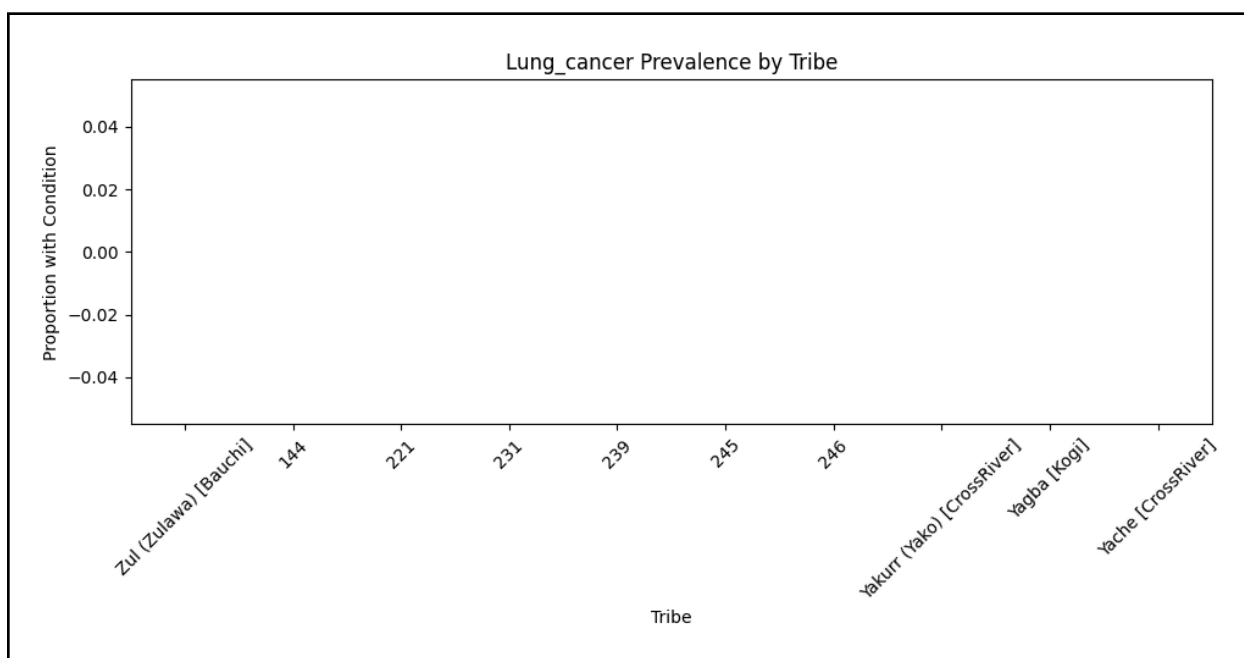
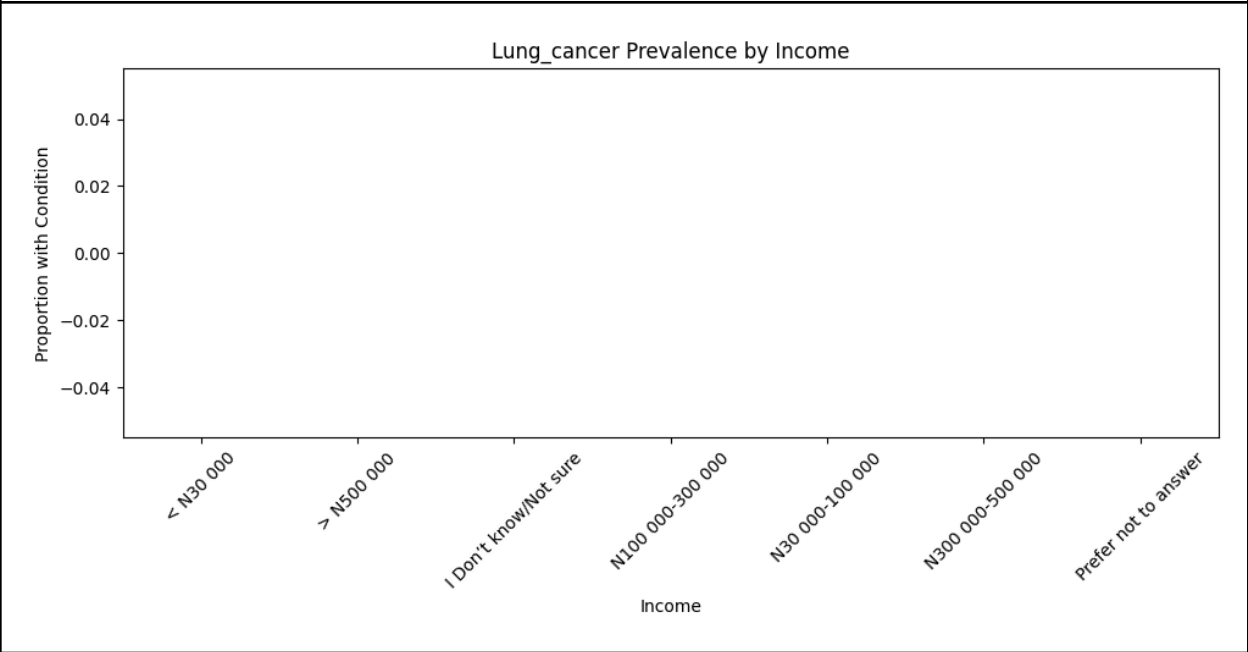
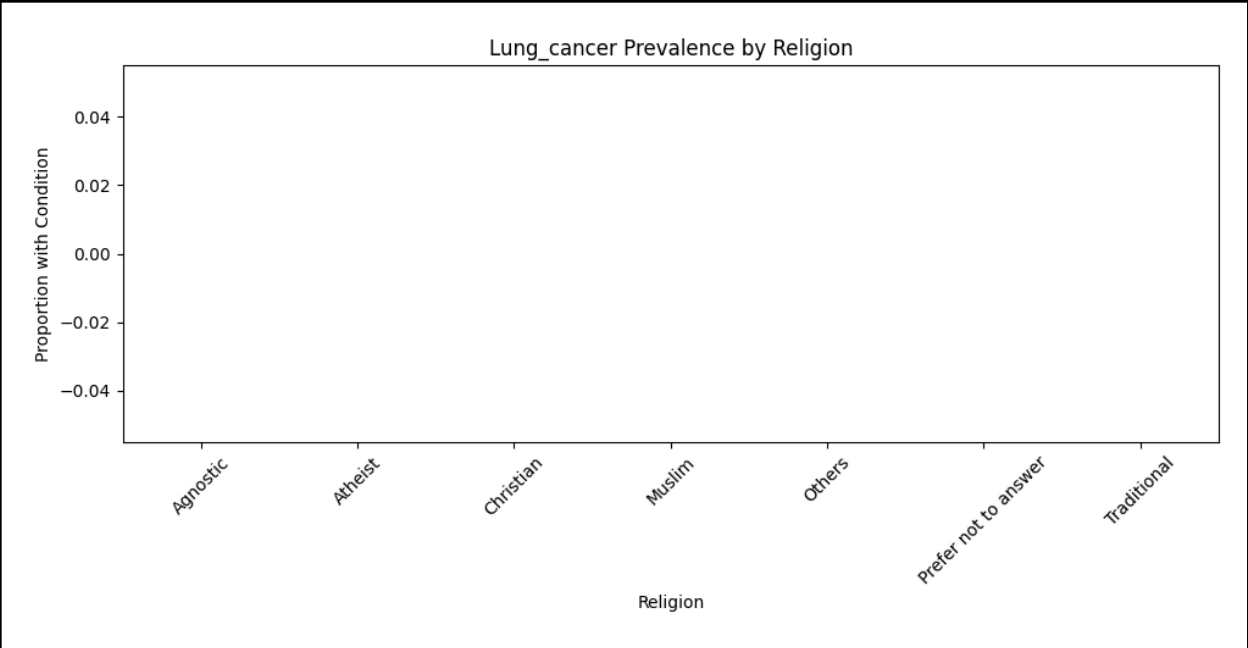


Figure 30 | Sociodemographic Distribution of Prostate Cancer Disease Prevalence in Nigeria. Bar plots displaying the prevalence of **Prostate Cancer** across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





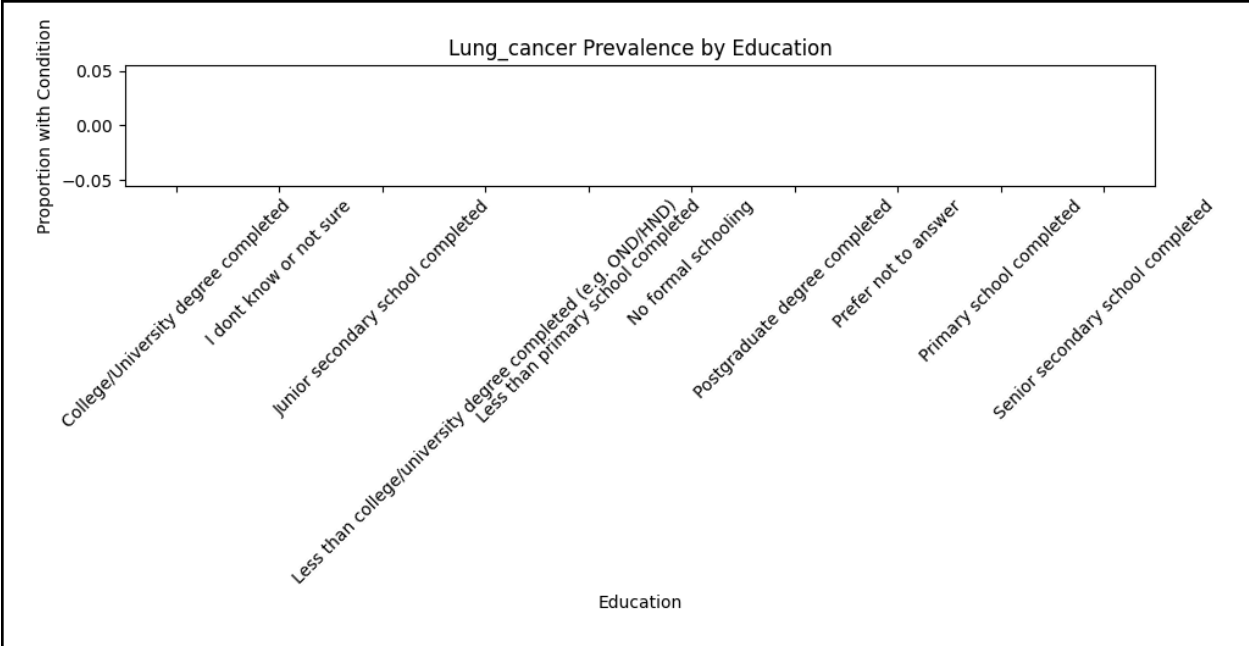
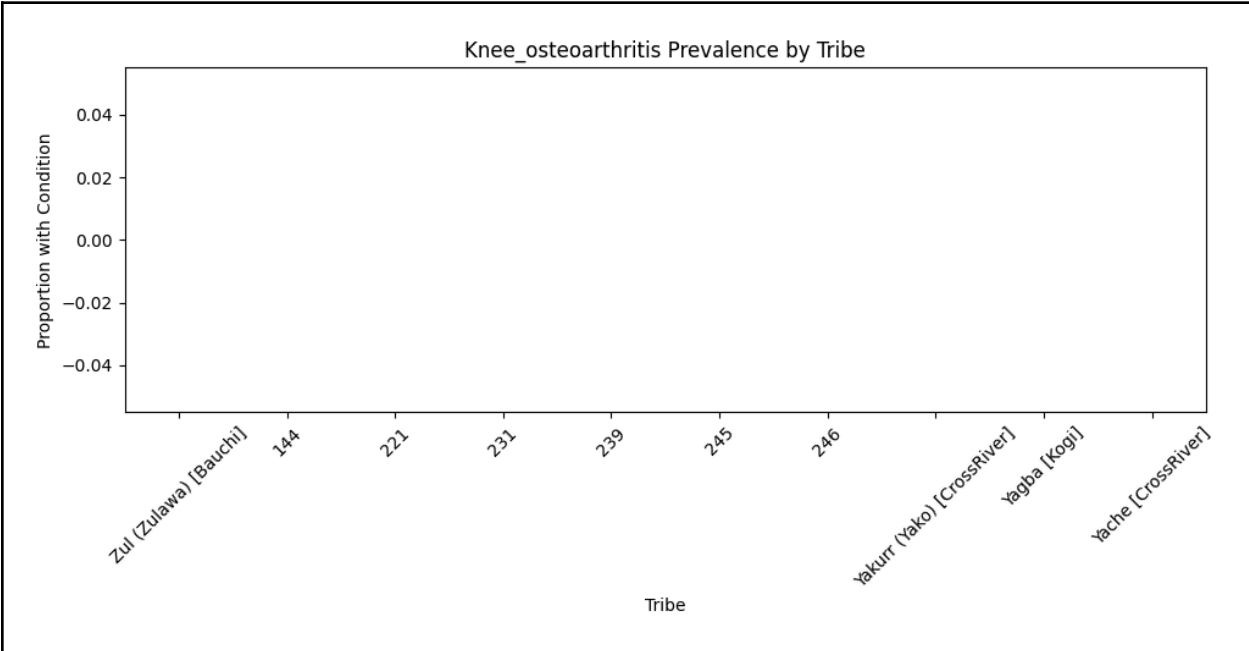
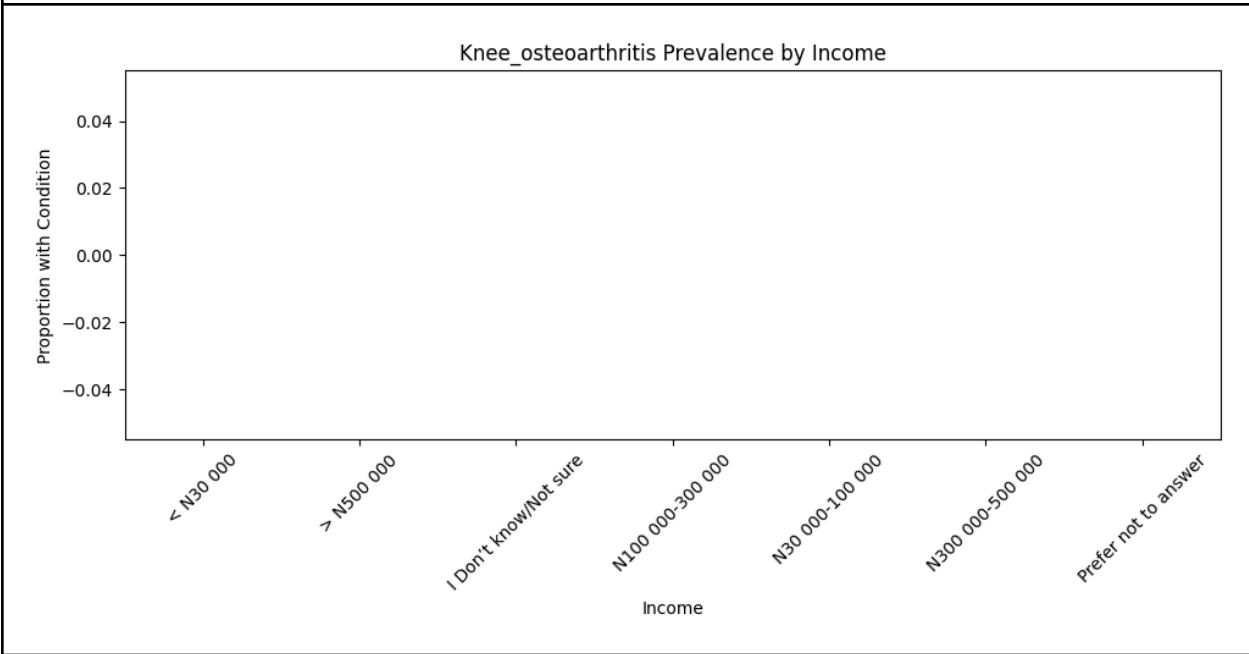
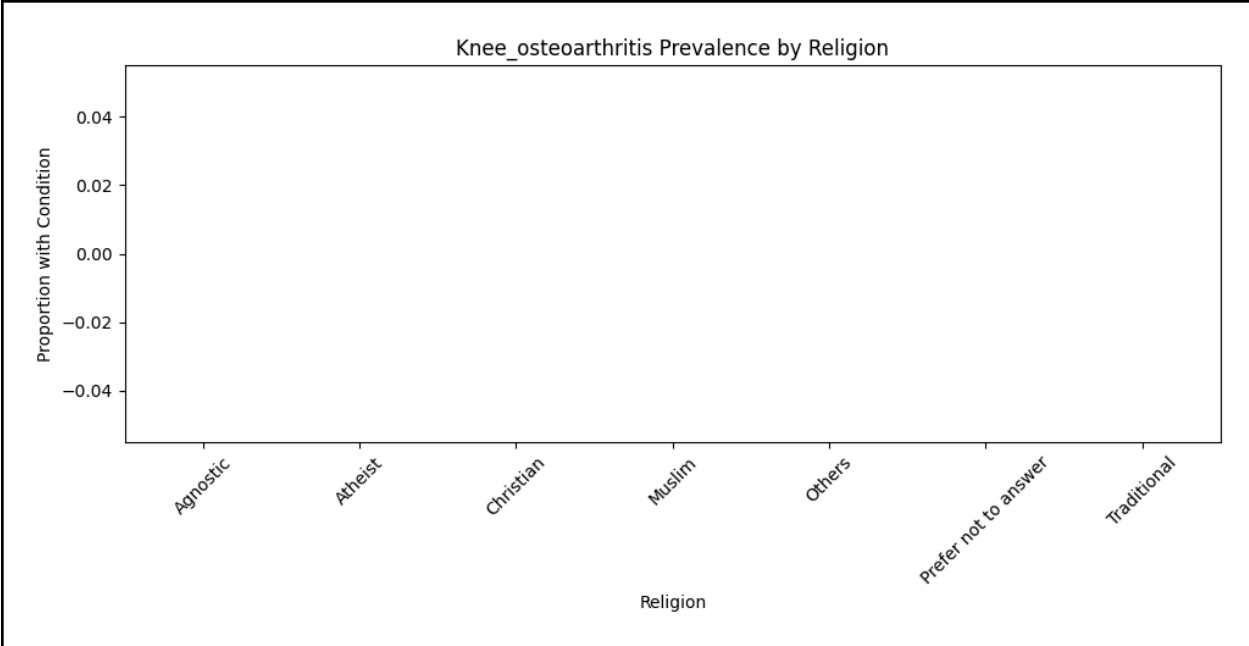


Figure 31 | Sociodemographic Distribution of Lung Cancer Disease Prevalence in Nigeria. Bar plots displaying the prevalence of Lung Cancer across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.





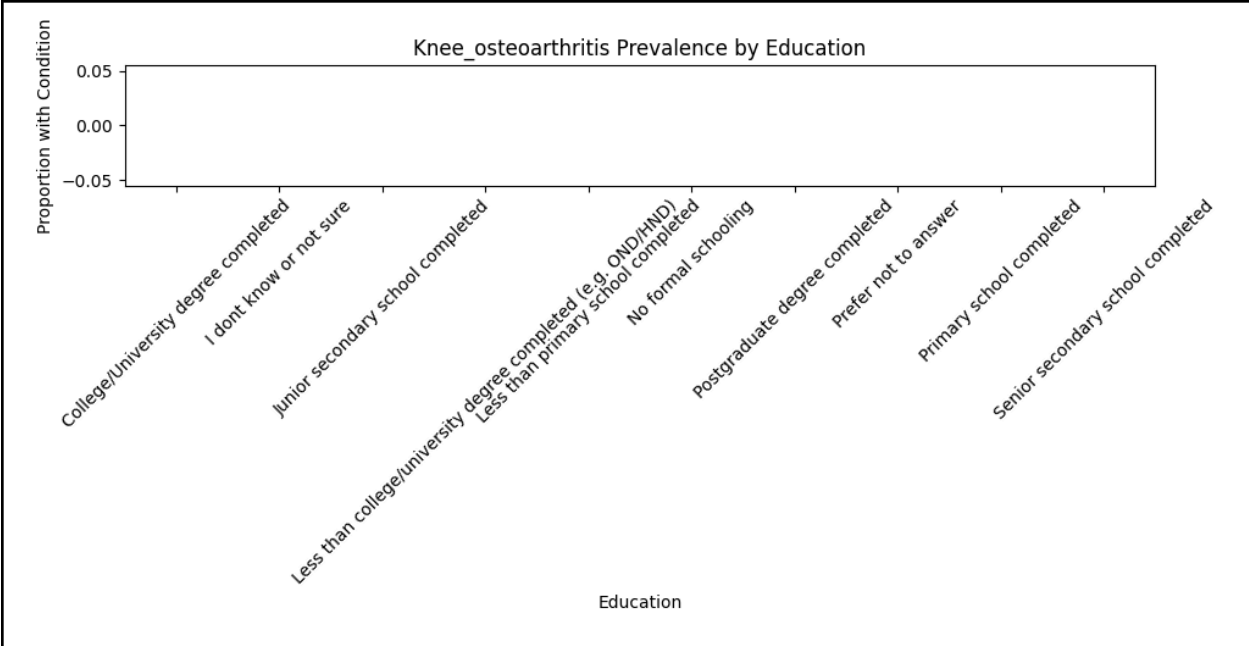


Figure 32 | Sociodemographic Distribution of Knee Osteoarthritis Disease Prevalence in Nigeria. Bar plots displaying the prevalence of Knee Osteoarthritis across different socio demographic groups: **Top-left: By Tribe; Top-right: By Religion; Bottom-left: By Income level; Bottom-right: By Education level.** No substantial prevalence is observed across any subgroup, suggesting either very low occurrence or insufficient reporting of *Angina pectoris* in the dataset.